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1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

CK APPD
DATE

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<ECO_DESCRIPTION>

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J41 MLB SCHEMATIC 6.6.0

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Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9795	1	SCHEM, MLB, J41	SCH	CRITICAL	
820-3435	1	PCBF, MLB, J41	PCB	CRITICAL	

DRAWING

TITLE=HLB

ABBREV=DRAWING

LAST MODIFIED=Thu Aug 8 10:45:45 2013

PRODUCT SAFETY REQUIREMENTS:

PCB, UL RECOGNIZED, MIN. 130-C TEMP. RATING AND V-0 FLAME RATING PER UL 796 & UL 94.

PCB TO BE SILK-SCREENED WITH UL/CUL RECOGNITION MARK, MANUFACTURER'S UL FILE NUMBER, UL PCB MATERIAL DESIGNATION, 130-C TEMP. RATING AND V-0 FLAME RATING.

DRAWING TITLE

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BOM Groups

BOM GROUP	BOM OPTIONS
MLB_COMMON	ALTERNATE, COMMON, MLB_MISC, MLB_DEBUG: ENG, MLB_PROGPARTS
MLB_MISC	PP5V5_DCIN:NO, TBTHV:P15V, EDP, CAM_XTAL:NO, CAM_WAKE:NO, APCLKRQ:ISOL, TPAD_INTWAKE:SHARED, USB_PWR:S3, SD_ON_MLB, VCORE_FETS
MLB_DEVEL: ENG	ALTERNATE, BKLT: ENG, XDP_CONN, DDRVREF_DAC, S0PGOOD_ISL, DBGLED, ISNS: ENG
MLB_DEVEL: PVT	XDP_CONN
MLB_DEBUG: ENG	DEVEL_BOM, XDP, LPCPLUS
MLB_DEBUG: PVT	DEVEL_BOM, BKLT: PROD, XDP, LPCPLUS, ISNS: PROD
MLB_DEBUG: PROD	BKLT: PROD, LPCPLUS, XDP, ISNS: PROD

Current Sensor Configuration

BOM GROUP	BOM OPTIONS
ISNS:ENG	CPU_HS_ISNS=YES,CPUVR_ISNS=YES,DRAM_ISNS=YES,P1V05_ISNS=YES,AIRPORT_ISNS=YES,S5D_ISNS=YES,LCDBKLT_ISNS=YES,P3V35S_ISNS=YES,3V350_ISNS=YES,OTHER_HS_ISNS=YES,CAP_ISNS=YES,CPUDDR_ISNS=YES,PANEL_ISNS=YES
ISNS:PROD	CPU_HS_ISNS=YES,CPUVR_ISNS=YES,DRAM_ISNS=YES,P1V05_ISNS=YES,AIRPORT_ISNS=NO,S5D_ISNS=YES,LCDBKLT_ISNS=NO,P3V35S_ISNS=NO,3V350_ISNS=NO,OTHER_HS_ISNS=NO,CAP_ISNS=NO,CPUDDR_ISNS=NO,PANEL_ISNS=NO

CPU DRAM SPD Straps

BOM GROUP	BOM OPTIONS
DDR3:HYNIX_4GB	RAMCFG0:L, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, DRAM_TYPE:HYNIX_4GB
DDR3:HYNIX_8GB	RAMCFG0:L, RAMCFG1:L, RAMCFG2:H, RAMCFG3:L, DRAM_TYPE:HYNIX_8GB
DDR3:SAMSUNG_4GB	RAMCFG0:L, RAMCFG1:H, RAMCFG2:L, RAMCFG3:L, DRAM_TYPE:SAMSUNG_4GB
DDR3:SAMSUNG_8GB	RAMCFG0:L, RAMCFG1:H, RAMCFG2:H, RAMCFG3:L, DRAM_TYPE:SAMSUNG_8GB
DDR3:ELPIDA_4GB	RAMCFG0:H, RAMCFG1:H, RAMCFG2:L, RAMCFG3:L, DRAM_TYPE:ELPIDA_4GB
DDR3:ELPIDA_8GB	RAMCFG0:H, RAMCFG1:H, RAMCFG2:H, RAMCFG3:L, DRAM_TYPE:ELPIDA_8GB
DDR3:MICRON_4GB	RAMCFG0:H, RAMCFG1:L, RAMCFG2:L, RAMCFG3:L, DRAM_TYPE:MICRON_4GB

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S0865	1	EEPROM,256KBIT,SPI,5MHZ,1.8V,2X3QFN	U2890	CRITICAL	TBTR0M:BLANK
341S3802	1	IC,EEPROM,C/R (V23.4) EVT,341/341	U2890	CRITICAL	TBTR0M:PROG
338S1159	1	IC,SMC12-A3,40MHZ/500MIPS MCU,9X9,157BGA	U5000	CRITICAL	SMC:BLANK
335S0809	1	64 MBIT SPI SERIAL DUAL I/O FLASH,8X6X8.8	U6100	CRITICAL	BOOTROM_MAC:BLANK
335S0803	1	64 MBIT SPI SERIAL DUAL I/O FLASH,8X6X8.8	U6100	CRITICAL	BOOTROM_NUM:BLANK
341S3809	1	IC,EPI ROM (V0071) DVT,341/343	U6100	CRITICAL	BOOTROM:PROG

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S4525	1	HSW, SR16M, PRQ, C0, 1.3, 15W, 2+3, 1.0, 3M, BGA	U0500	CRITICAL	CPU:1.3GHZ
337S4526	1	HSW, SR16L, PRQ, C0, 1.4, 15W, 2+3, 1.1, 3M, BGA	U0500	CRITICAL	CPU:1.4GHZ
337S4528	1	HSW, SR16M, PRQ, C0, 1.7, 15W, 2+3, 1.1, 4M, BGA	U0500	CRITICAL	CPU:1.7GHZ
338S1113	1	IC, TBT, CR -4C, B1, PRQ, C10, 288, 12X12 FC-CSP	U2800	CRITICAL	
338S1186	1	IC, BCM1570BA2, S2 PCIe CAMERA PROCESSOR	U3900	CRITICAL	
607-6811	1	ASSEMBLY, SUBASSY, PCBA, HALL EFFECT, K99	J6955	CRITICAL	J41_MLB
946-3892	1	311/313 MLB DYNAM ADHESIVE 29993-SC 0.4G	GLUE	CRITICAL	
825-7670	1	LABEL, TEXT, MLB, K21/K78	LABEL		
376S0964	2	MOSFET, N-CH, 25V, 30A, 9.6M, 8P 3.3X3.3 DFN	Q7310, Q7320	CRITICAL	VCORE_FET:REN
376S1104	2	MOSFET, N-CH, 25V, 30A, 6.1M, 8P 3.3X3.3 DFN	Q7311, Q7321	CRITICAL	VCORE_FET:REN
376S1173	2	MOSFET, N-CH, 30V, 15.3A, 12M, 8P 3.3X3.3 DFN	Q7310, Q7320	CRITICAL	VCORE_FET:VSHY
376S1174	2	MOSFET, N-CH, 30V, 22A, 6.0M, 8P 3.3X3.3 DFN	Q7311, Q7321	CRITICAL	VCORE_FET:VSHY
900-0090	1		SOLDERPASTE	CRITICAL	

DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0677	4	IC, SDRAM, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE:HYNIX_4Gb
333S0681	4	IC, SDRAM, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE:HYNIX_8Gb
333S0676	4	IC, SDRAM, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE:SAMSUNG_4Gb
333S0680	4	IC, SDRAM, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE:SAMSUNG_8Gb
333S0678	4	IC, SDRAM, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE:ELPIDA_4Gb
333S0666	4	IC, SDRAM, 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE:ELPIDA_8Gb
333S0679	4	IC, SDRAM, 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	DRAM_TYPE:MICRON_4Gb

Alternate Parts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
376S1032	376S0855		ALL	Toshiba alt for Diodes dual
376S1129	376S0855		ALL	NDX alt for Diodes dual
376S1089	376S1128		ALL	NDX alt for Diodes single
138S0684	138S0660		ALL	Murata alt to Taiyo Yuden
138S0703	138S0648		ALL	Murata alt to Taiyo Yuden
152S0586	152S1301		ALL	Dale/Vishay alt to Cyntec
372S0186	372S0185		ALL	NDX alt to Diodes
197S0479	197S0478		ALL	200uW Epsom alt to NDK
376S1053	376S0604		ALL	Diodes alt to Fairchild
371S0713	371S0558		ALL	Diodes alt to ST Micro
128S0371	128S0376		ALL	Kemet alt to Sanyo
128S0394	128S0415		ALL	NEC alt to Sanyo
152S1821	152S1757		ALL	Cyntec alt to NEC
197S0480	197S0343		ALL	NDK crystal alt to TXC
197S0481	197S0343		ALL	Epsom crystal alt to TXC
107S0254	107S0241		ALL	Cyntec sense R alt to TFT
353S3452	353S1286		ALL	Maxim alt to Microchip
128S0386	128S0284		ALL	Kemet alt to Sanyo
128S0397	128S0325		ALL	Kemet alt to Sanyo
377S0155	377S0104		ALL	OnSemi alt to Infineon
128S0398	128S0220		ALL	Kemet alt to Sanyo
197S0542	197S0544		ALL	NDK alt to TXC
197S0545	197S0544		ALL	Epsom alt to TXC
138S0681	138S0638		ALL	Taiyo alt to Samsung
138S0841	138S0638		ALL	Murata alt to Samsung
376S1180	376S0761		ALL	Renesas alt to Vishay
152S1876	152S1804		ALL	TDK alt to Toko
107S0255	107S0240		ALL	Cyntec alt to TFT
107S0250	107S0248		ALL	Cyntec alt to TFT

CPU DRAM CFG Chart

	VENDOR	CFG 1	CFG 0
	HYNIX	0	0
	SAMSUNG	1	0
	MICRON	0	1
	ELPIDA	1	1

SIZE	CFG 2
4GB	0
8GB	1

DIE REV	CFG 3
A	0
B	1

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BOM Variants

BOM NUMBER	BOM NAME	BOM OPTIONS
639-4118	PCBA, MLB, BEST, HY 4GB, J41:	MLB_CMNPTS, CPU: 1.7GHZ, DDR3: HYNIX_4GB
639-4274	PCBA, MLB, BEST, HY 8GB, J41	MLB_CMNPTS, CPU: 1.7GHZ, DDR3: HYNIX_8GB
639-4275	PCBA, MLB, BEST, EL 4GB, J41	MLB_CMNPTS, CPU: 1.7GHZ, DDR3: ELPIDA_4GB
639-4276	PCBA, MLB, BEST, EL 8GB, J41	MLB_CMNPTS, CPU: 1.7GHZ, DDR3: ELPIDA_8GB
639-4702	PCBA, MLB, BEST, MI 4GB, J41	MLB_CMNPTS, CPU: 1.7GHZ, DDR3: MICRON_4GB
639-4434	PCBA, MLB, BETTER, HY 4GB, J41	MLB_CMNPTS, CPU: 1.3GHZ, DDR3: HYNIX_4GB
639-4435	PCBA, MLB, BETTER, HY 8GB, J41	MLB_CMNPTS, CPU: 1.3GHZ, DDR3: HYNIX_8GB
639-4436	PCBA, MLB, BETTER, EL 4GB, J41	MLB_CMNPTS, CPU: 1.3GHZ, DDR3: ELPIDA_4GB
639-4437	PCBA, MLB, BETTER, EL 8GB, J41	MLB_CMNPTS, CPU: 1.3GHZ, DDR3: ELPIDA_8GB
639-4703	PCBA, MLB, BETTER, MI 4GB, J41	MLB_CMNPTS, CPU: 1.3GHZ, DDR3: MICRON_4GB
685-0024	CMN PTS, PCBA, MLB, J41	MLB_COMMON, J41_MLB
985-0017	J41 MLB DEVELOPMENT BOM	MLB_DEVEL:ENG
685-0062	VCORE FET, REN, J41	VCORE_FET:REN
685-0063	VCORE FET, VSHY, J41	VCORE_FET:VSHY

BOM Groups

BOM GROUP	BOM OPTIONS
MLB_PROGPARTS	BOOTROM:PROG, SMC:PROG, TBTR0M:PROG

Programmable Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S3757	1	IC, SMC-A3 SCPL, EXT, V22. 12A18, PROTO 1, J41	U5000	CRITICAL	SMC:PROG

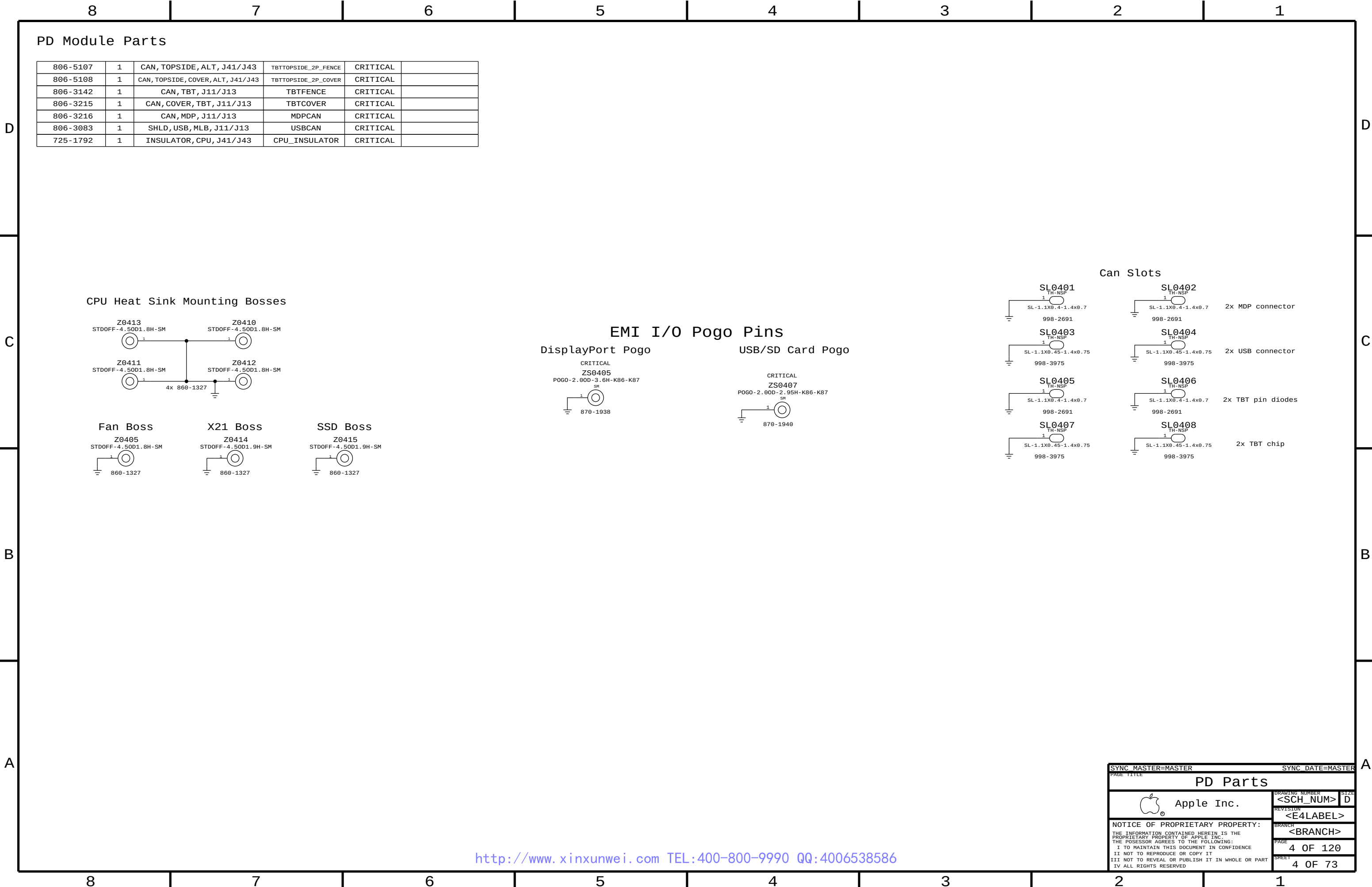
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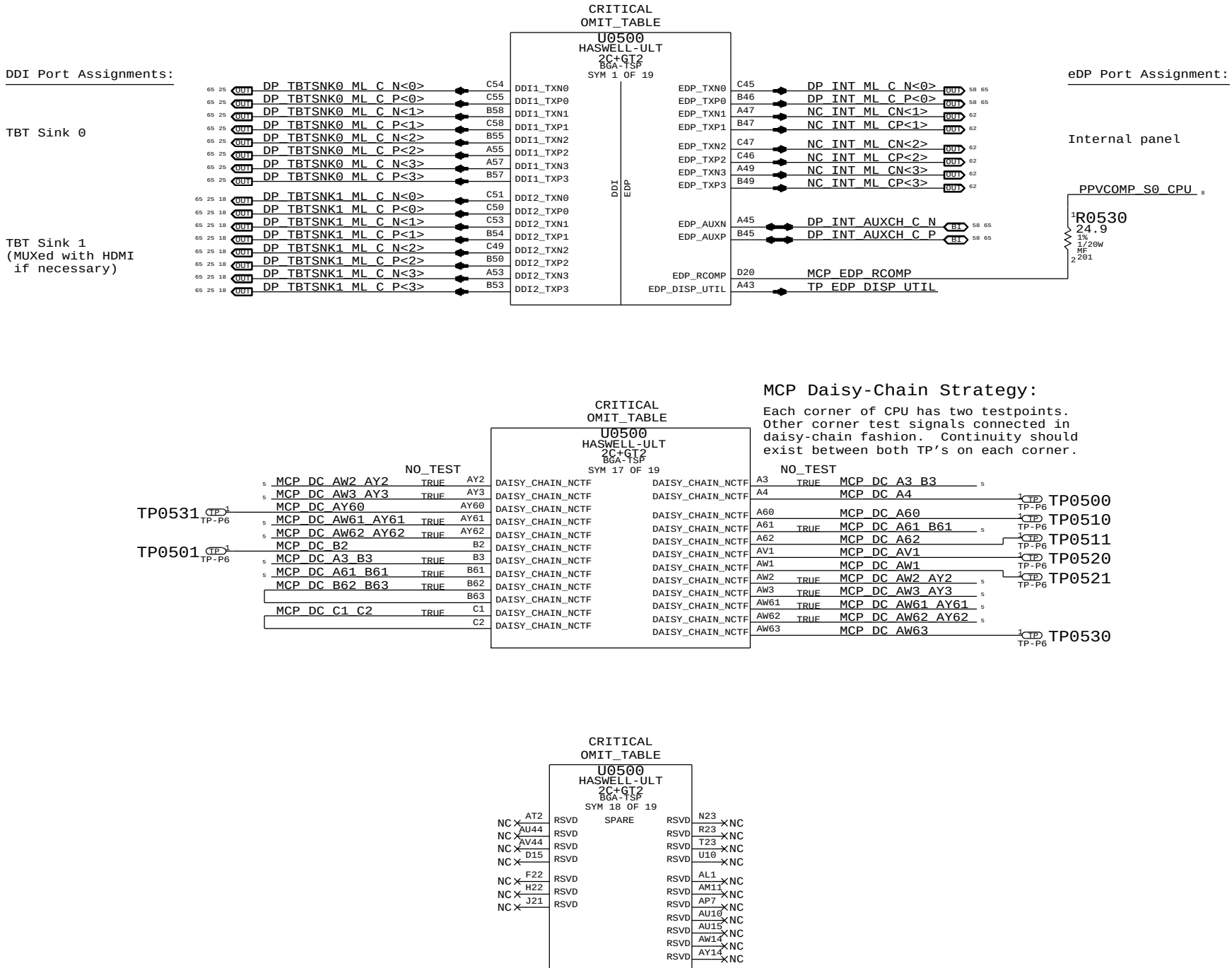
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
985-0017	1	J41 MLB DEVELOPMENT BOM	DEVEL	CRITICAL	DEVEL_BOM
685-0024	1	CMN PTS,PCBA,MLB,J41	CMNPTS	CRITICAL	MLB_CMNPTS
685-0063	1	VCORE FET,VSHY,J41	VCOREFETS	CRITICAL	VCORE_FETS

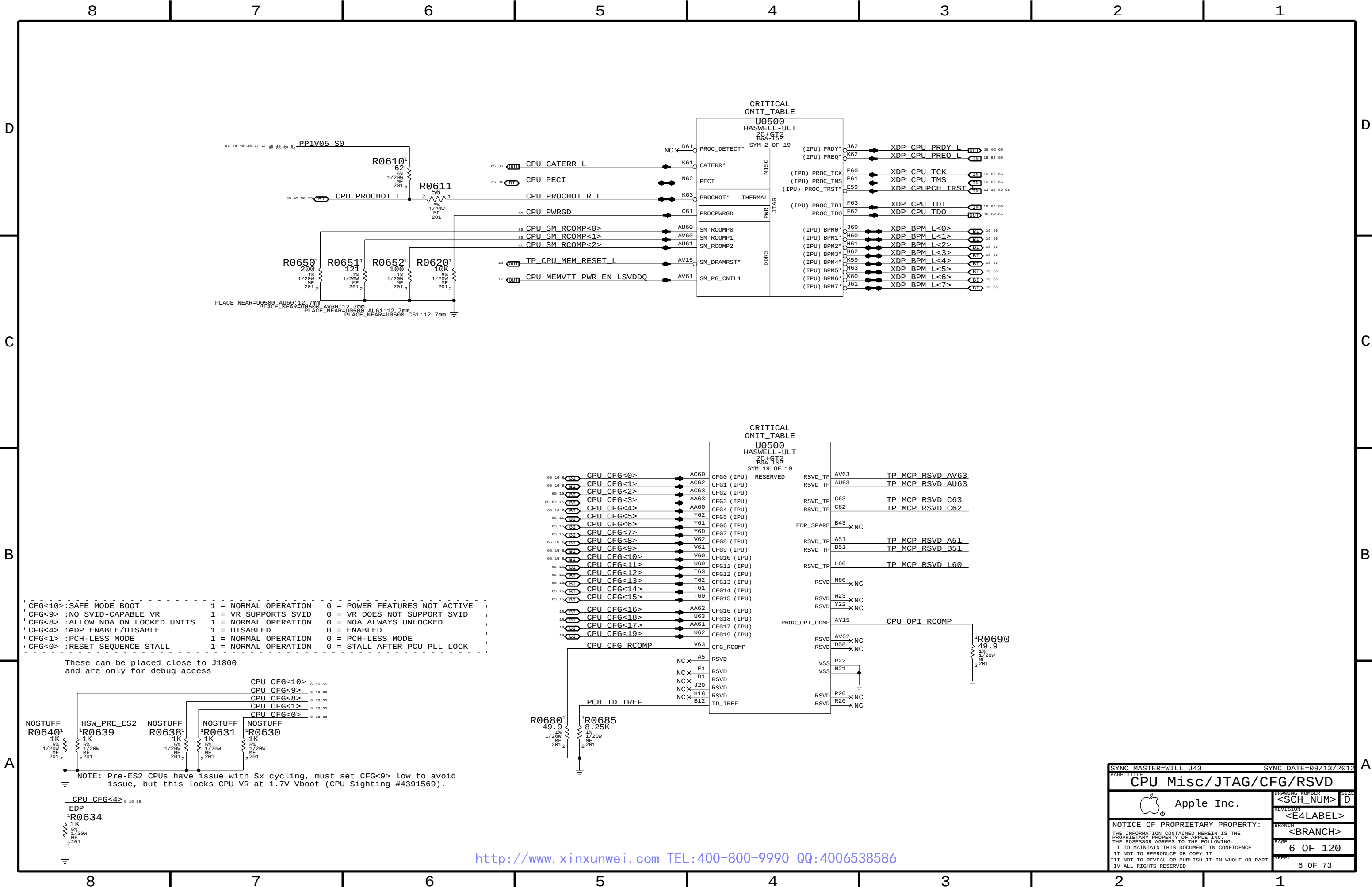
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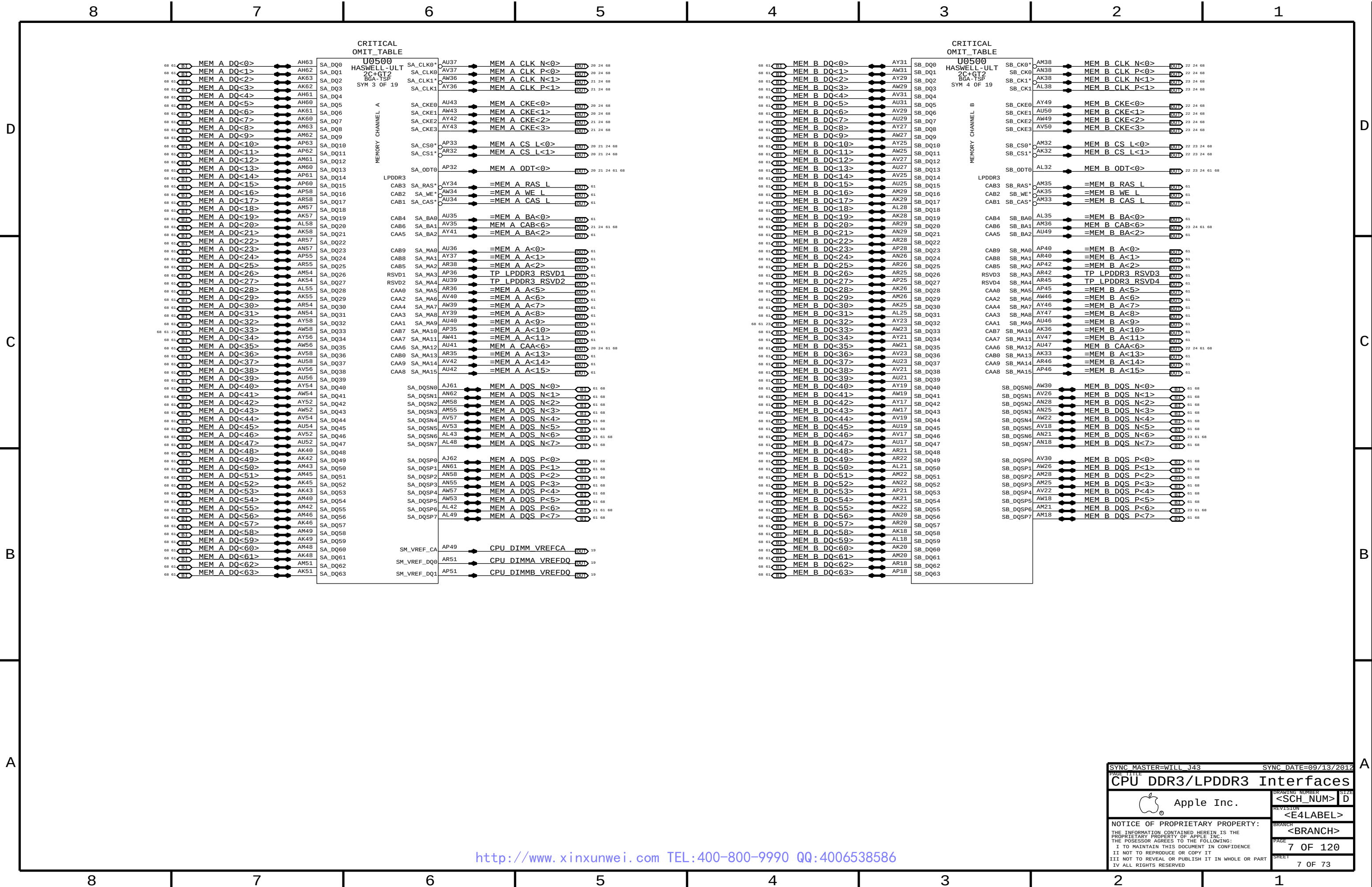
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-0062	685-0063		ALL	Renesas alt to Vishay

333S0704	333S0700		ALL	Elpida CAM DRAM alt to Hynix
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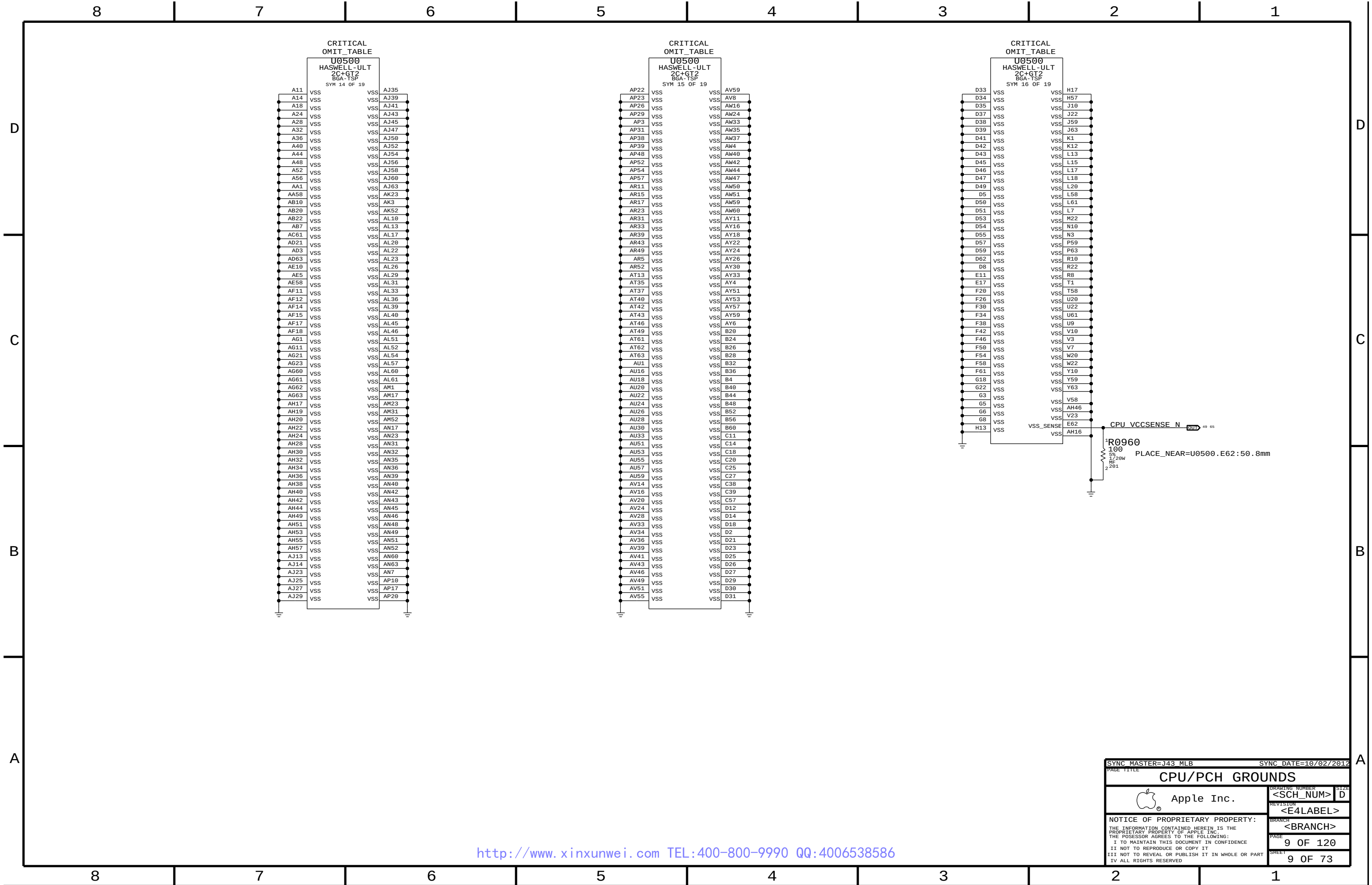


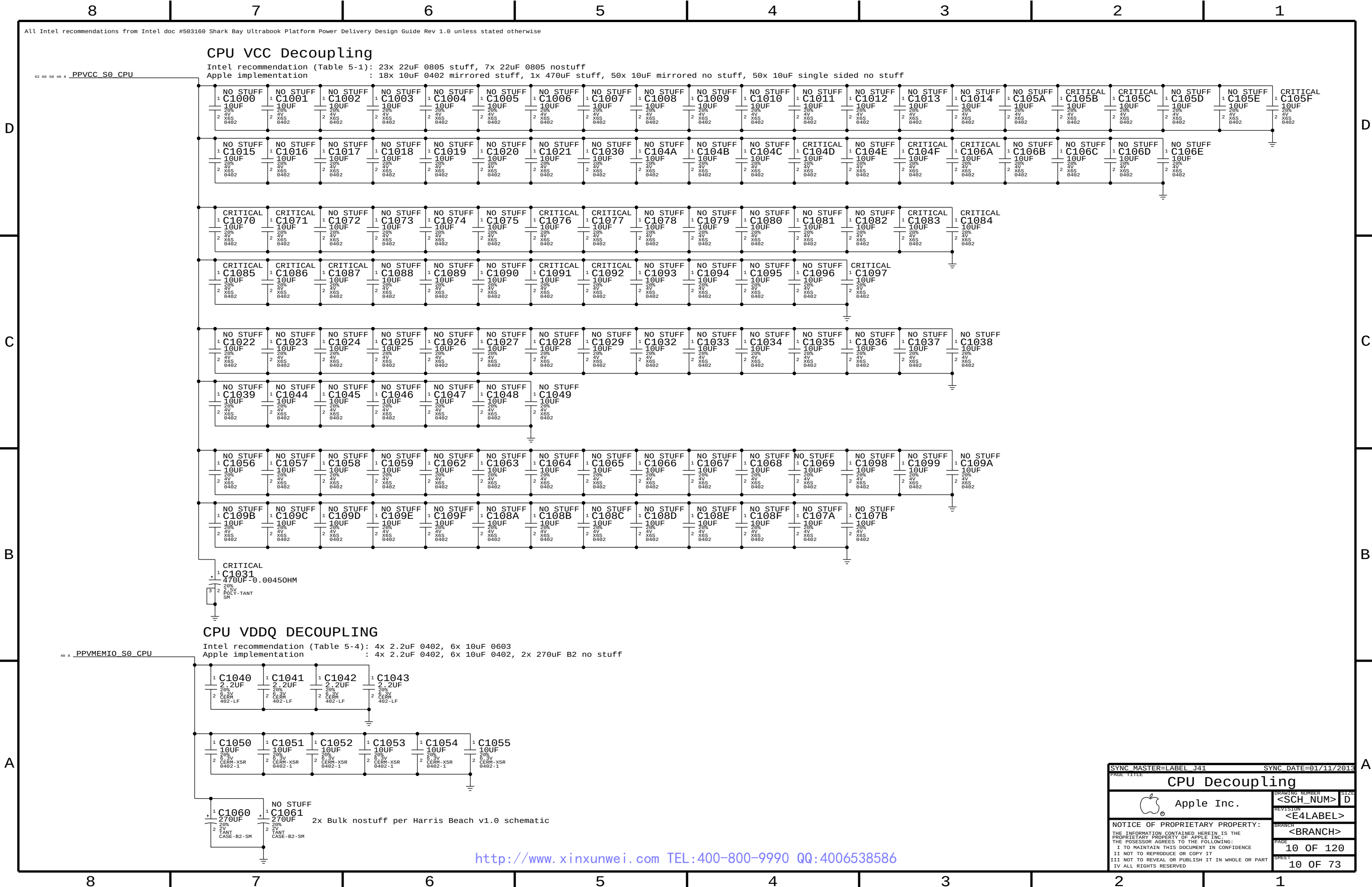




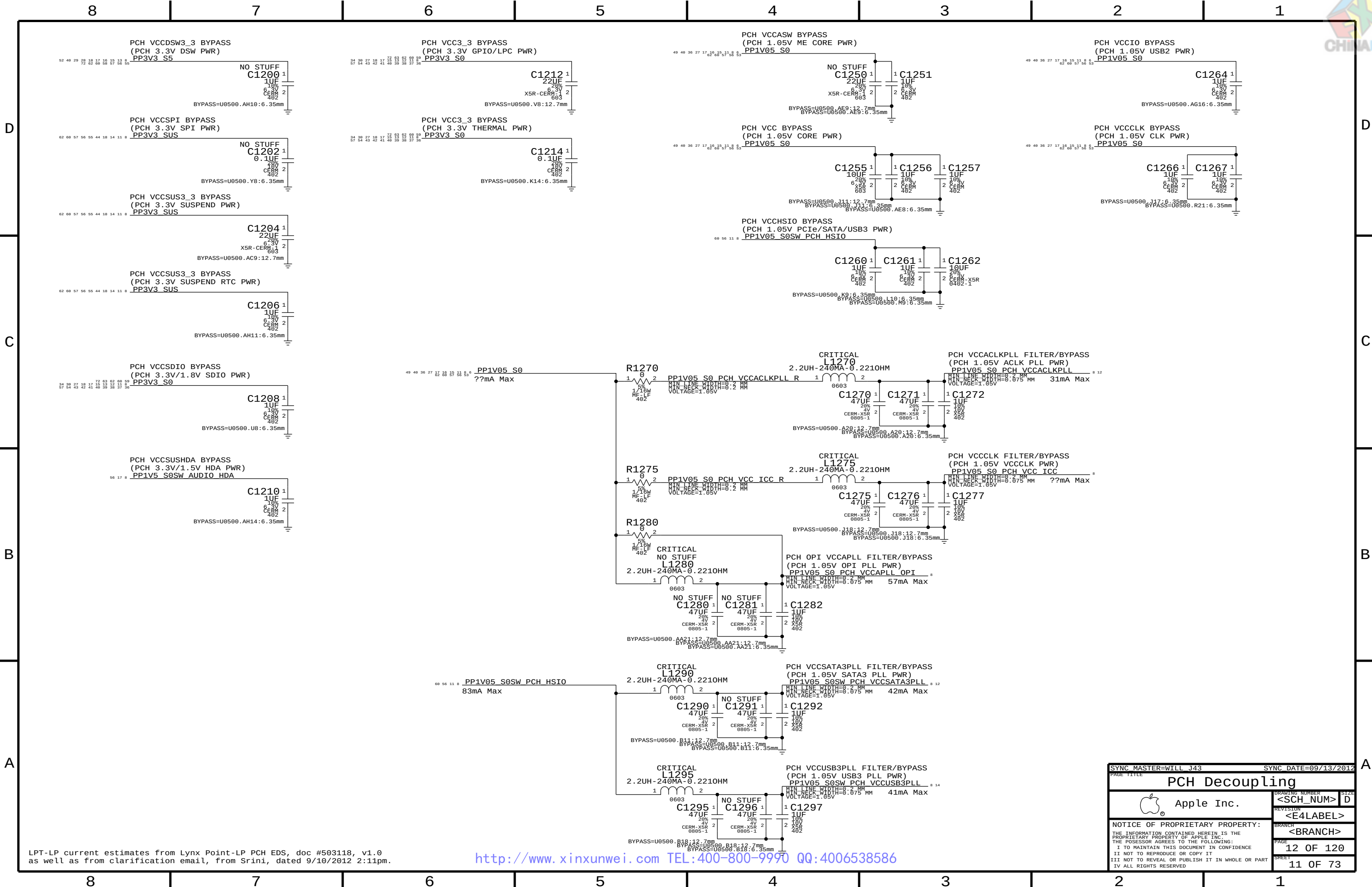
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LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0.
Note [1] current numbers from clarification email, from Srini, dated 9/10/2012 2:11pm.







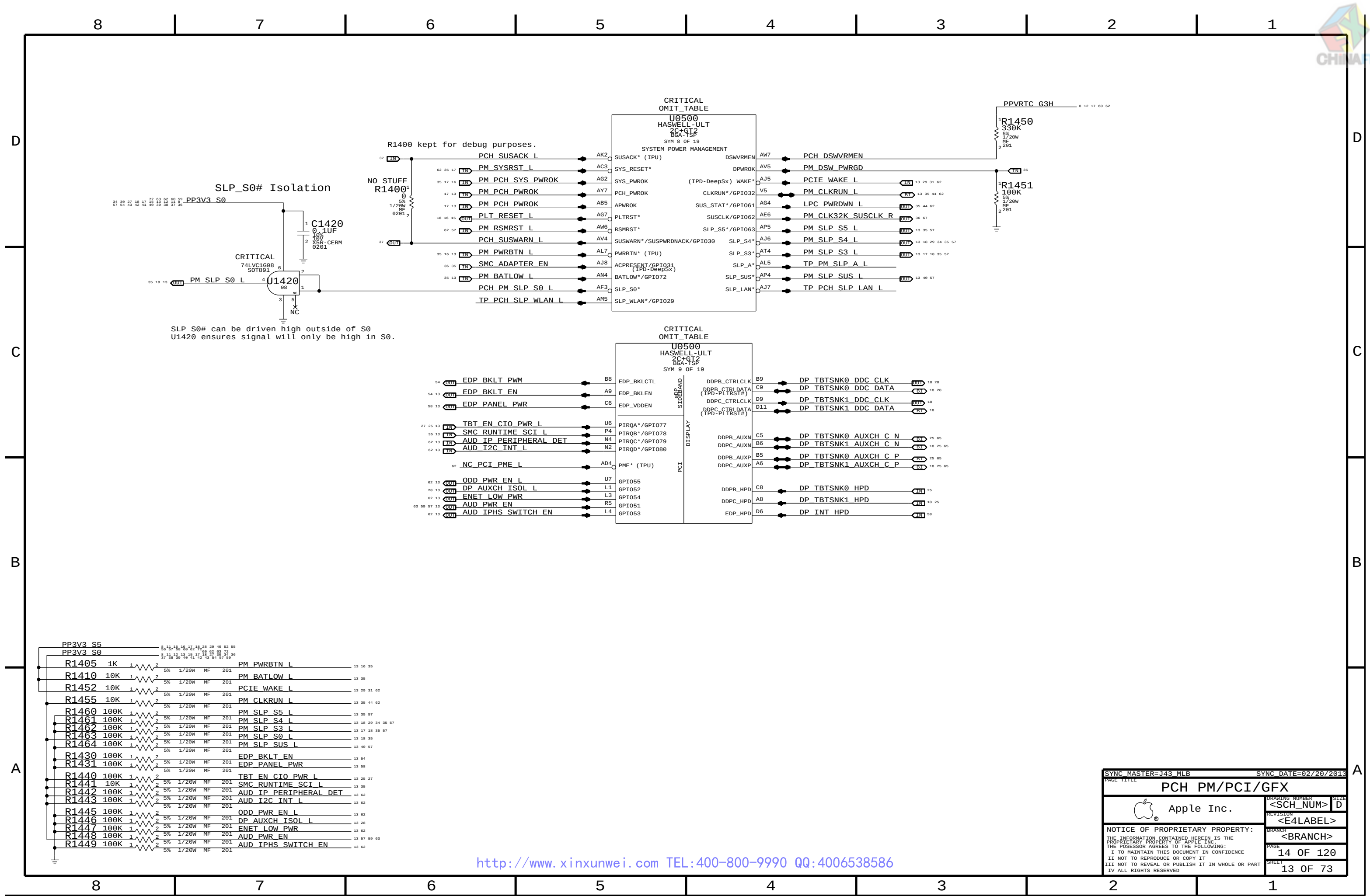
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LPT-LP current estimates from Lynx Point-LP PCH EDS, doc #503118, v1.0 as well as from clarification email, from Srin, dated 9/10/2012 2:11pm.

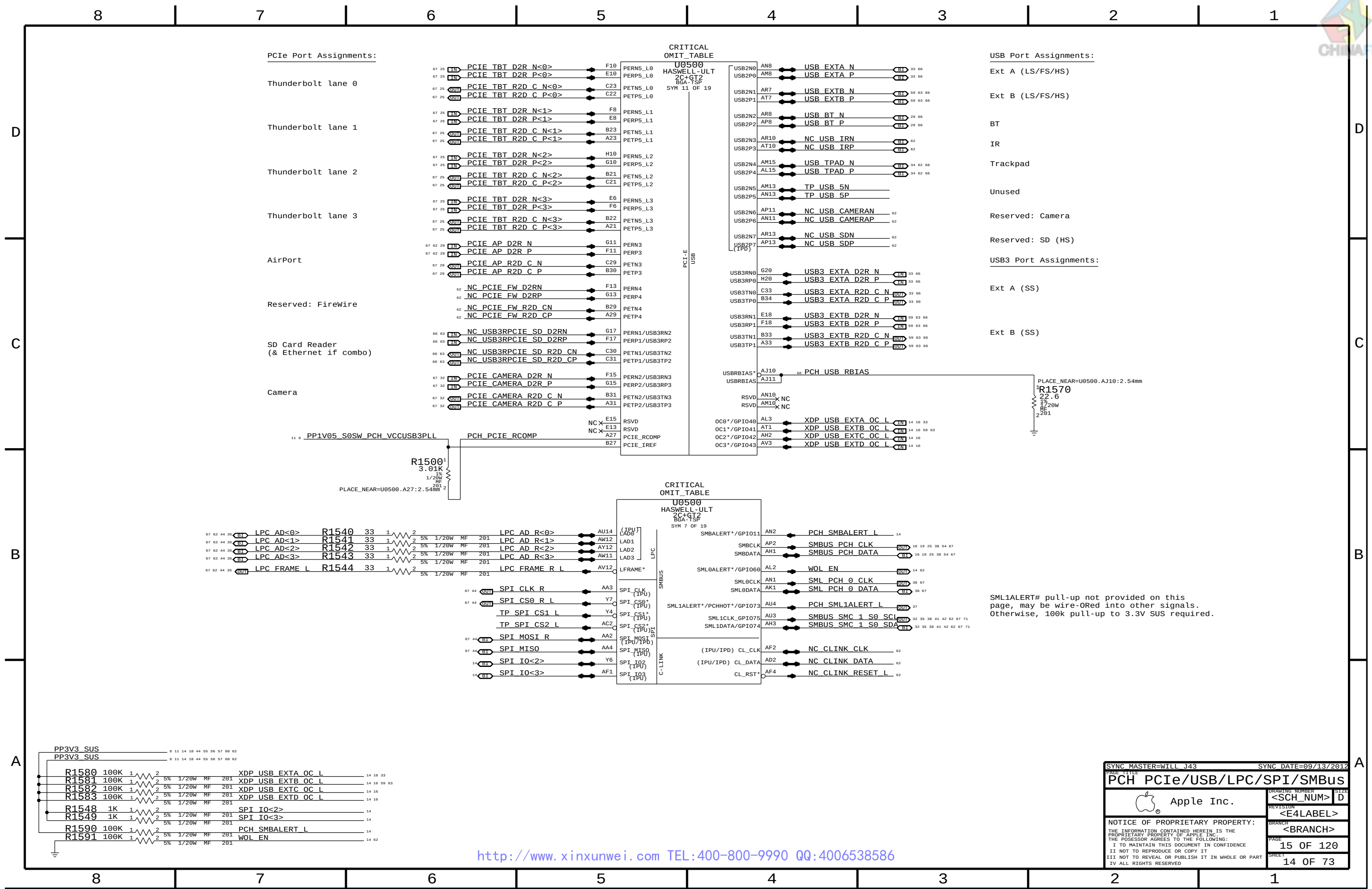
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SYNC MASTER=J43 MLB		SYNC DATE=02/20/2013	
PAGE TITLE			
PCH PM/PCI/GFX			
Apple Inc.		DRAWING NUMBER	SIZE
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PCIe Port Assignments:

Thunderbolt lane 0

Thunderbolt lane 1

Thunderbolt lane 2

Thunderbolt lane 3

AirPort

Reserved: FireWire

SD Card Reader

(& Ethernet if combo)

Camera

CRITICAL OMIT_TABLE

U0500
HASWELL-ULT
2C+GT2
BGA-TSP
SYM 11 OF 19

USB Port Assignments:

Ext A (LS/FS/HS)

Ext B (LS/FS/HS)

BT

IR

Trackpad

Unused

Reserved: Camera

Reserved: SD (HS)

USB3 Port Assignments:

Ext A (SS)

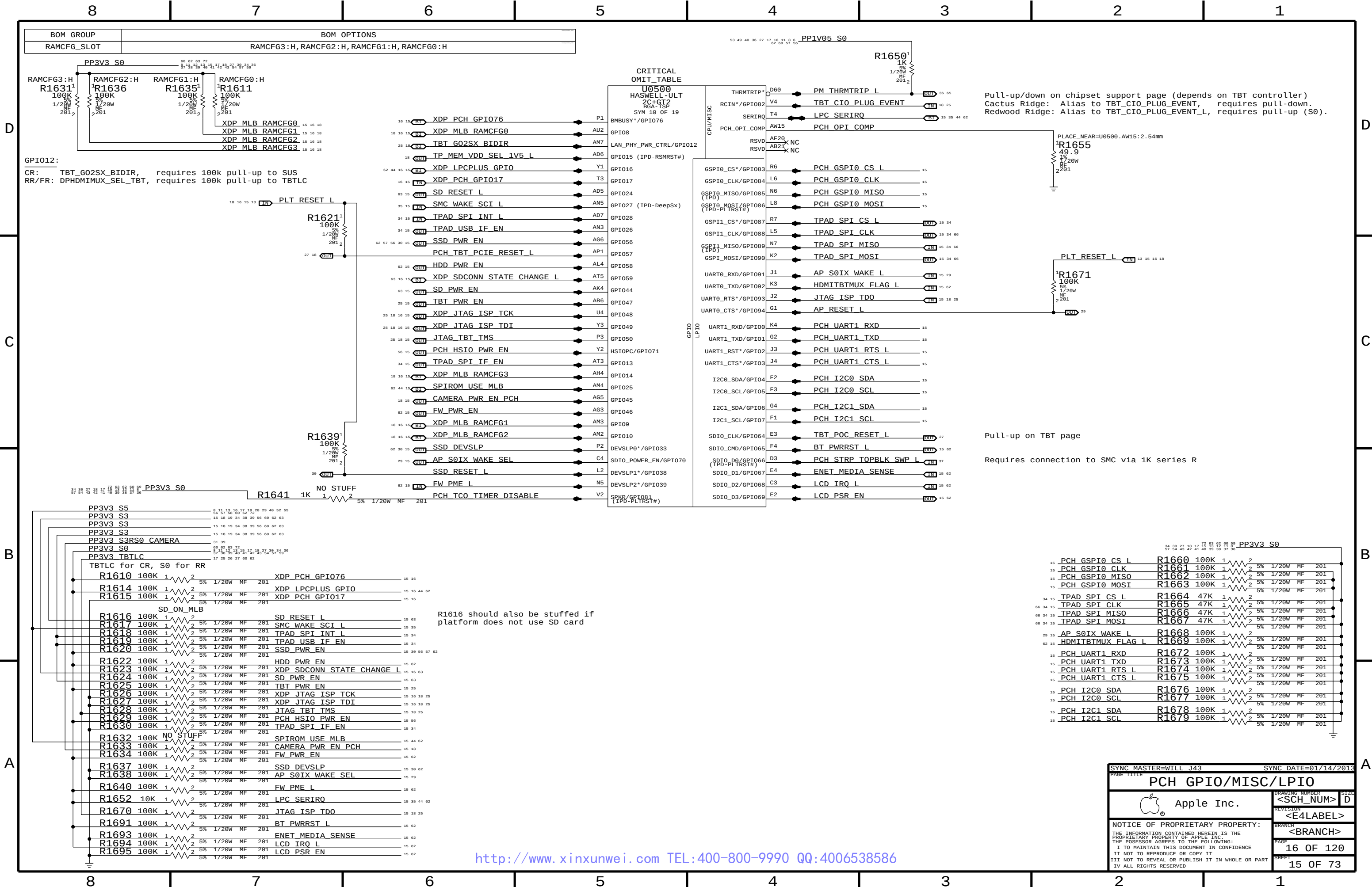
Ext B (SS)

CRITICAL OMIT_TABLE

U0500
HASWELL-ULT
2C+GT2
BGA-TSP
SYM 7 OF 19

SML1ALERT# pull-up not provided on this page, may be wire-0Red into other signals. Otherwise, 100k pull-up to 3.3V SUS required.

SYNC MASTER=WILL J43		SYNC DATE=09/13/2012	
PAGE TITLE			
PCH PCIe/USB/LPC/SPI/SMBus			
		DRAWING NUMBER <SCH_NUM>	SIZE D
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Pull-up/down on chipset support page (depends on TBT controller)
Cactus Ridge: Alias to TBT_CIO_PLUG_EVENT, requires pull-down.
Redwood Ridge: Alias to TBT_CIO_PLUG_EVENT_L, requires pull-up (S0).

Pull-up on TBT page

Requires connection to SMC via 1K series R

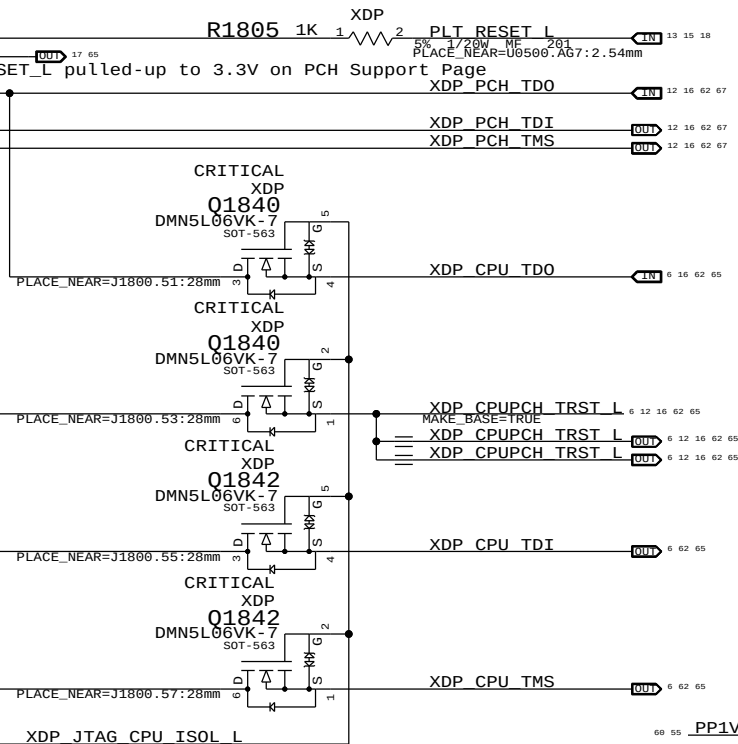
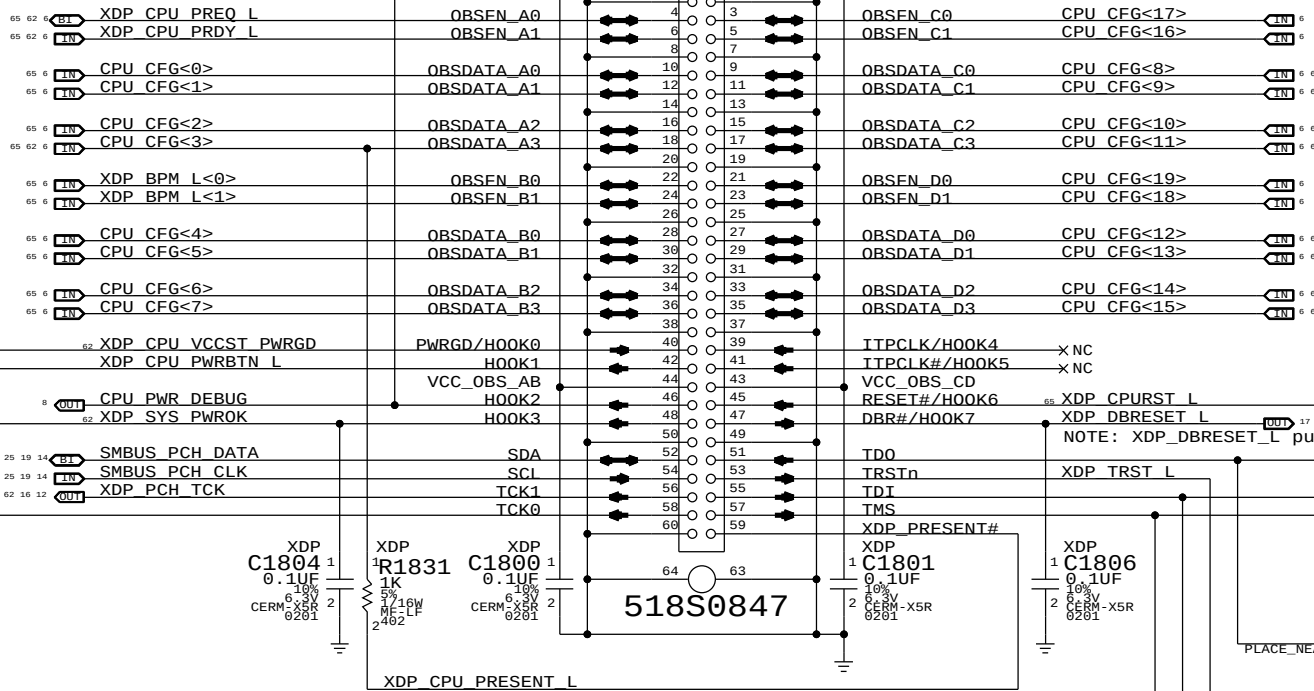
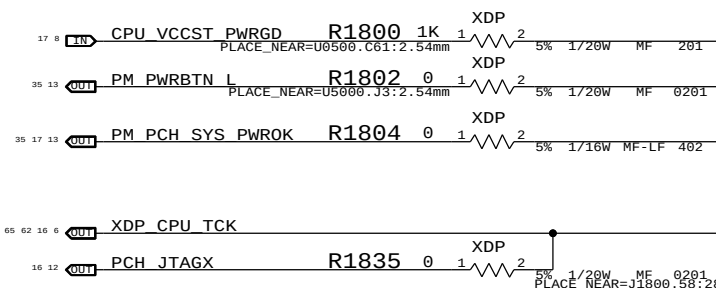
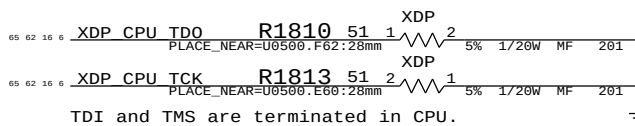
R1616 should also be stuffed if platform does not use SD card

Extra BPM Testpoints

IN	XDP_BPM_L<2>	TP1802
IN	XDP_BPM_L<3>	TP1803
IN	XDP_BPM_L<4>	TP1804
IN	XDP_BPM_L<5>	TP1805
IN	XDP_BPM_L<6>	TP1806
IN	XDP_BPM_L<7>	TP1807

Merged (CPU/PCH) Micro2-XDP

NOTE: This is not the standard XDP pinout.
Use with 921-0133 Adapter Flex to
support chipset debug.



PCH XDP Signals

These signals do not connect to XDP connector in this architecture, only accessible via Top-Side Probe. Nets are listed here to show XDP associations and to make clear what restrictions exist on PCH GPIOs when Top-Side Probe is used for PCH debug.

PCH/XDP Signals

IN	XDP_MLB_RAMCFG0	TP1870
OUT	XDP_USB_EXT_A_OC_L	MAKE_BASE=TRUE
OUT	XDP_USB_EXT_B_OC_L	MAKE_BASE=TRUE
OUT	XDP_USB_EXT_C_OC_L	TP1873
IN	XDP_USB_EXT_D_OC_L	TP1874
OUT	XDP_SDCONN_STATE_CHANGE_L	MAKE_BASE=TRUE
IN	XDP_MLB_RAMCFG1	TP1876
IN	XDP_MLB_RAMCFG2	TP1877
IN	XDP_MLB_RAMCFG3	TP1878

Non-XDP Signals

OUT	XDP_USB_EXT_A_OC_L	14 16 33
OUT	XDP_USB_EXT_B_OC_L	14 16 59 63
OUT	XDP_SDCONN_STATE_CHANGE_L	15 16 63

IN	XDP_JTAG_ISP_TCK	MAKE_BASE=TRUE
OUT	XDP_SSD_PCIE3_SEL_L	R1881 1K
OUT	XDP_SSD_PCIE2_SEL_L	R1882 1K
OUT	XDP_SSD_PCIE1_SEL_L	R1883 1K
OUT	XDP_SSD_PCIE0_SEL_L	R1884 1K
IN	XDP_LPCPLUS_GPIO	MAKE_BASE=TRUE
OUT	XDP_PCH_GPIO17	TP1886
IN	XDP_PCH_GPIO76	TP1887
IN	XDP_JTAG_ISP_TDI	MAKE_BASE=TRUE

NOTE: Must not short XDP pins together!

OUT	SSD_PCIE_SEL_L	38 62
IN	XDP_LPCPLUS_GPIO	15 16 44 62
OUT	XDP_JTAG_ISP_TDI	15 16 18 25

Unused & MLB_RAMCFGx GPIOs have TPs.

USB Overcurrents are aliased, do not cause USB OC# events during PCH debug.

SDCONN_STATE_CHANGE_L is aliased, do not plug/unplug SD Cards during PCH debug.

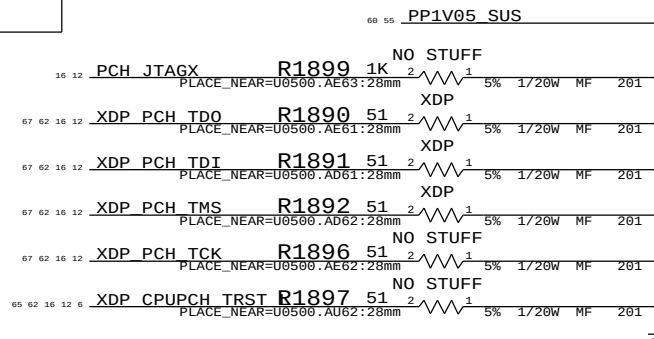
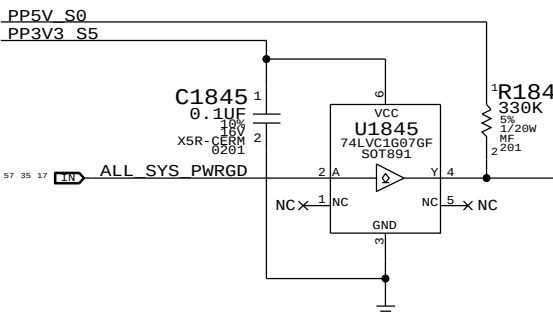
JTAG_ISP (non-TMS) nets are aliased, do not attempt bit-banged JTAG during PCH debug.

NOTE: Should force PCH GPIO47 high to ensure TBT router powered to avoid leakage/clamping of signals.

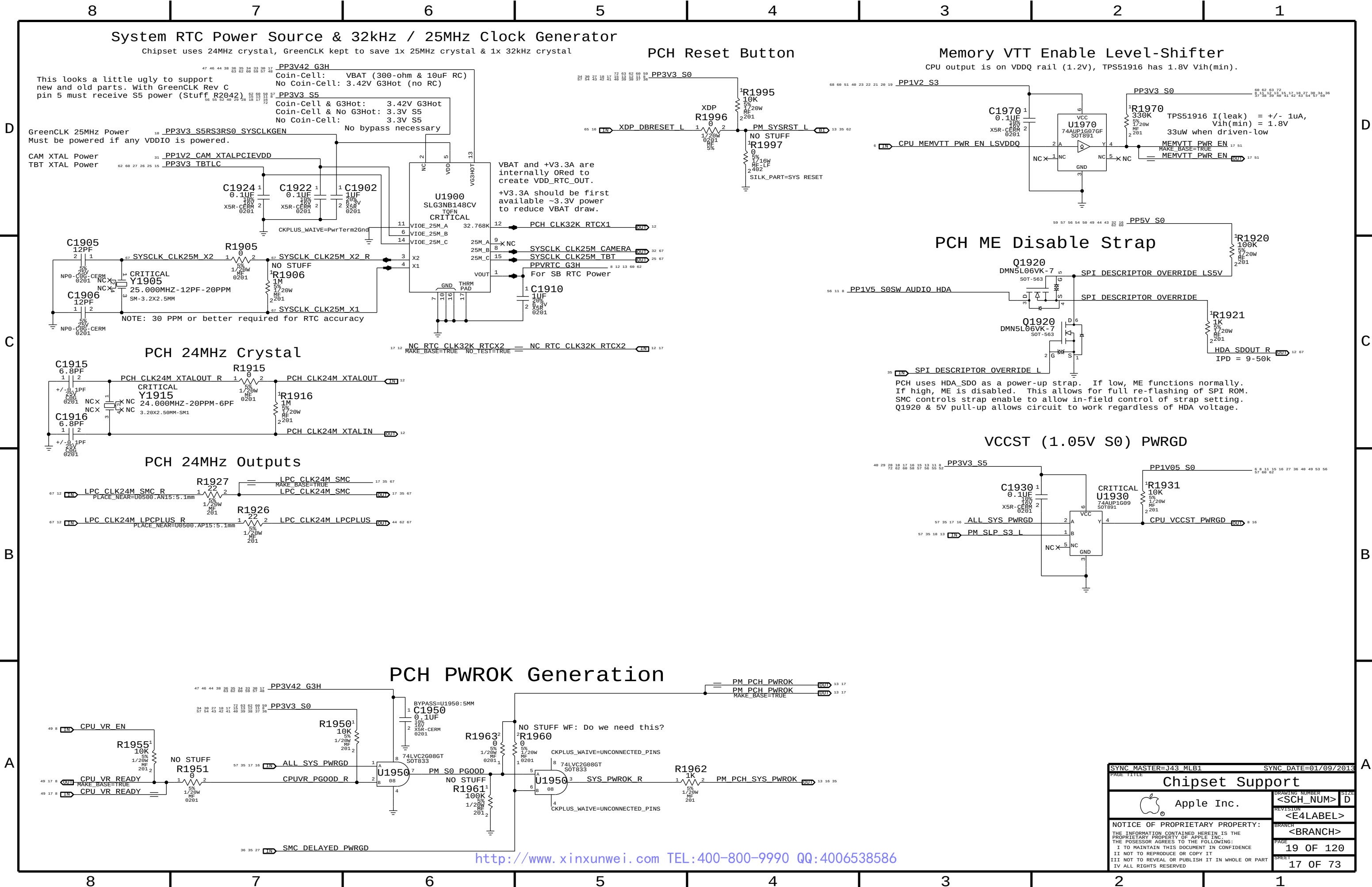
SSD_PCIEx_SEL_L straps are connected via 1K to common net.

LPCPLUS_GPIO is aliased, do not attempt use during PCH debug.

CPU JTAG Isolation



SYNC MASTER=WILL J43		SYNC DATE=12/17/2012	
PAGE TITLE		CPU/PCH Merged XDP	
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System RTC Power Source & 32kHz / 25MHz Clock Generator

Chipset uses 24MHz crystal, GreenCLK kept to save 1x 25MHz crystal & 1x 32kHz crystal

This looks a little ugly to support new and old parts. With GreenCLK Rev C pin 5 must receive S5 power (Stuff R2042)

GreenCLK 25MHz Power Must be powered if any VDDIO is powered.

CAM XTAL Power
TBT XTAL Power

PP3V42 G3H

Coin-Cell: VBAT (300-ohm & 10uF RC)
No Coin-Cell: 3.42V G3Hot (no RC)

PP3V3 S5

Coin-Cell & G3Hot: 3.42V G3Hot
Coin-Cell & No G3Hot: 3.3V S5
No Coin-Cell: 3.3V S5

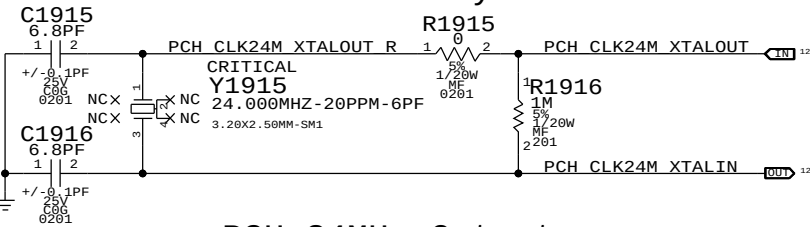
No bypass necessary

C1924 0.1uF
C1922 0.1uF
C1902 1uF

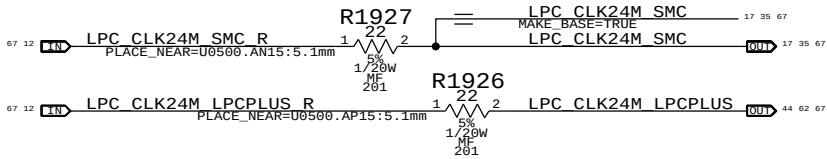
VBAT and +V3.3A are internally 0Red to create VDD_RTC_OUT.
+V3.3A should be first available ~3.3V power to reduce VBAT draw.

NOTE: 30 PPM or better required for RTC accuracy

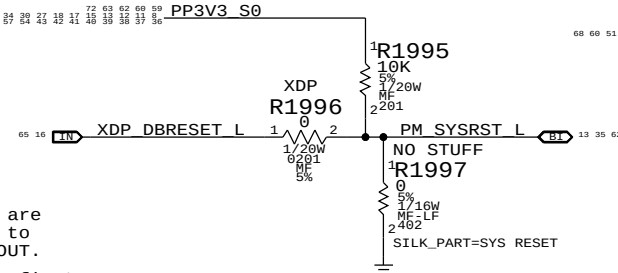
PCH 24MHz Crystal



PCH 24MHz Outputs

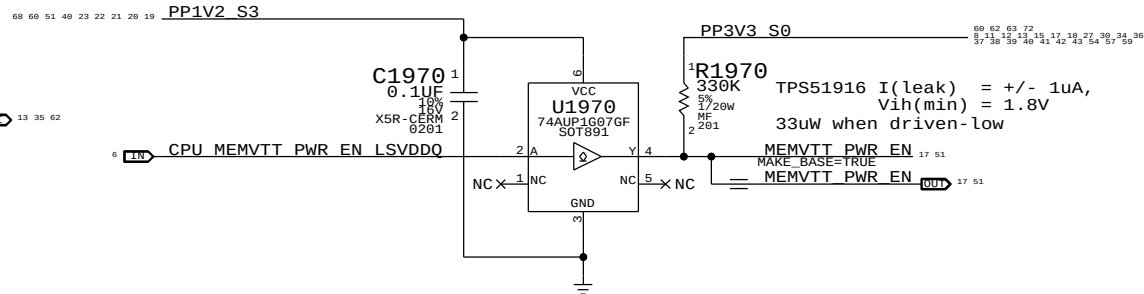


PCH Reset Button

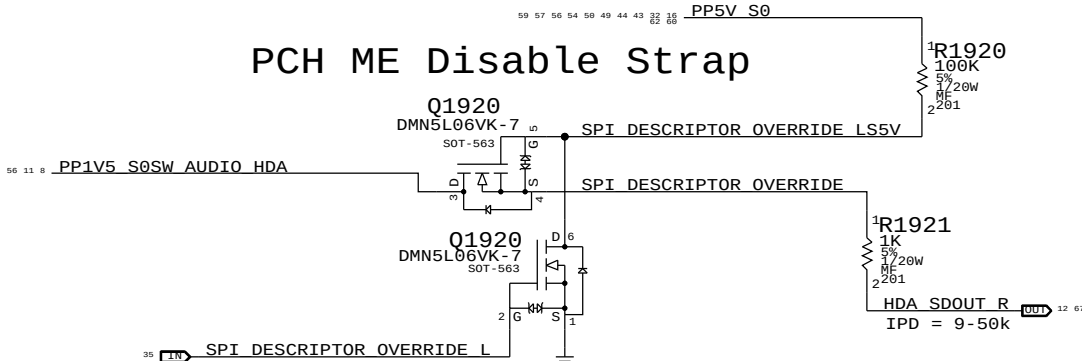


Memory VTT Enable Level-Shifter

CPU output is on VDDQ rail (1.2V), TPS51916 has 1.8V Vih(min).

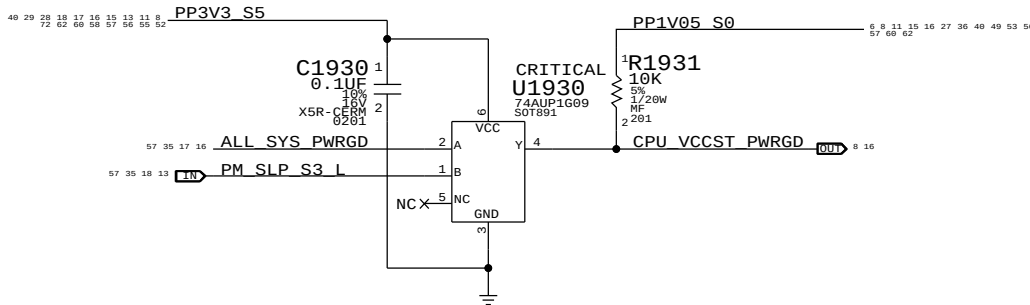


PCH ME Disable Strap

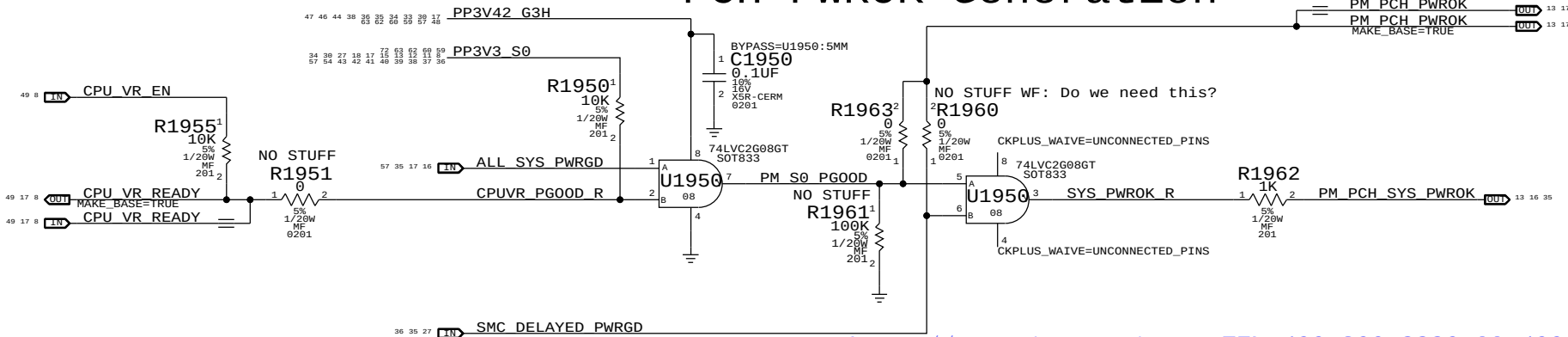


PCH uses HDA SD0 as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. Q1920 & 5V pull-up allows circuit to work regardless of HDA voltage.

VCCST (1.05V S0) PWRGD



PCH PWR0K Generation



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SYNC MASTER=J43 MLB1		SYNC DATE=01/09/2013	
PAGE TITLE			
Chipset Support			
 Apple Inc.		DRAWING NUMBER	SIZE
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```
Power aliases required by this page:
- =PP3V3_S3_VREFMRGN
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:
- =I2C_VREFDAC5_SCL
- =I2C_VREFDAC5_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:
- DDRVREF_DAC - Stuffs DAC margining circuit.
```

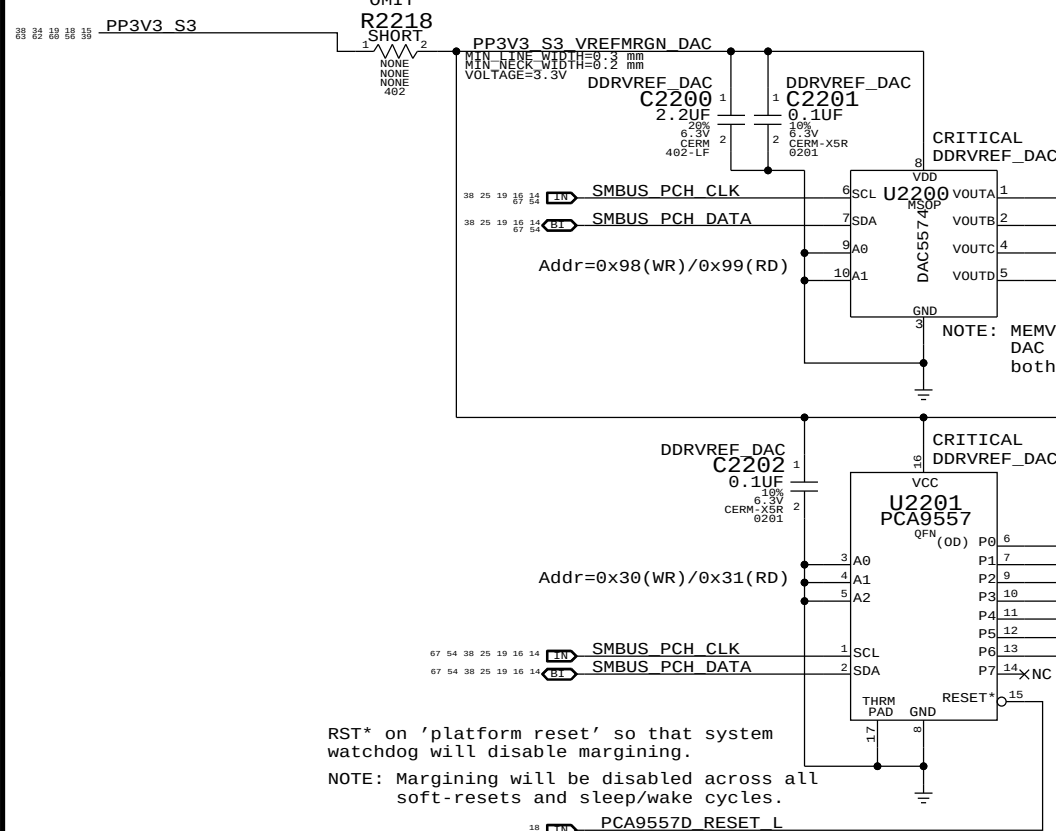
FETs for CPU isolation during DAC margining

NOTE: CPU DAC output step sizes:

DDR3	(1.5V)	7.70mV	per step
DDR3L	(1.35V)	6.99mV	per step
LPDDR3	(1.2V)	?.??mV	per step

NOTE: CPU has single output for VREFCA. Split into two signals for independent DAC margining support. When DAC margining VREFCA ensure VREFMRGN_CPU_EN is low to remove short due to CPU.

DAC sets voltage level, PCA9557 & FETs enable outputs and disables margining after platform reset.

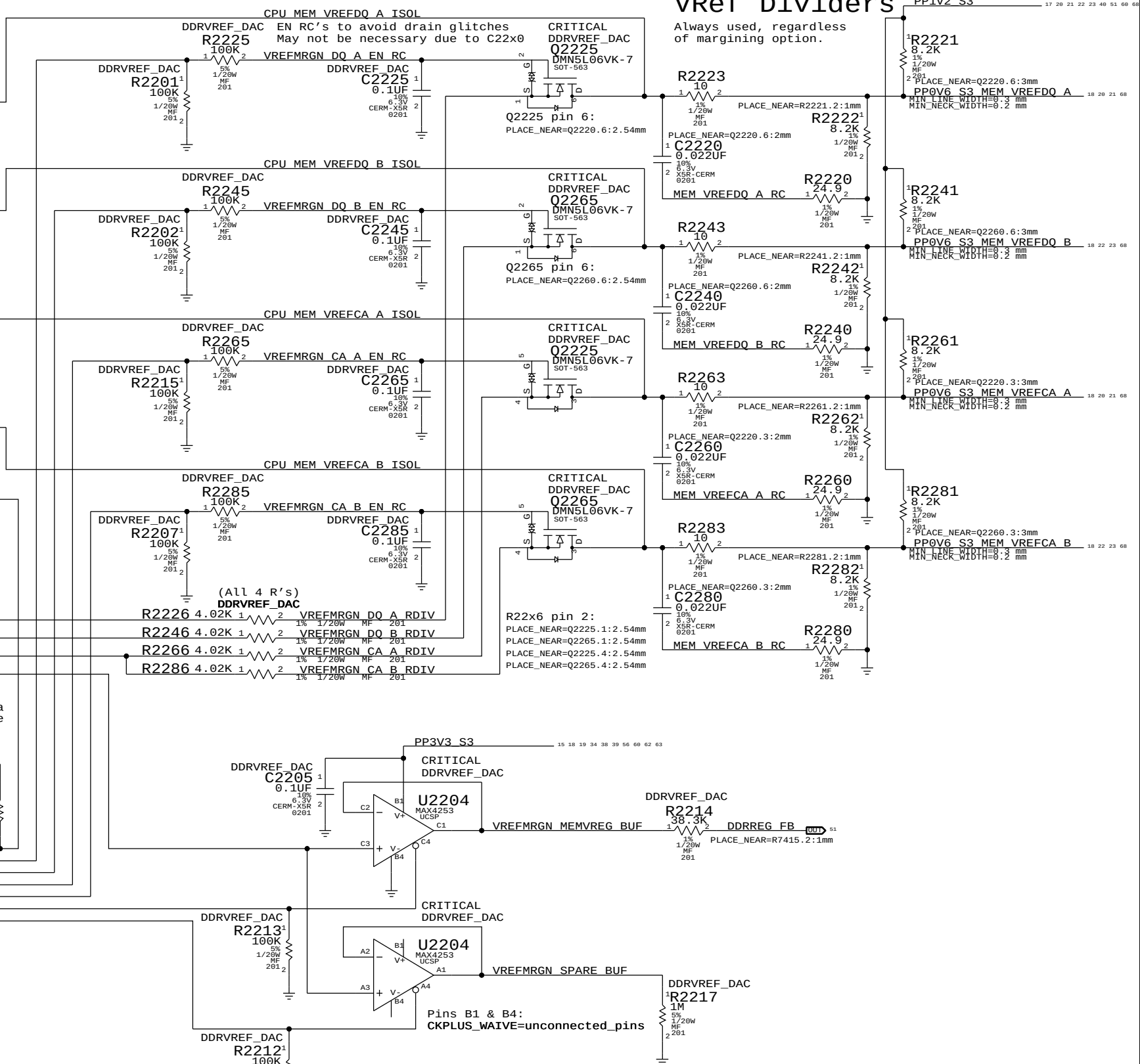


RST* on 'platform reset' so that system watchdog will disable margining.

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG
DAC Channel:	A	B	C	C	D
PCA9557D Pin:	1	2	3	4	5
	LPDDR3 (1.2V)	DDR3L (1.35V)	LPDDR3 (1.2V)	DDR3L (1.35V)	NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider DDR3L assumes TPS51916 supply with 19.6k/57.6k divider
Nominal value	0.600V (DAC: 0x2E.5)	0.675V (DAC: 0x34)	1.200V (DAC: 0x5D)	1.343V (DAC: 0x68)	
Margined target:	0.300V - 0.900V (+/- 300mV)	0.337V - 1.013V (+/- 337.5mV)	0.800V - 1.600V (+/- 400mV)	0.972V - 1.714V (+/- 371mV)	
DAC range:	0.000V - 1.199V (0x00 - 0x5D)	0.000V - 1.354V (0x00 - 0x69)	0.000V - 2.317V (0x00 - 0xBA)	0.000V - 2.694V (0x00 - 0xD1)	
VRef current:	+73uA - -73uA (- = sourced)	+82uA - -82uA (- = sourced)	+21uA - -21uA (- = sourced)	+25uA - -25uA (- = sourced)	
DAC step size:	6.36mV / step @ output	6.36mV / step @ output	4.28mV / step @ output	3.53mV / step @ output	

Always used, regardless of margining option.



SYNC MASTER-WILL J43		SYNC DATE=02/04/2013	
PAGE TITLE		DRAWING NUMBER	
DDR3 VREF MARGINING		SIZE	
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D



B

A

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C

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B

A

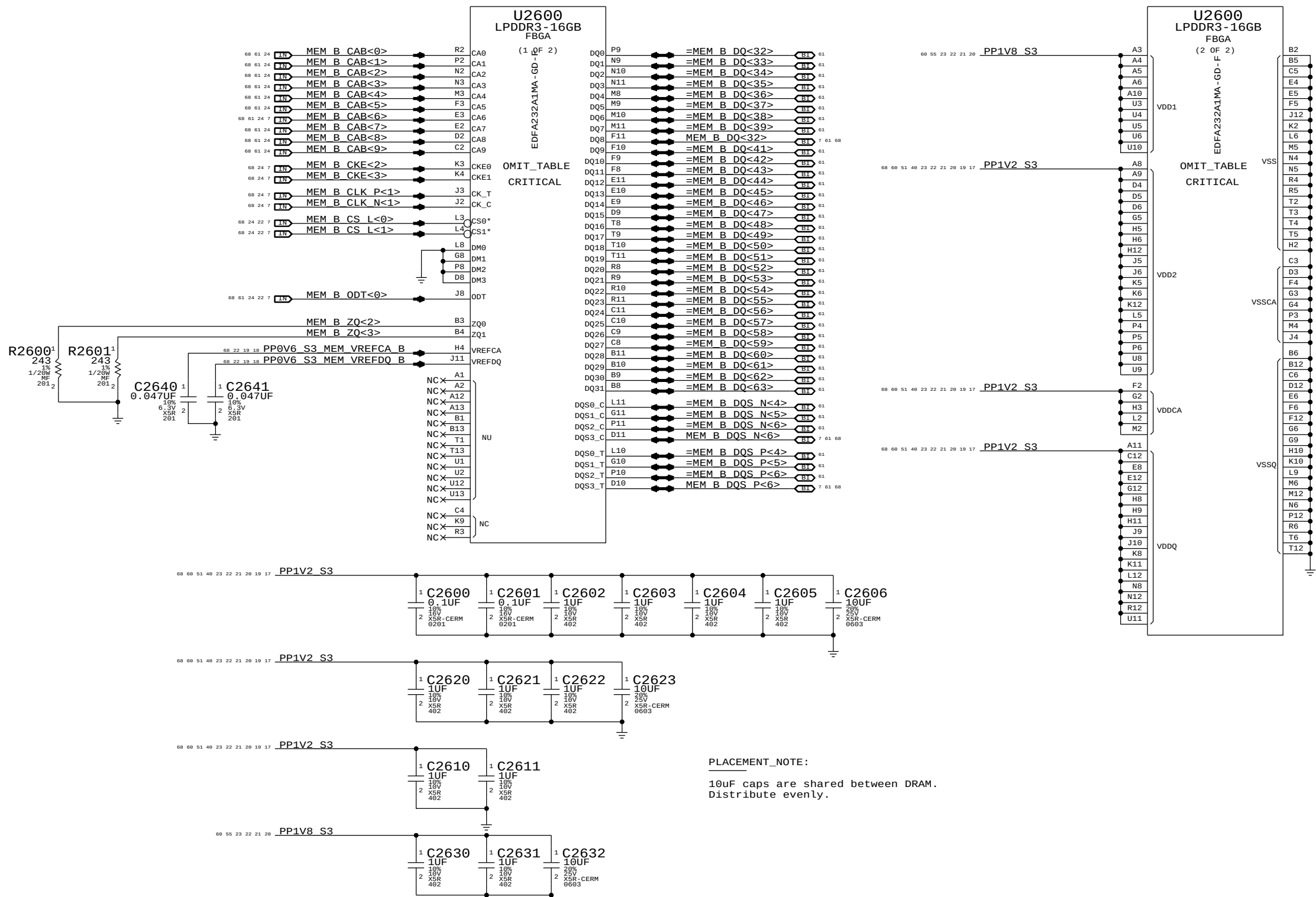
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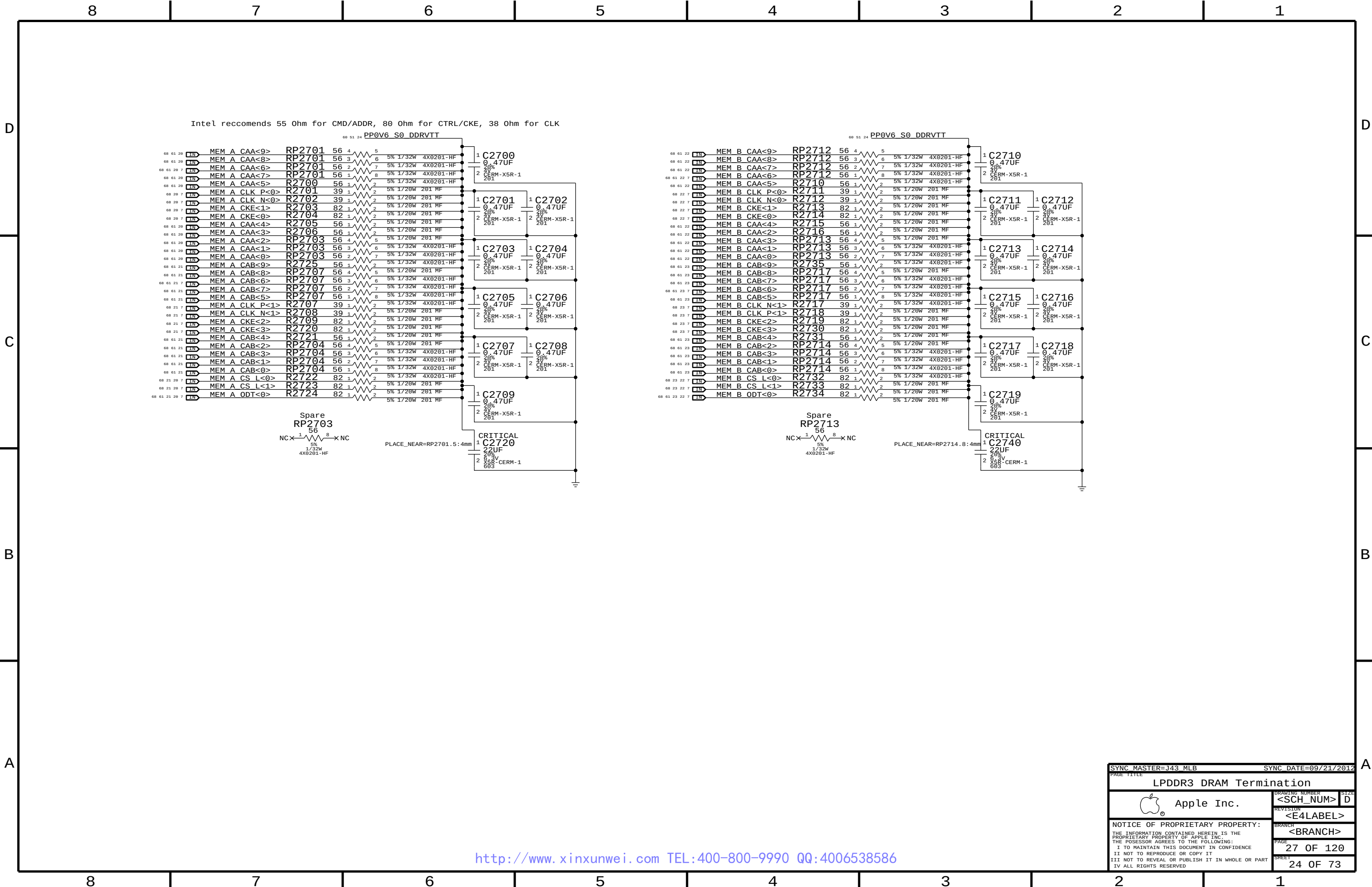
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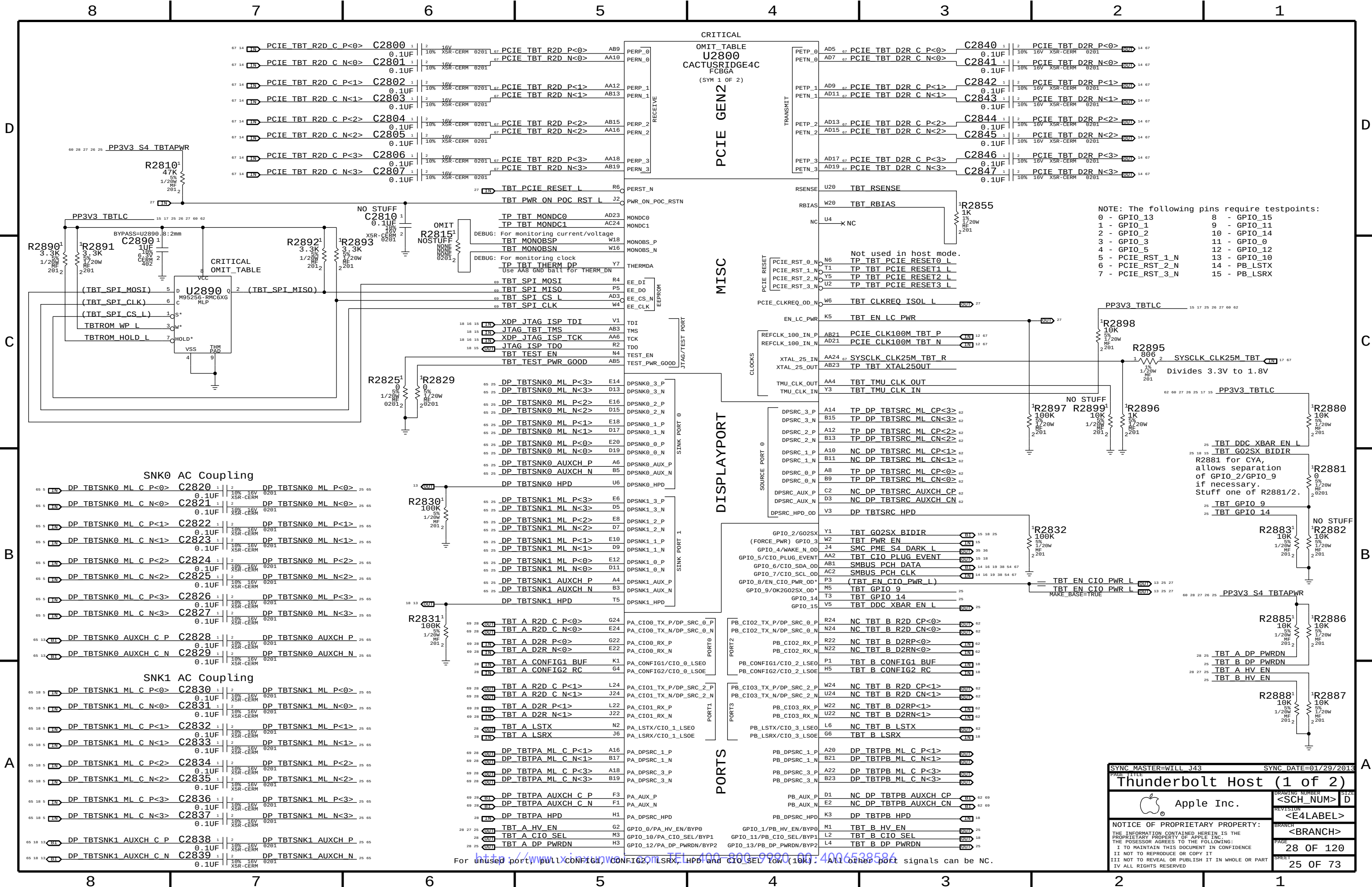
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LPDDR3 CHANNEL B (32-63)



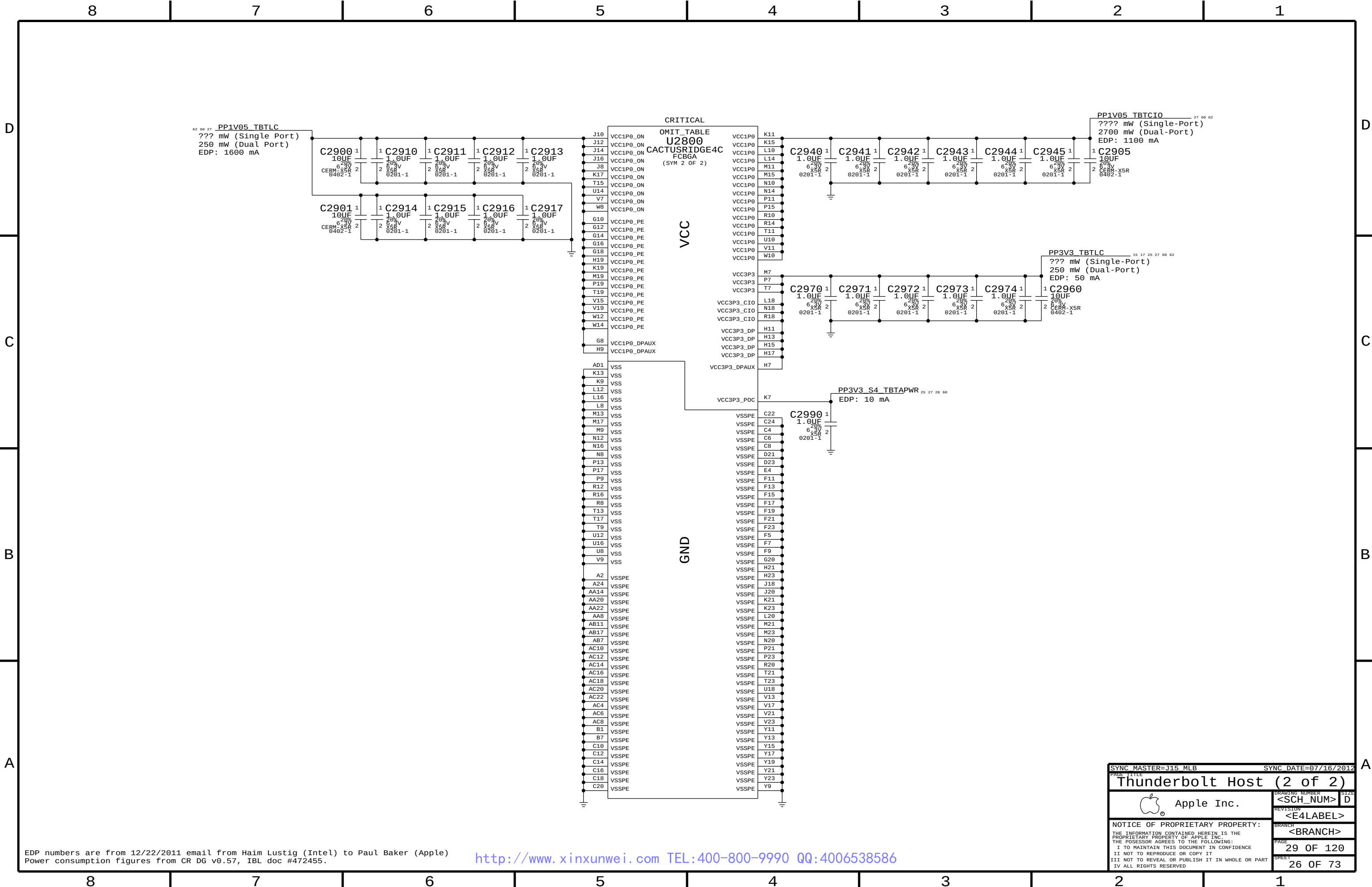
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LPDDR3 DRAM Channel B		(32-63)	
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		PAGE	26 OF 120
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NOTE: The following pins require testpoints:

0 - GPIO_13	8 - GPIO_15
1 - GPIO_1	9 - GPIO_11
2 - GPIO_2	10 - GPIO_14
3 - GPIO_3	11 - GPIO_0
4 - GPIO_5	12 - GPIO_12
5 - PCIE_RST_1_N	13 - GPIO_10
6 - PCIE_RST_2_N	14 - PB_LSTX
7 - PCIE_RST_3_N	15 - PB_LSRX



EDP numbers are from 12/22/2011 email from Haim Lustig (Intel) to Paul Baker (Apple)
Power consumption figures from CR DG v0.57, IBL doc #472455.

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SYNC MASTER=J15 MLB		SYNC DATE=07/16/2012	
PAGE TITLE		Thunderbolt Host (2 of 2)	
		DRAWING NUMBER	SIZE
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		<BRANCH>	
		PAGE	29 OF 120
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Page Notes

Power aliases required by this page:

- PPVIN_SW_TBTBST (8-13V Boost Input)
- PP15V_TBT_REG (15V Boost Output)
- PP3V3_TBT_P3V3TBTFT (3.3V FET Input)
- PP3V3_TBT_FET (3.3V FET Output)
- PP3V3_S0_TBT_PWRCTL (1.05V FET Input)
- PP1V05_TBT_P1V05TBTFT (1.05V FET Input)
- PP1V05_TBT_FET (1.05V FET Output)

Signal aliases required by this page:

- TBT_CLKREQ_L
- TBT_RESET_L

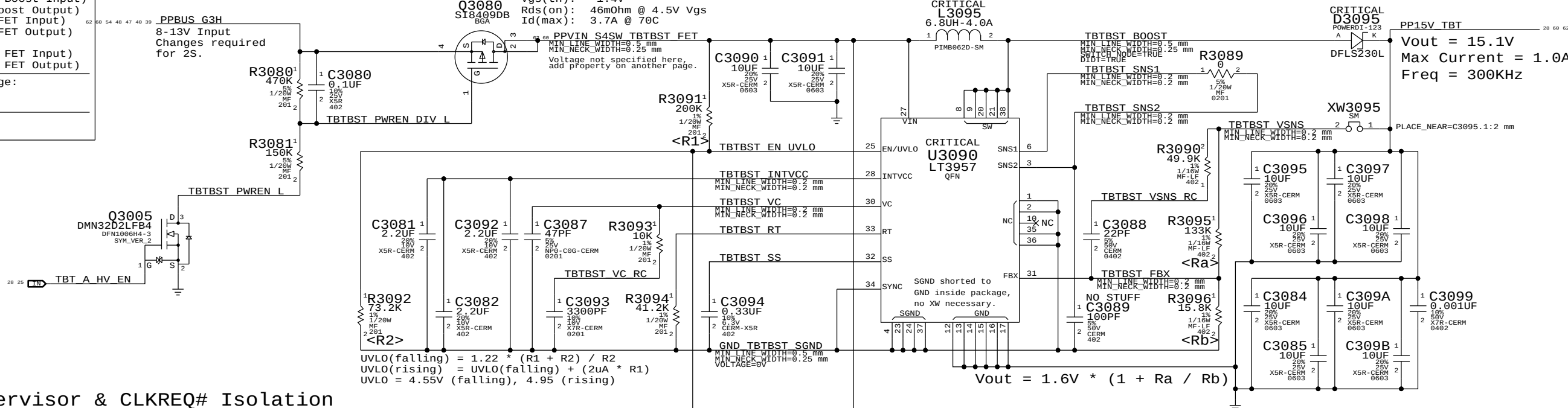
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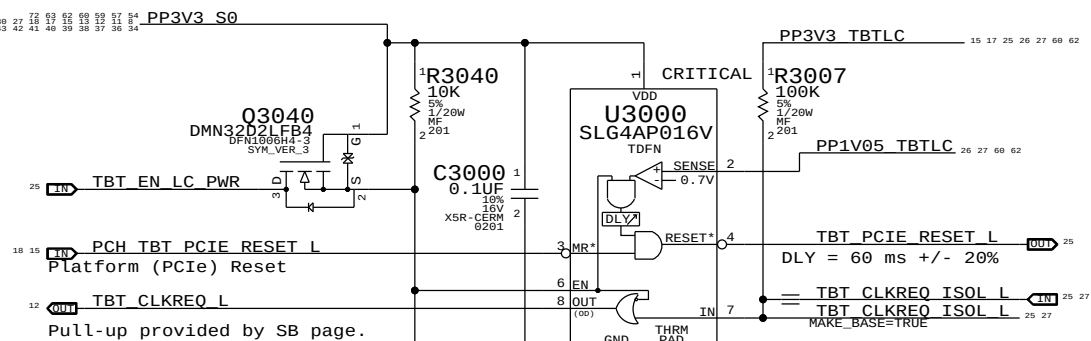
TBT 15V Boost Regulator

SI8409DB:
Vds(max): -30V
Vgs(max): +/-12V
Vgs(th): -1.4V
Rds(on): 46mOhm @ 4.5V Vgs
Id(max): 3.7A @ 70C

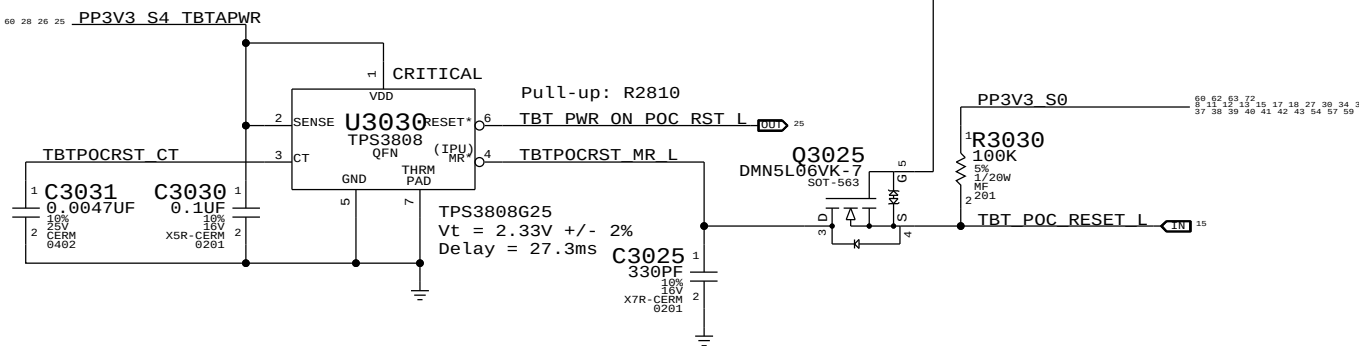
PPVIN S4SW TBTBST FET
Voltage not specified here,
add property on another page.



Supervisor & CLKREQ# Isolation



TBT "POC" Power-up Reset



3.3V TBT "LC" Switch

U3010
Max Current = 2A (85C)

Part	TPS22924C
Type	Load Switch
R(on)	18.5 mOhm Typ 25.8 mOhm Max
@ 2.5V	

1.05V TBT "LC" Switch

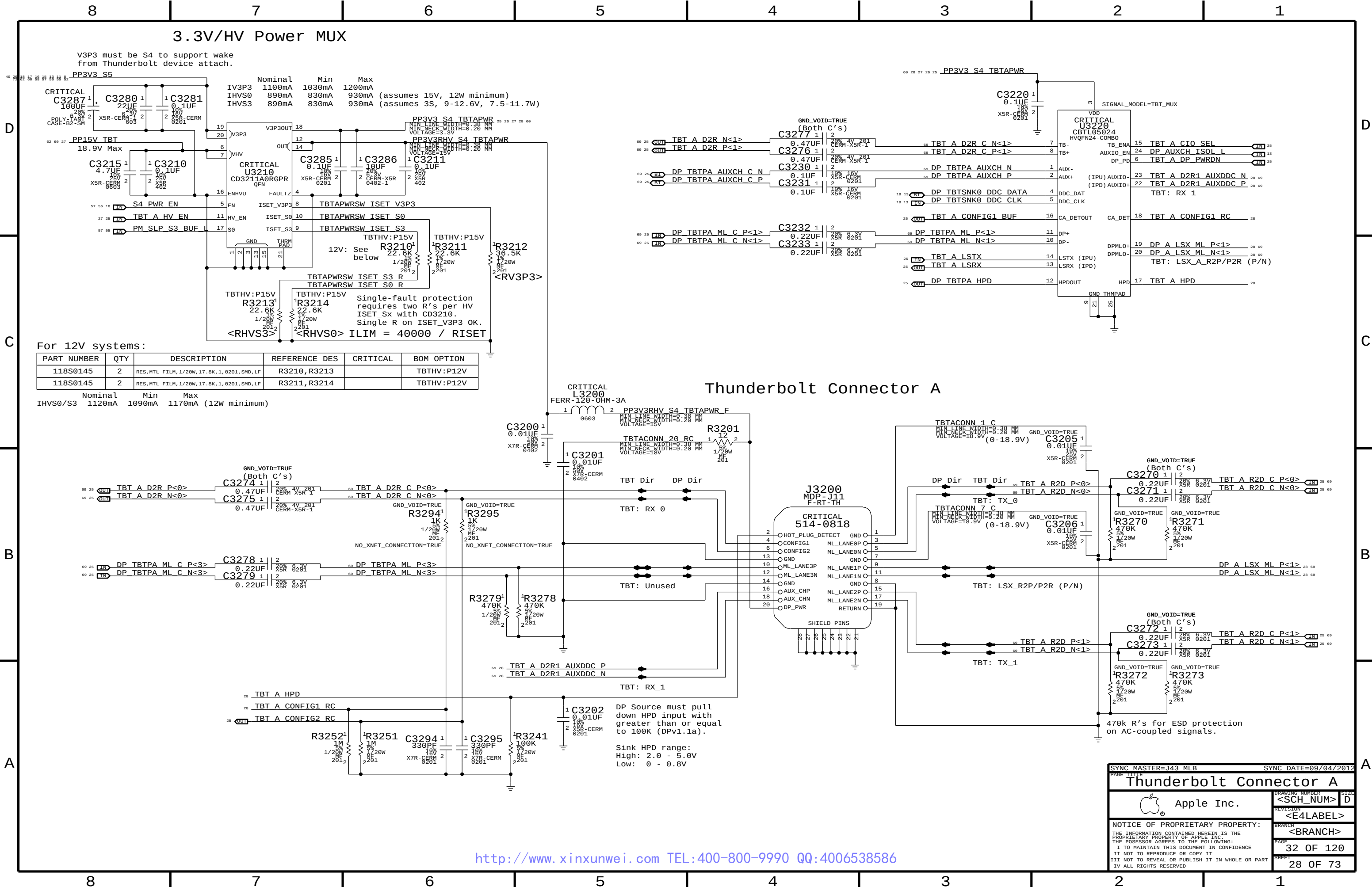
U3015
Max Current = 4A (85C)

Part	TPS22920
Type	Load Switch
R(on)	6.1 mOhm Typ 10.4 mOhm Max
@ 1.05V	

1.05V TBT "CIO" Switch

U3020
Max Current = 4A (85C)

Part	TPS22920
Type	Load Switch
R(on)	6.1 mOhm Typ 10.4 mOhm Max
@ 1.05V	



Thunderbolt Connector A

SYNC MASTER=J43 MLB

SYNC DATE=09/04/2012

Thunderbolt Connector A

Apple Inc.

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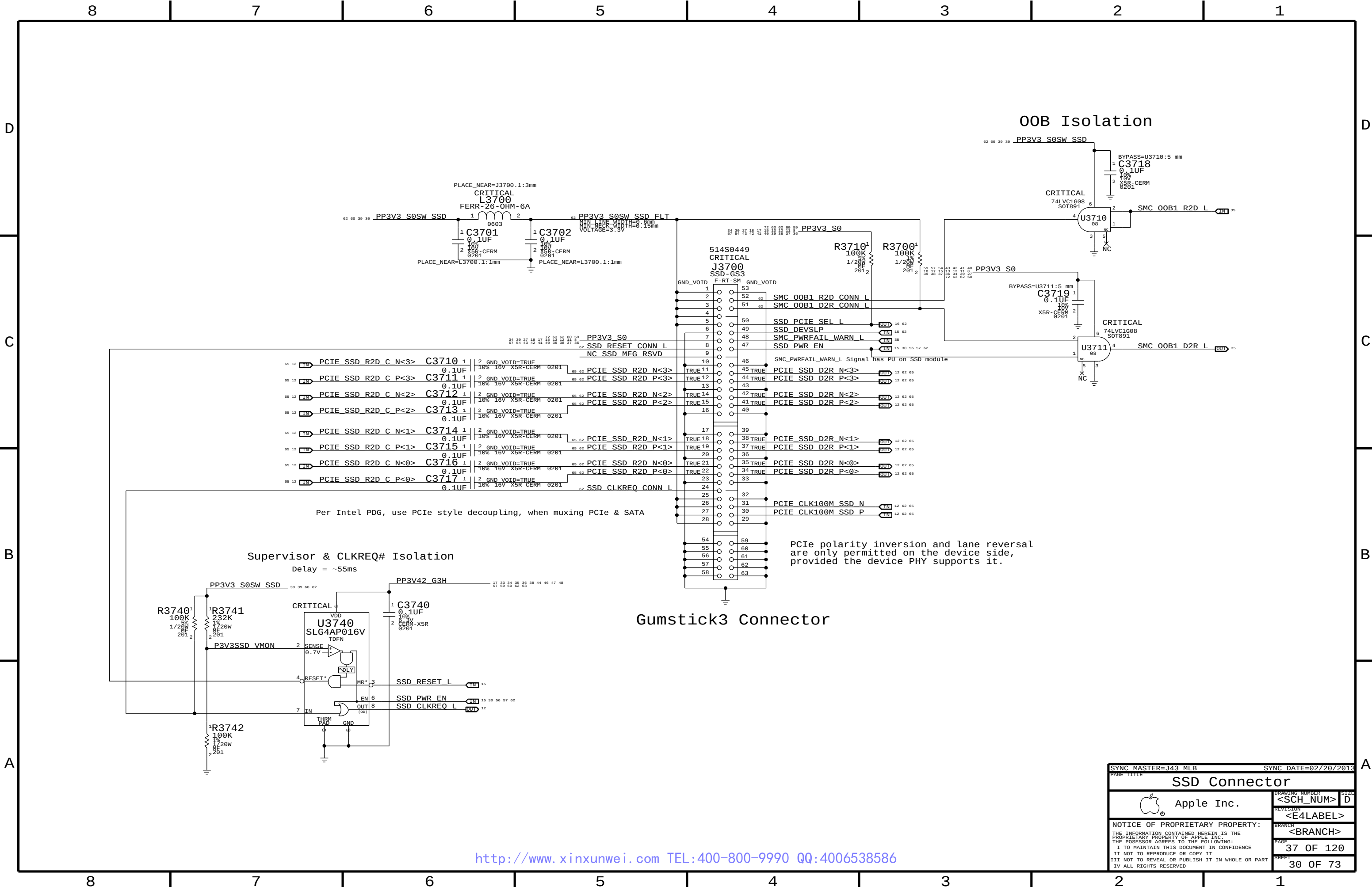
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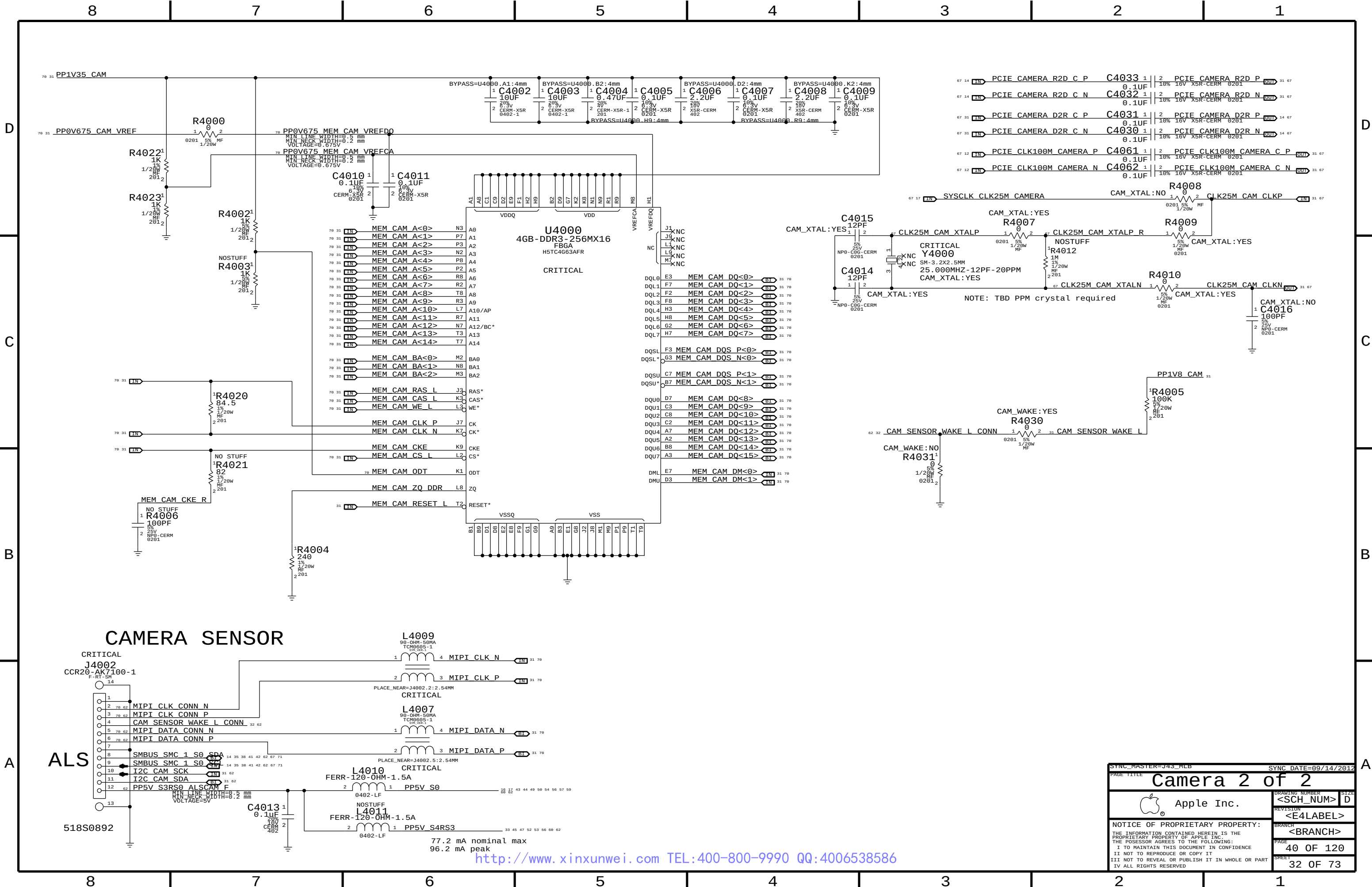
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SSD Connector			
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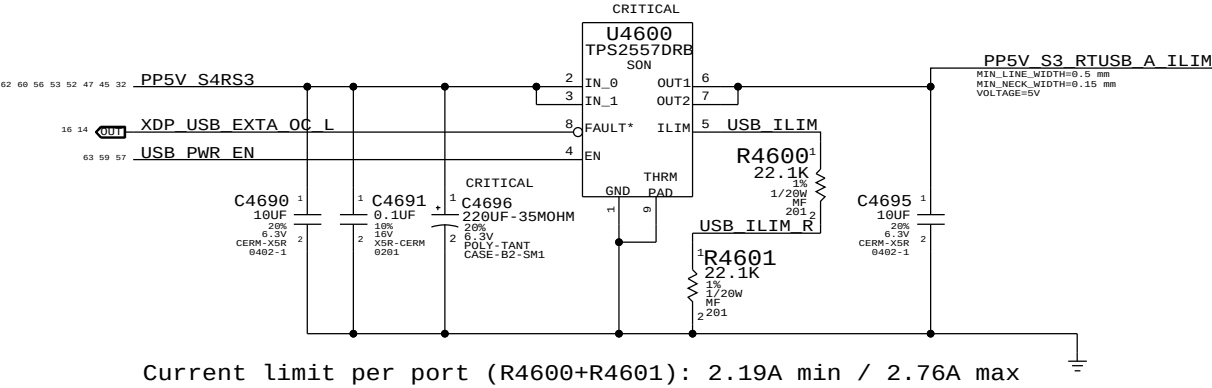


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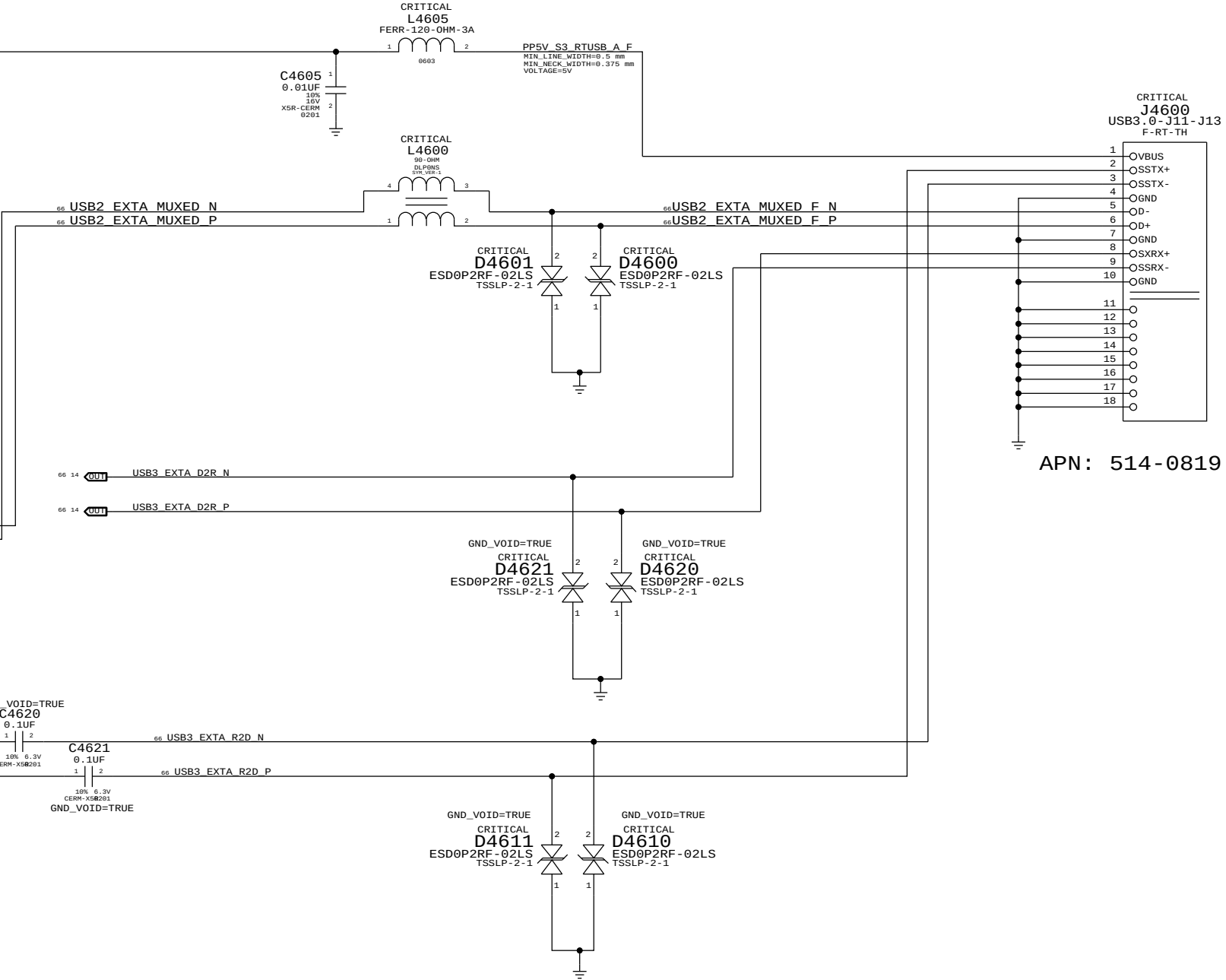
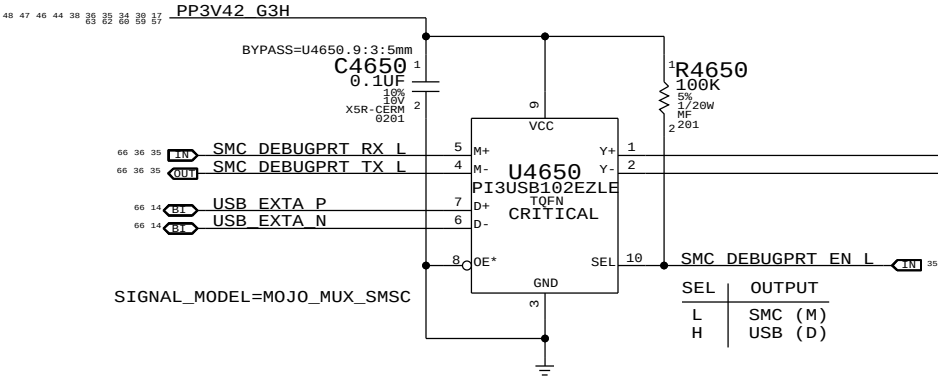
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Right USB Port A

USB Port Power Switch



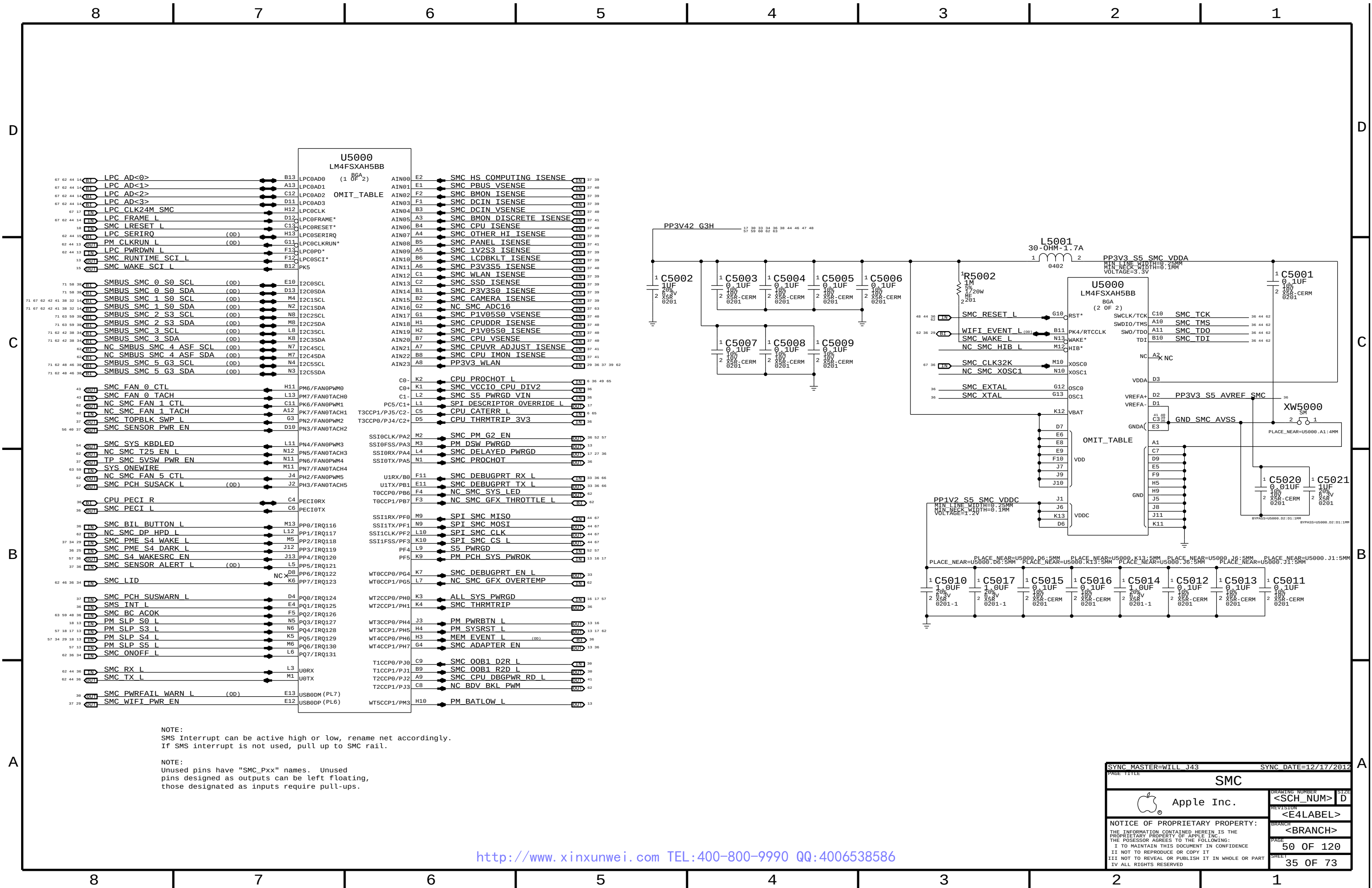
Mojo SMC Debug Mux

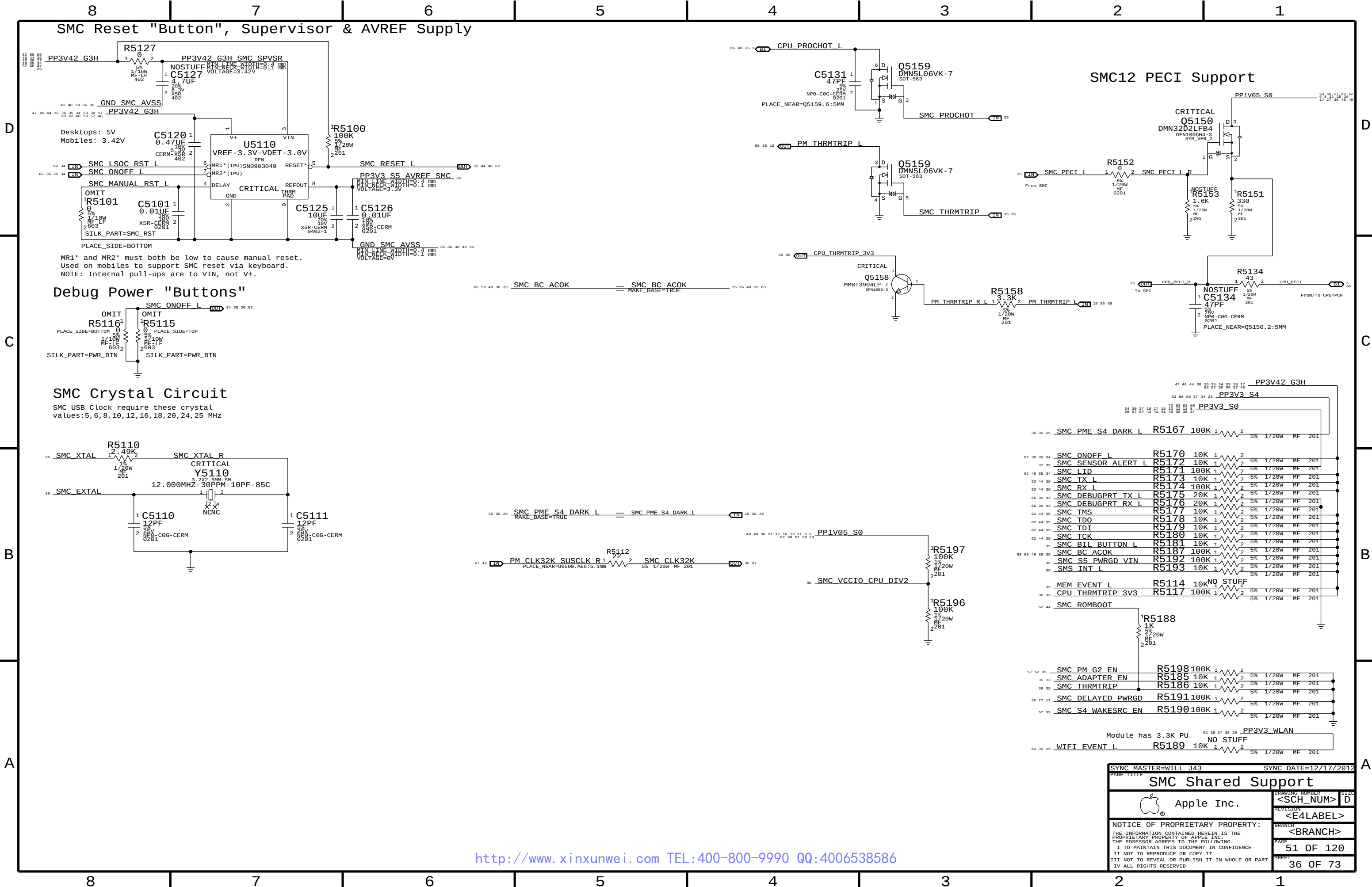




A horizontal number line with a vertical tick mark at the center. The number '1' is written above the tick mark, and the number '2' is written below the tick mark. A circle is drawn on the number line to the right of the tick mark.







SMC Reset "Button", Supervisor & AVREF Supply

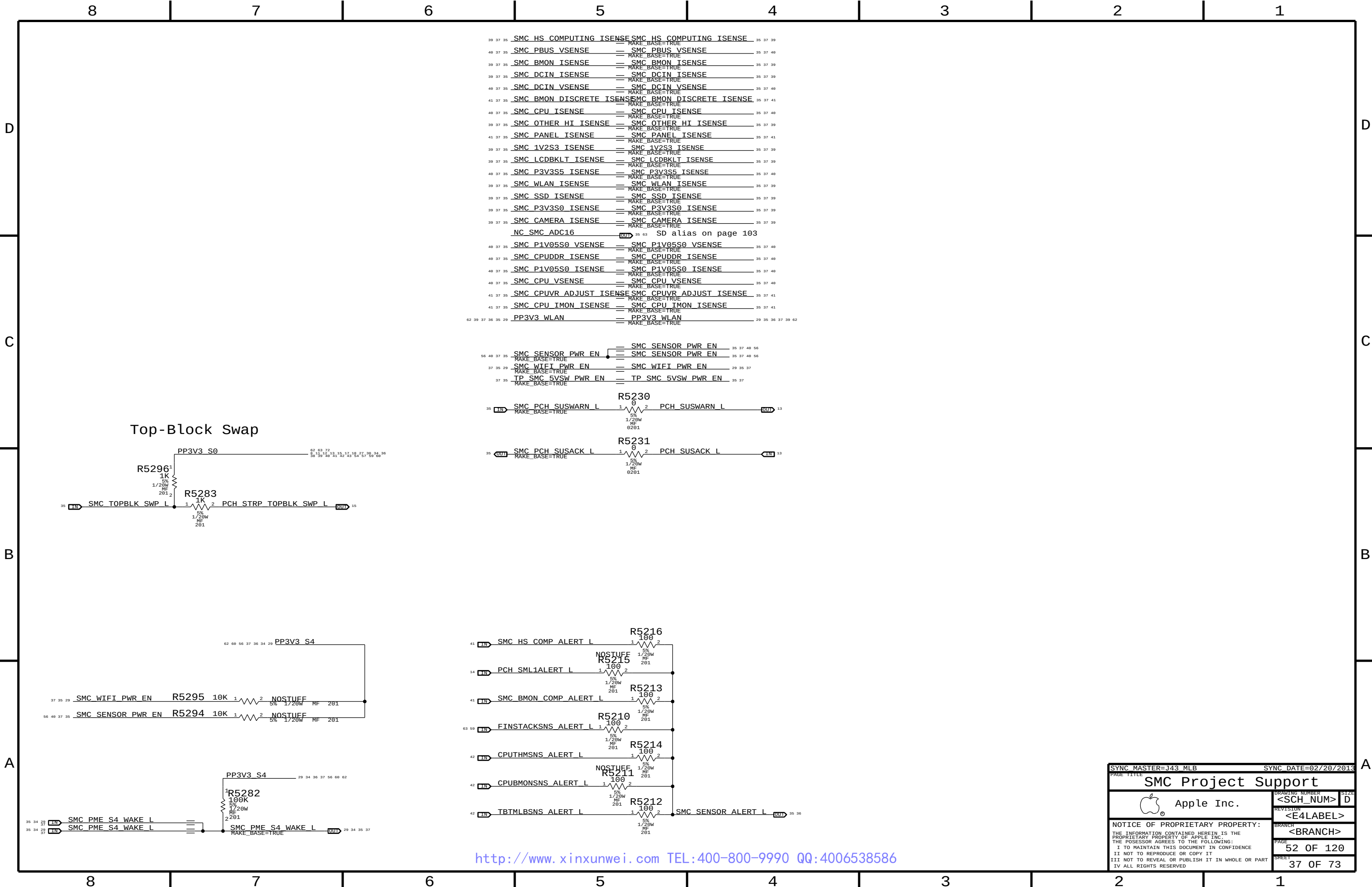
SMC12 PECl Support

Debug Power "Buttons"

SMC Crystal Circuit

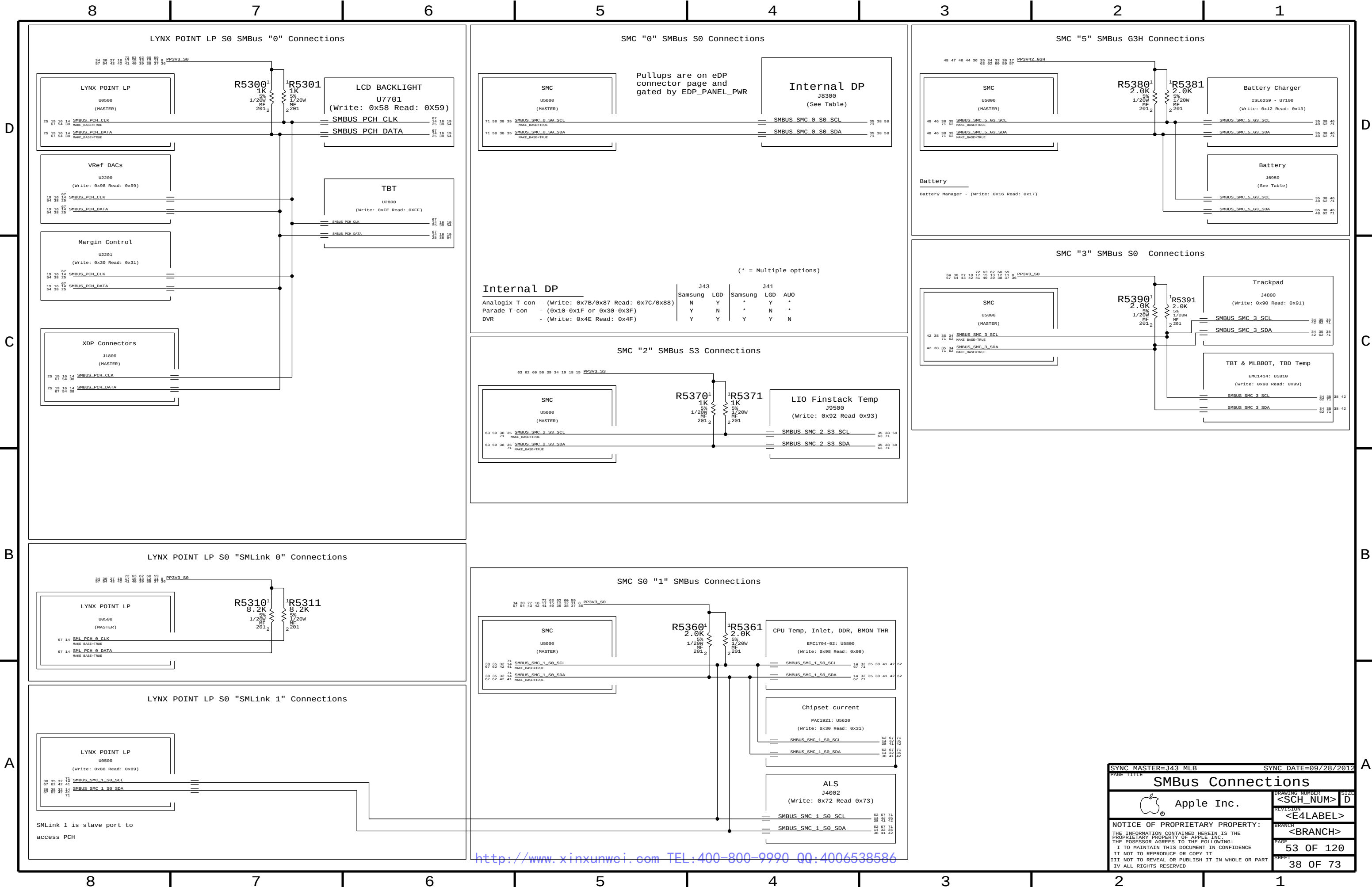
- PP3V42_G3H
- PP3V3_S4
- PP3V3_S0
- SMC PME S4 DARK L
- SMC ONOFF L
- SMC SENSOR ALERT L
- SMC LID
- SMC TX L
- SMC RX L
- SMC DEBUGPRT TX L
- SMC DEBUGPRT RX L
- SMC TMS
- SMC TDO
- SMC TDI
- SMC TCK
- SMC BIL BUTTON L
- SMC BC ACOK
- SMC S5 PWRGD VIN
- SMC INT L
- MEM EVENT L
- CPU THRMTRIP 3V3
- SMC ROMBOOT
- SMC PM G2 EN
- SMC ADAPTER EN
- SMC THRMTRIP
- SMC DELAYED PWRGD
- SMC S4 WAKESRC EN
- PP3V3 WLAN
- WIFI EVENT L

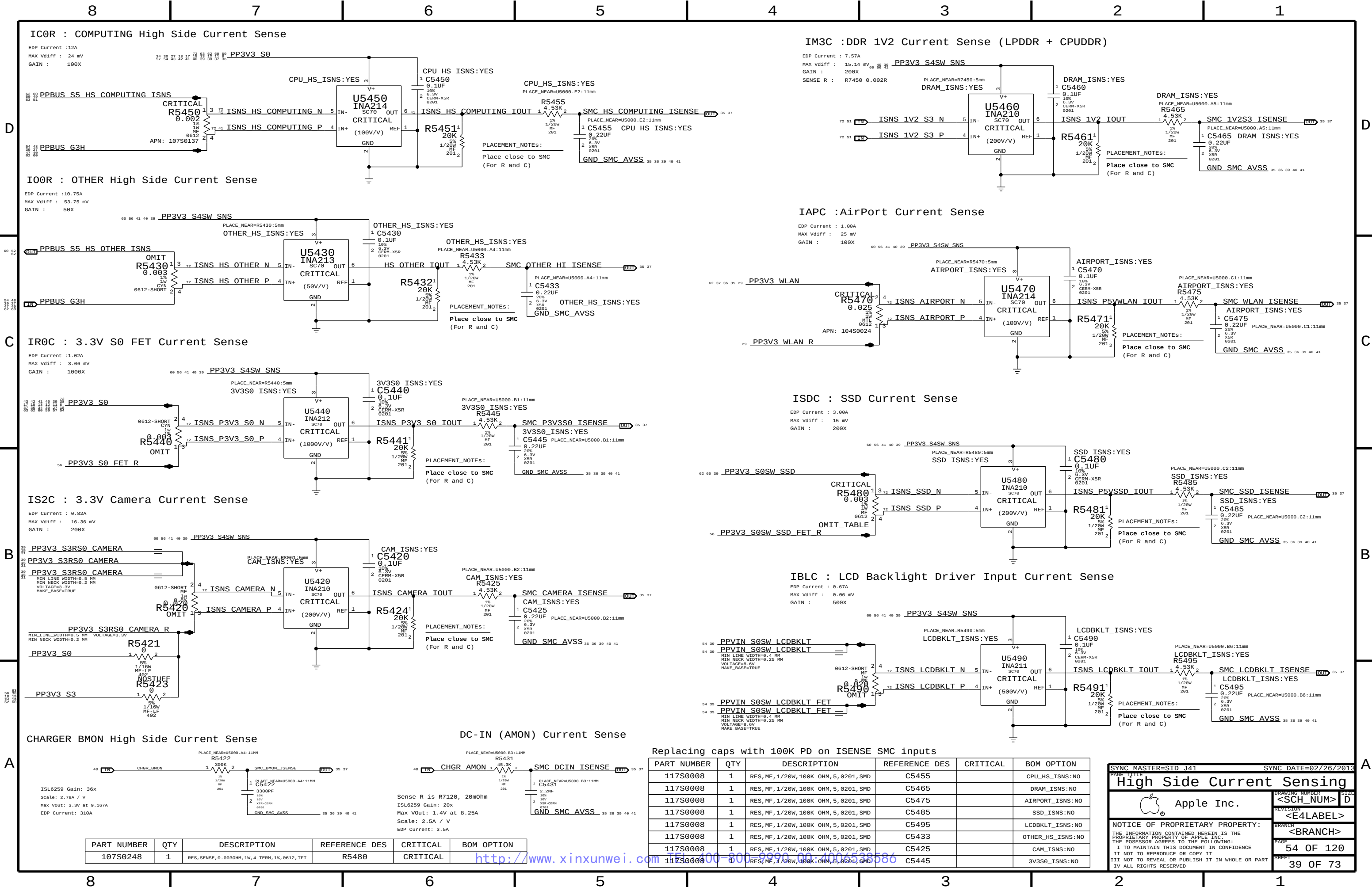
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SYNC MASTER=J43 MLB		SYNC DATE=02/20/2013	
PAGE TITLE			
SMC Project Support			
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SYNC MASTER=SID 341

SYNC DATE=02/26/2013

High Side Current Sensing

Apple Inc.

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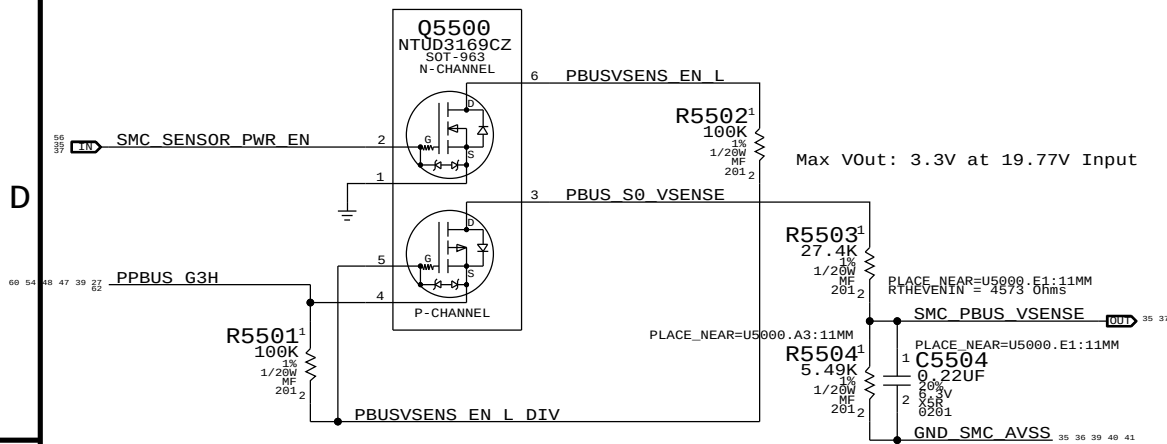
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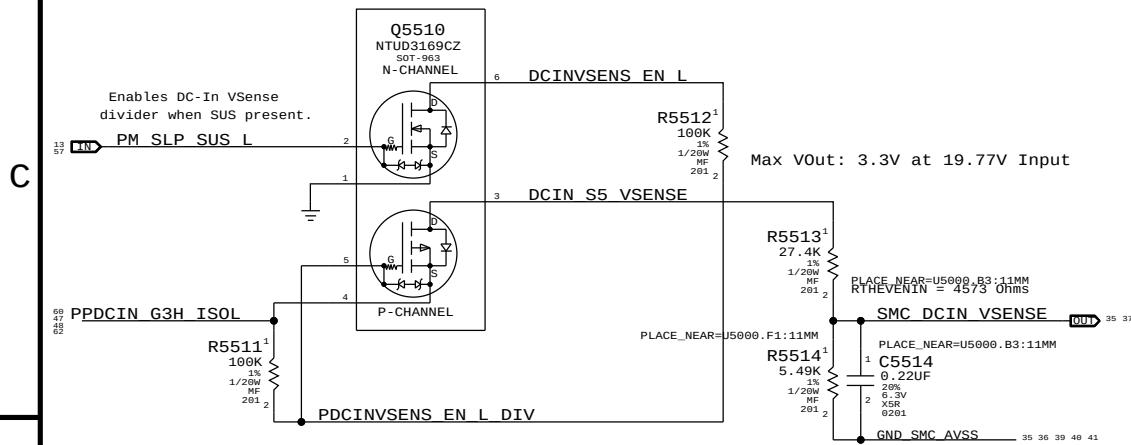
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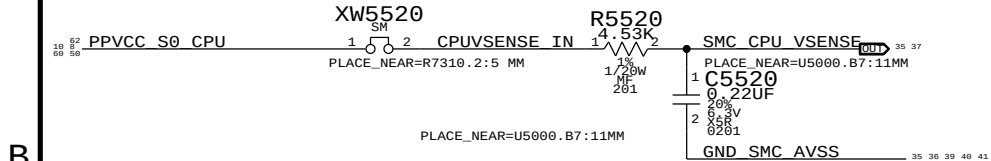
VP0R: PBUS Voltage Sense Enable & Filter



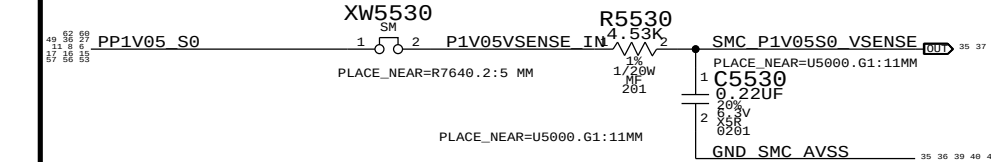
VD0R: DC-In Voltage Sense Enable & Filter



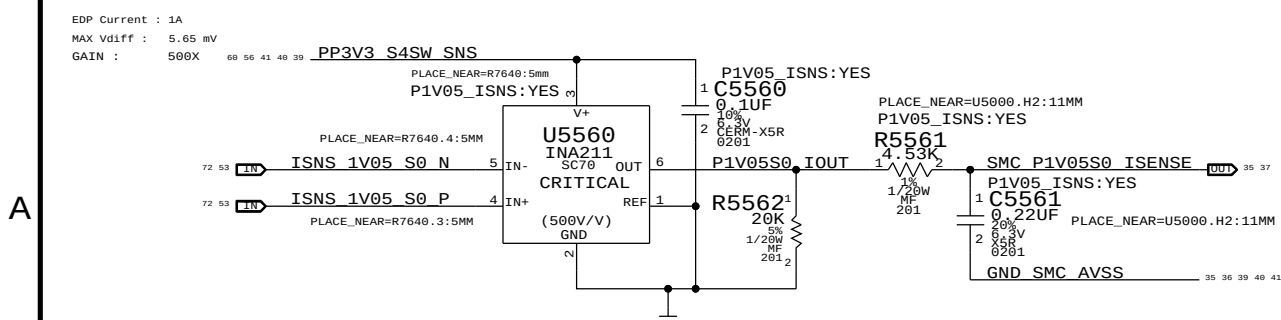
CPU Vcore Voltage Sense / Filter



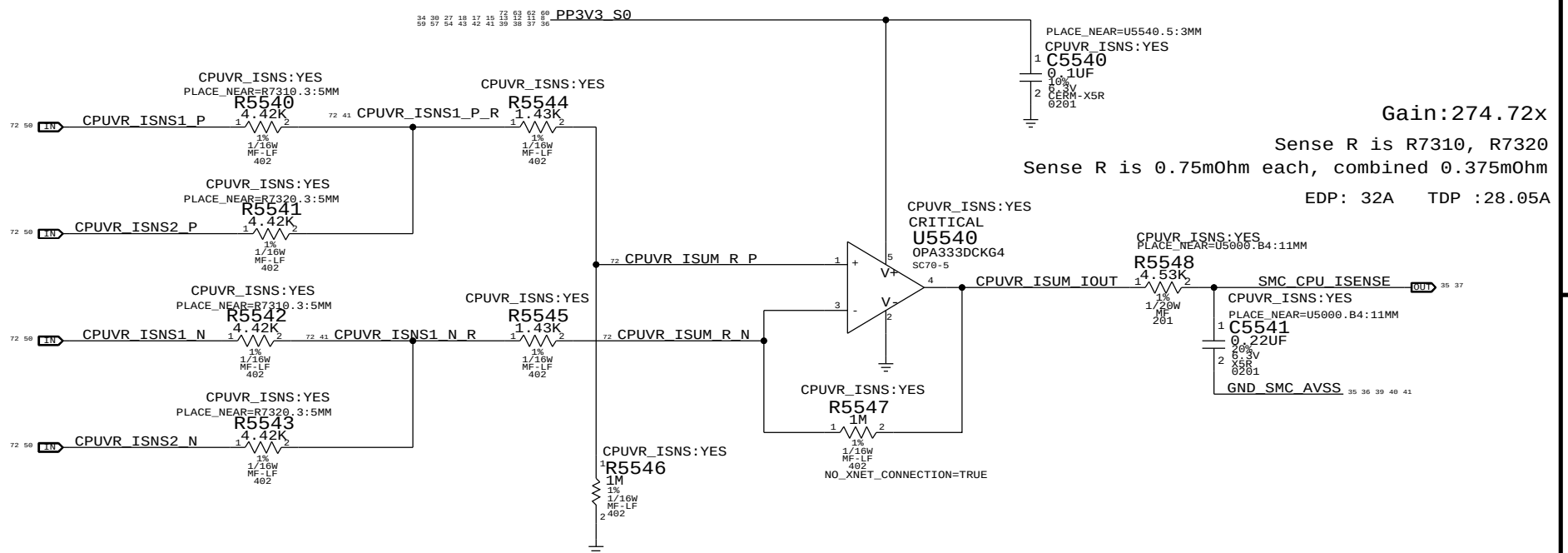
1.05V Voltage Sense / Filter



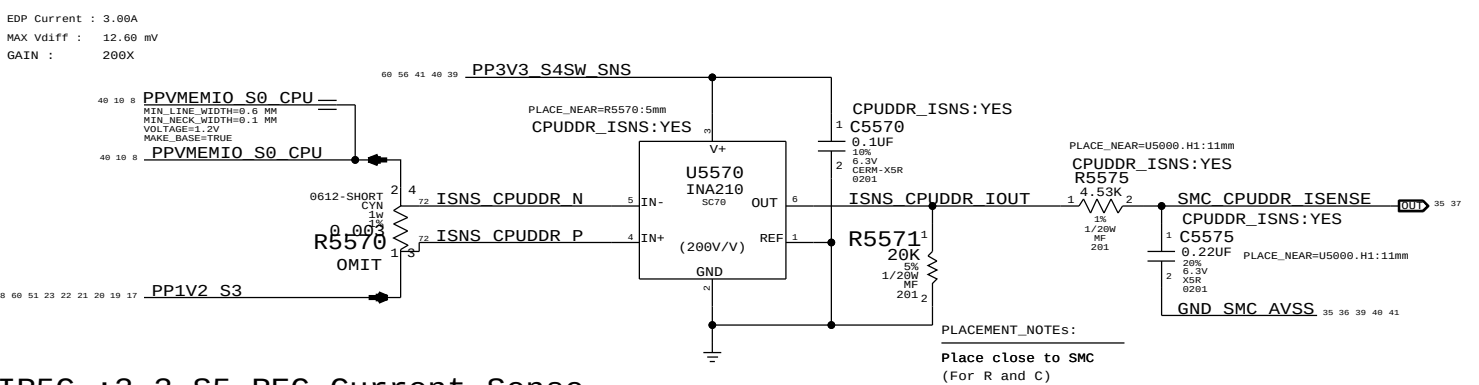
IC1C: 1.05V S0 CURRENT SENSE / FILTER



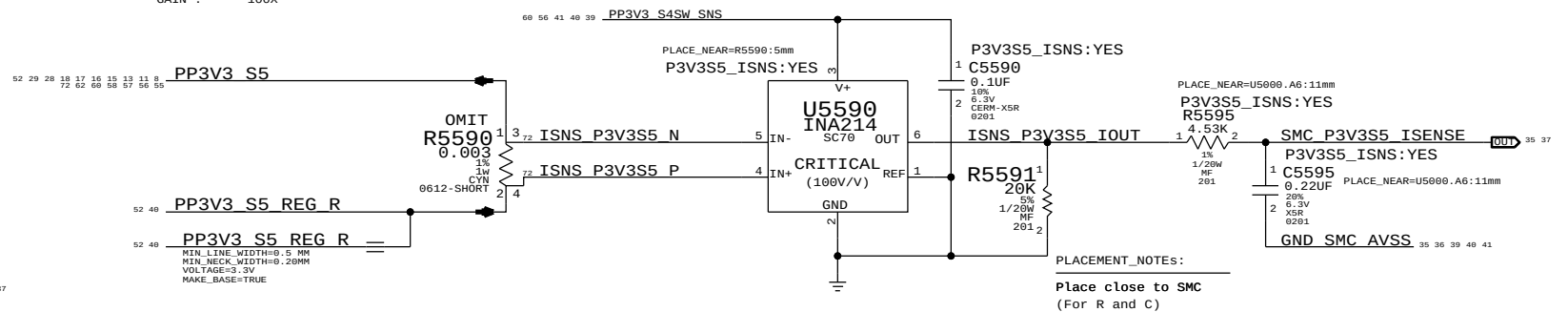
ICS0 : CPU VCore Load Side Current Sense



IM0C : CPU DDR Current Sense



IR5C :3.3 S5 REG Current Sense



Replacing caps with 100K PD on ISENSE SMC inputs

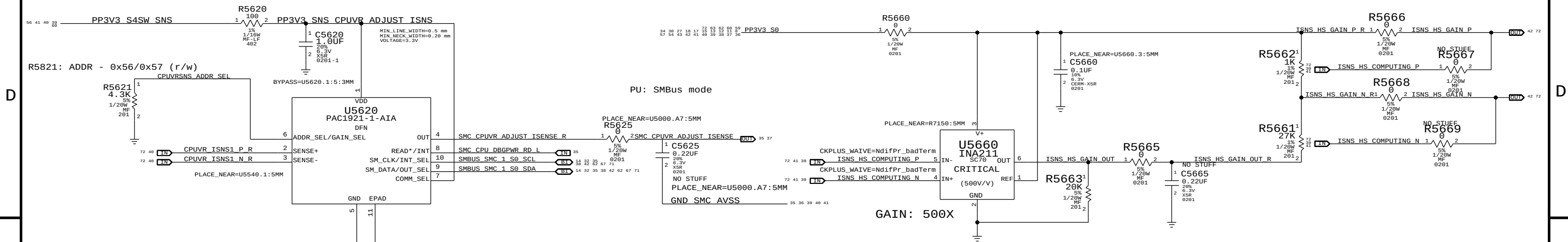
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5541		CPUVR_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5561		P1V05_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5595		P3V3S5_ISNS:NO
117S0008	1	RES,MF,1/20W,100K OHM,5,0201,SMD	C5575		CPUDDR_ISNS:NO

Voltage & Load Side Current Sensing

SYNC MASTER=SID_J41		SYNC DATE=02/26/2013	
Apple Inc.		DRAWING NUMBER	SIZE
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		REVISION	
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		PAGE	55 OF 120
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ICS3 : Adjustable Gain CPU VR Current

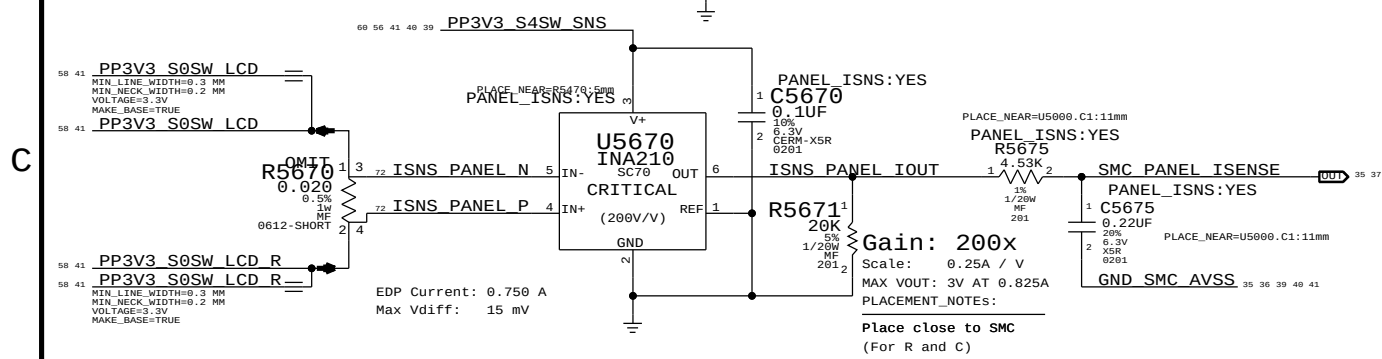
Sense Pins gain stage for U5800 (EMC1704)



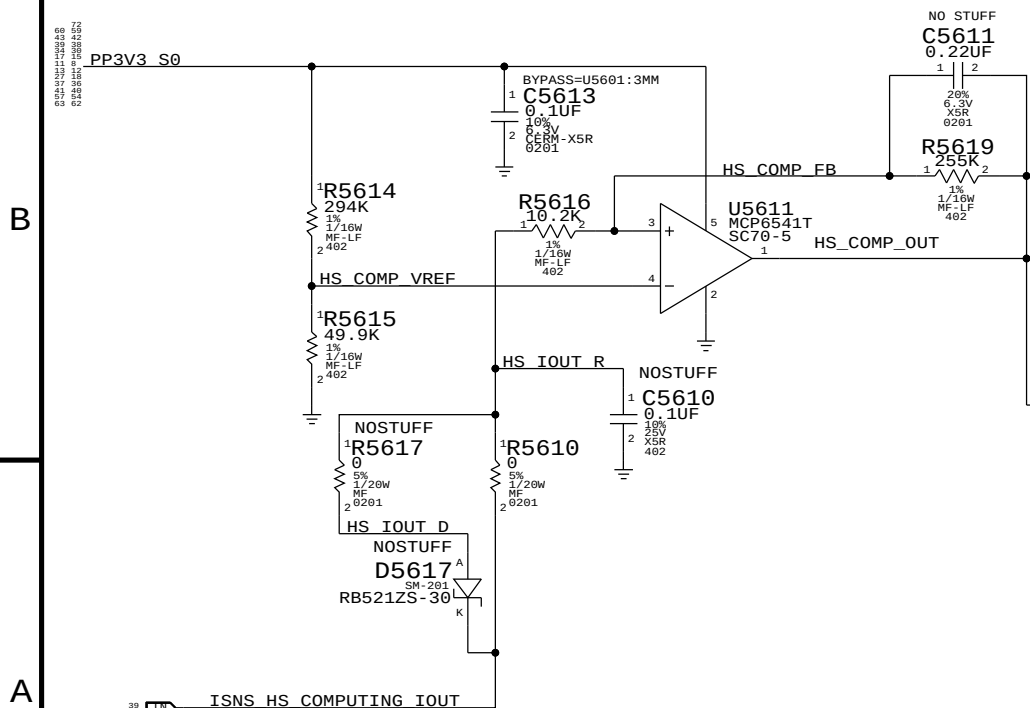
In battery discharge scenario negative voltage will be present on IN+/- pins with INA output voltage decreasing from 3.3V with increasing discharge current.

With 100mA battery current, Will have 10.2mV difference going into sense pins of U5800. This will set the minimum current threshold at 0.100mA

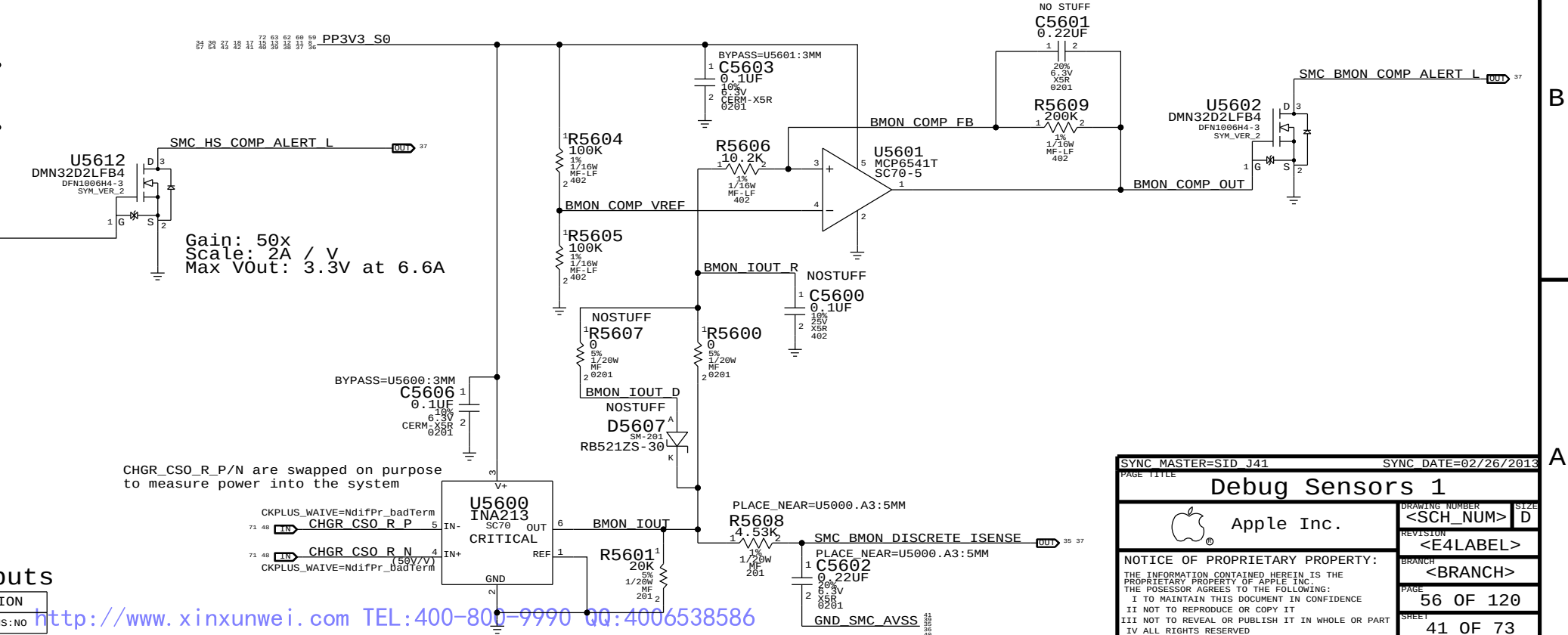
ILDC :LCD Panel Current Sense / Filter



Discrete High side Current threshold



BMON : Discrete BMON Current Sense / Filter



Replacing caps with 100K PD on ISENSE SMC inputs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	1	RES, MF, 1/20W, 100K 0HM, 5, 0201, SMD	C5675		PANEL_ISNS:NO

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SYNC MASTER=SID J41

SYNC DATE=02/26/2013

Debug Sensors 1

Apple Inc.

<SCH_NUM> D

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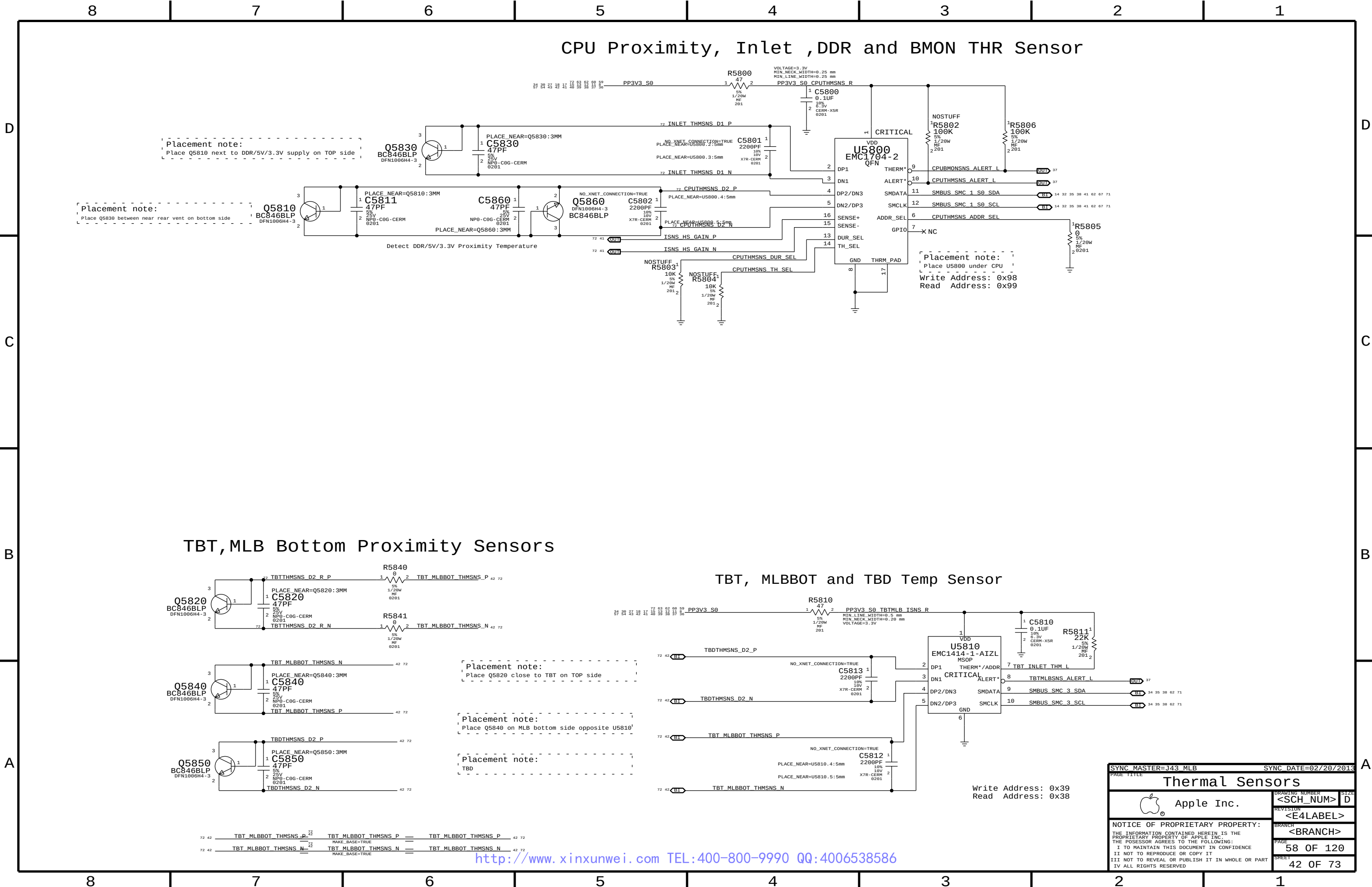
<E4LABEL>

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Placement note:
Place Q5810 next to DDR/5V/3.3V supply on TOP side

Placement note:
Place Q5830 between rear rear vent on bottom side

Placement note:
Place U5800 under CPU
Write Address: 0x98
Read Address: 0x99

TBT,MLB Bottom Proximity Sensors

TBT, MLBBOT and TBD Temp Sensor

Placement note:
Place Q5820 close to TBT on TOP side

Placement note:
Place Q5840 on MLB bottom side opposite U5810

Placement note:
TBD

Write Address: 0x39
Read Address: 0x38

SYNC MASTER=J43 MLB

SYNC DATE=02/20/2013

Thermal Sensors

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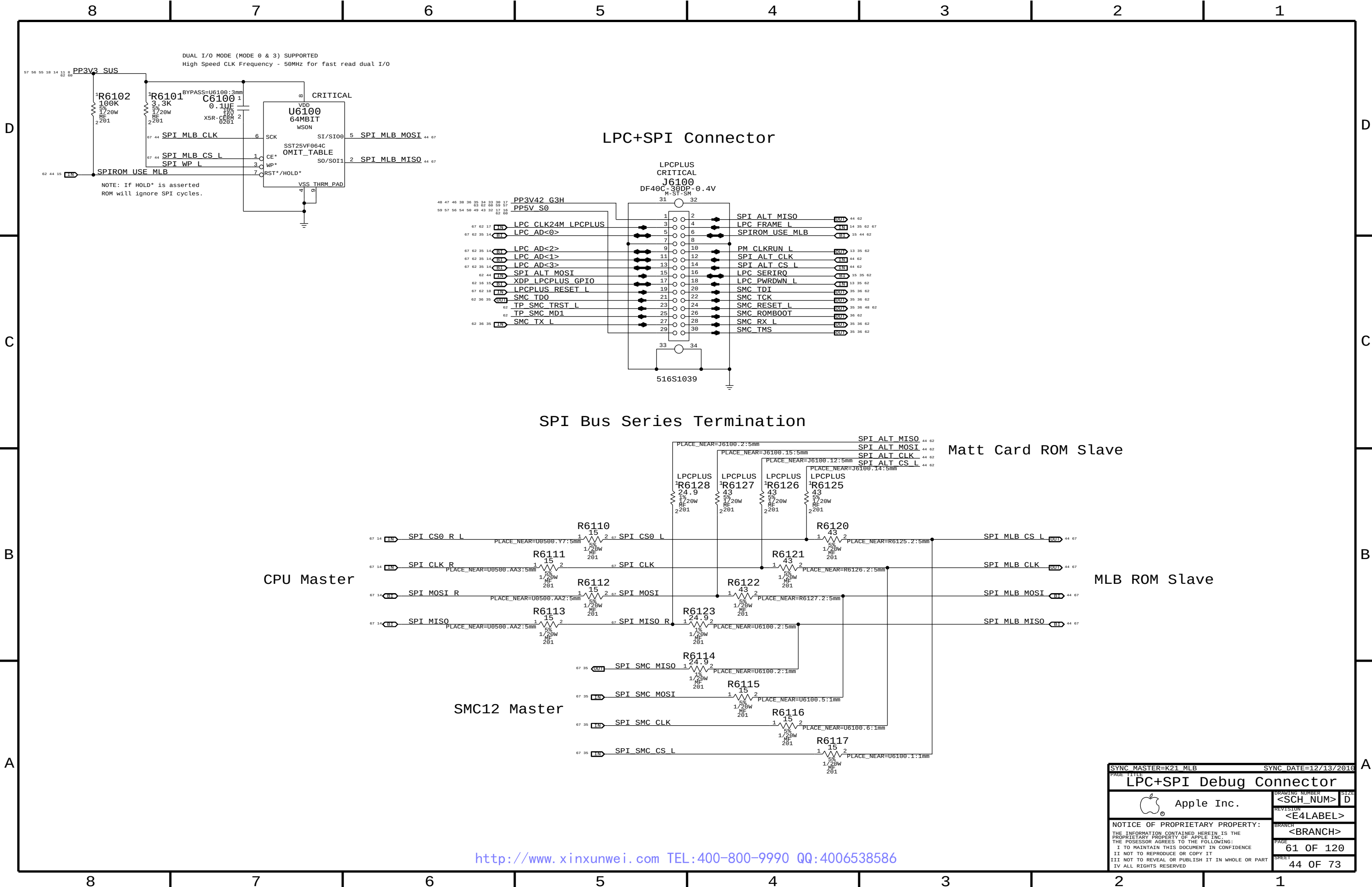
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D

A



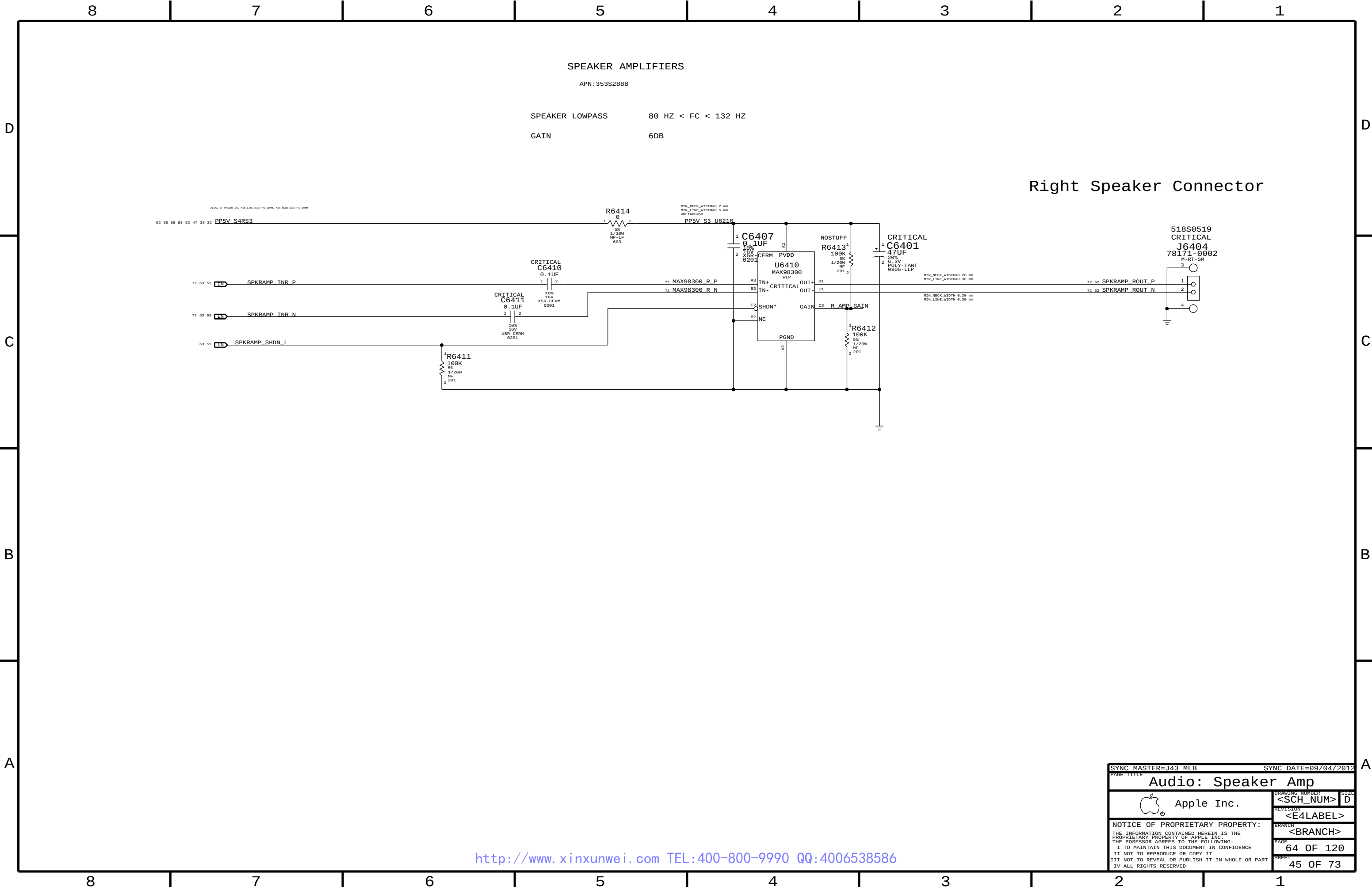
LPC+SPI Connector

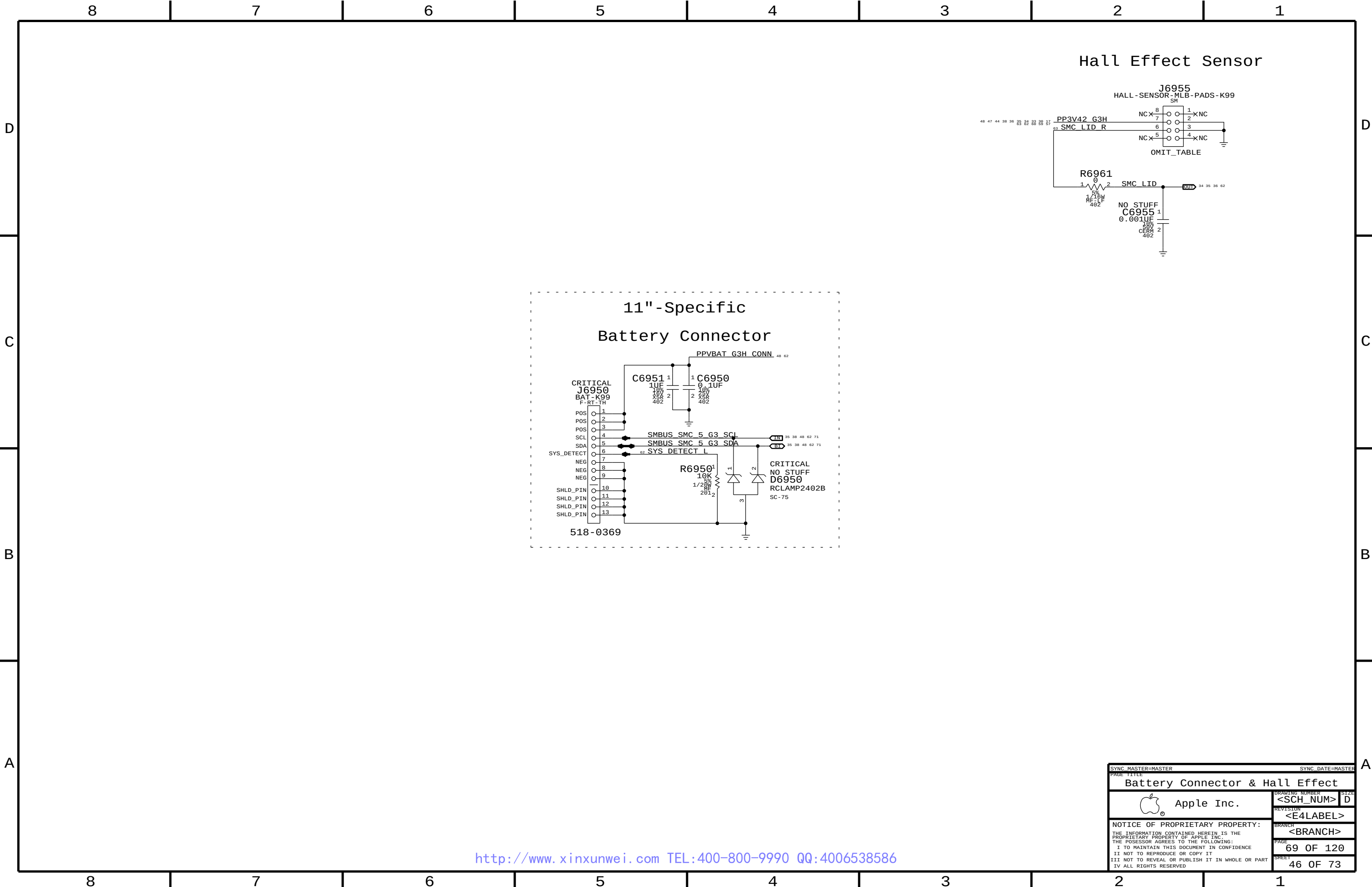
SPI Bus Series Termination

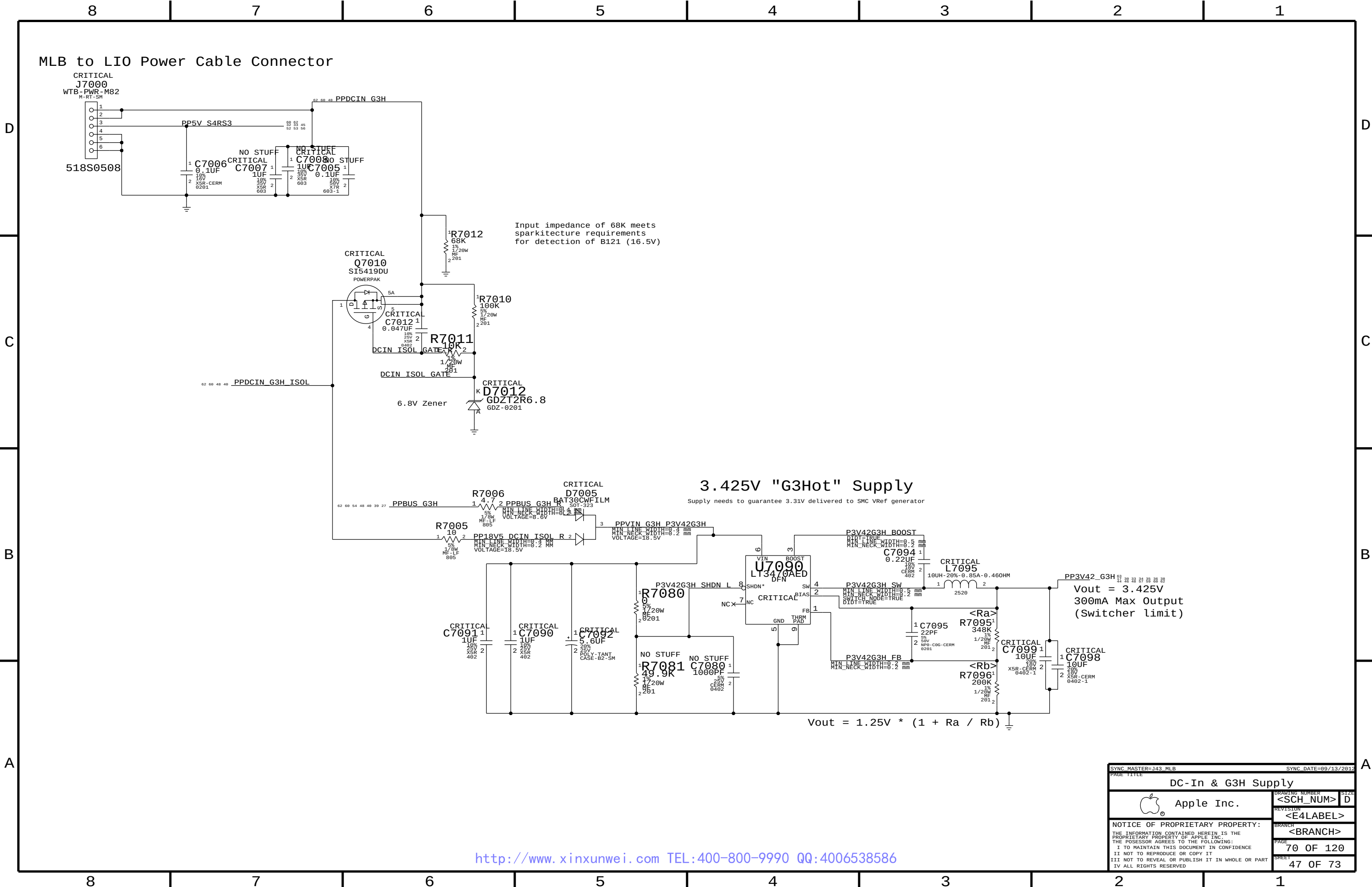
Matt Card ROM Slave

MLB ROM Slave

PAGE TITLE		PAGE	
LPC+SPI Debug Connector		61 OF 120	
Apple Inc.		44 OF 73	
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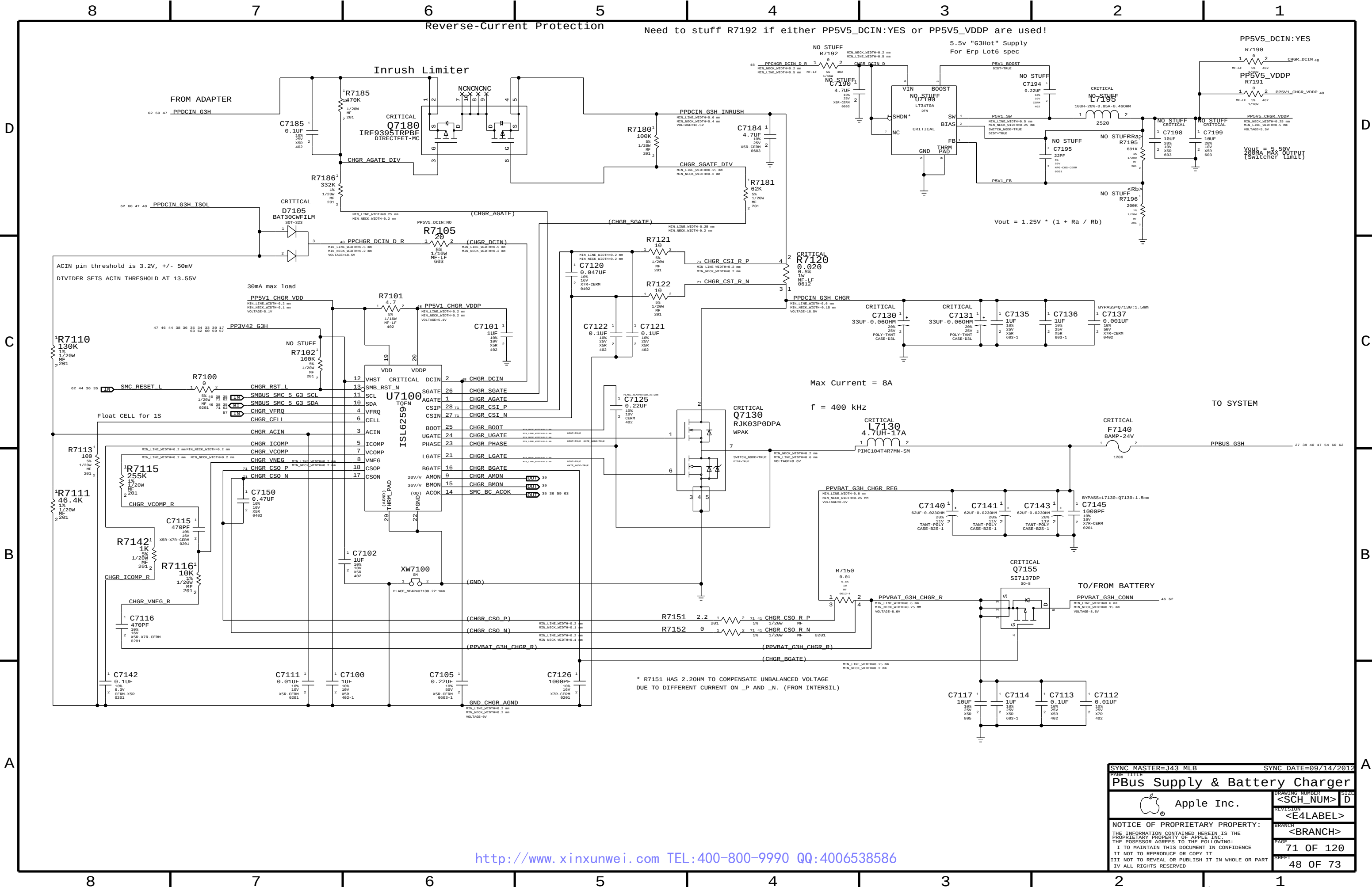
MLB to LIO Power Cable Connector

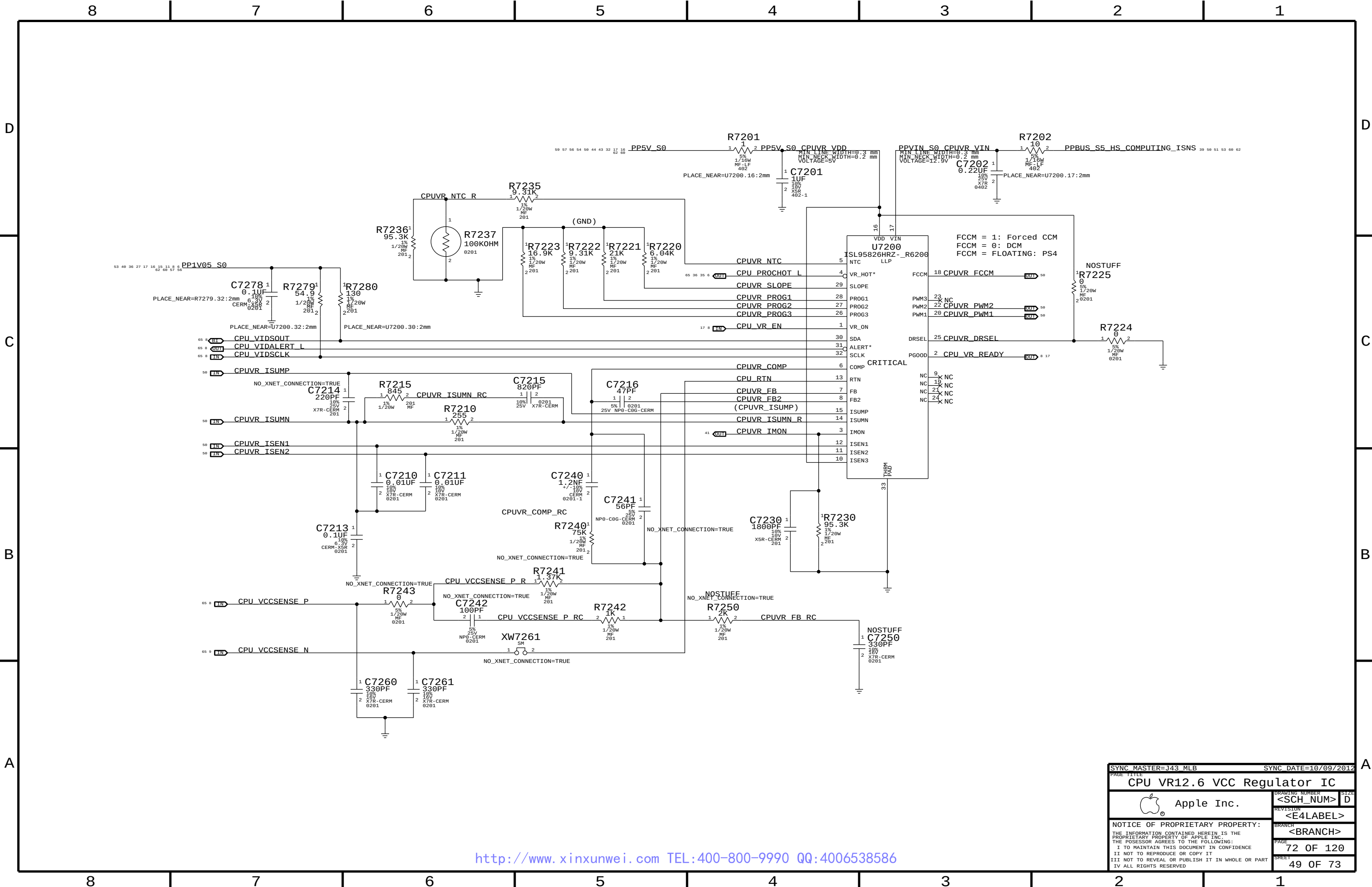
3.425V "G3Hot" Supply

Vout = 3.425V
300mA Max Output
(Switcher limit)

$$V_{out} = 1.25V * (1 + R_a / R_b)$$

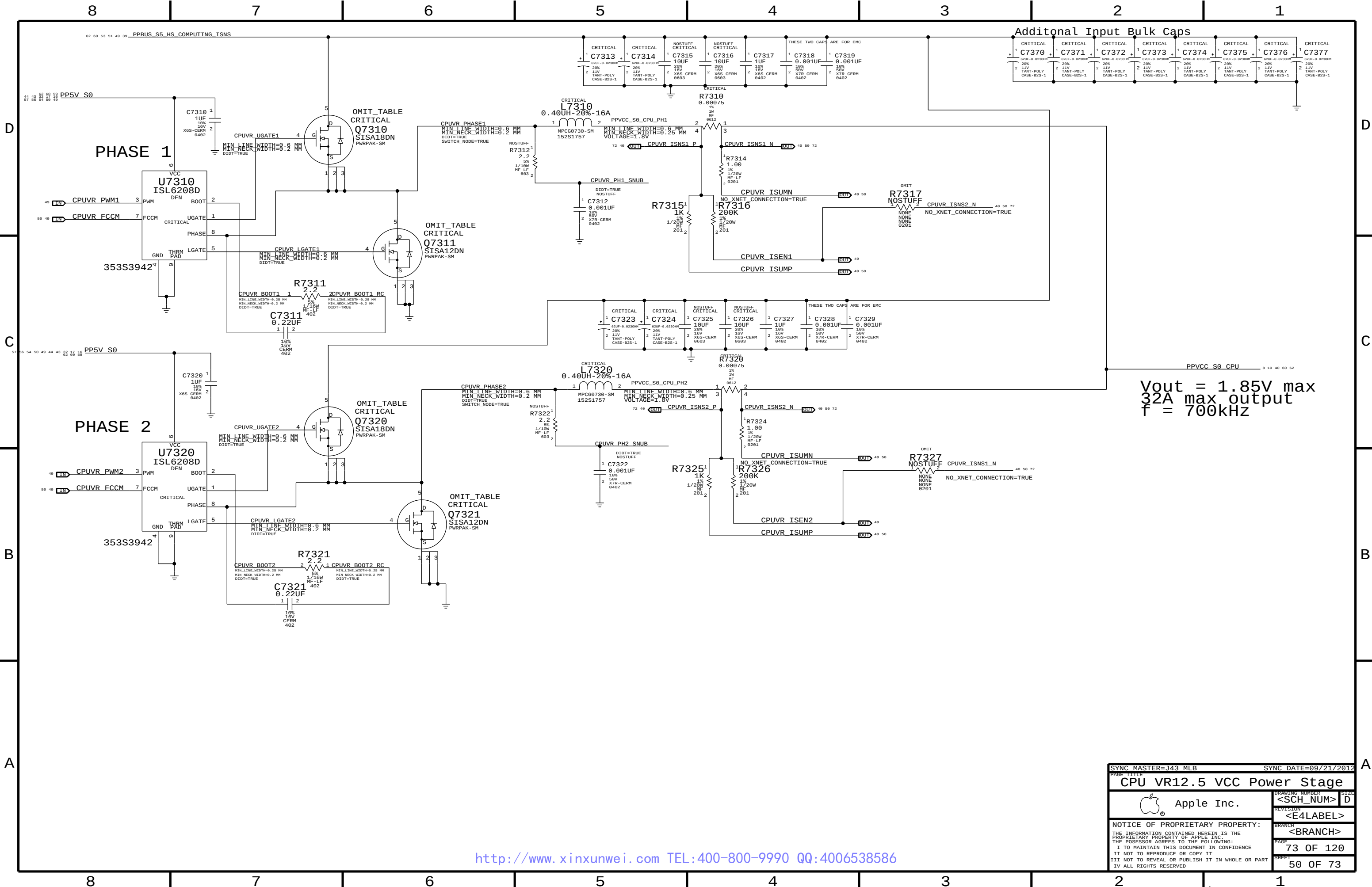
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DC-In & G3H Supply			
	DRAWING NUMBER		SIZE
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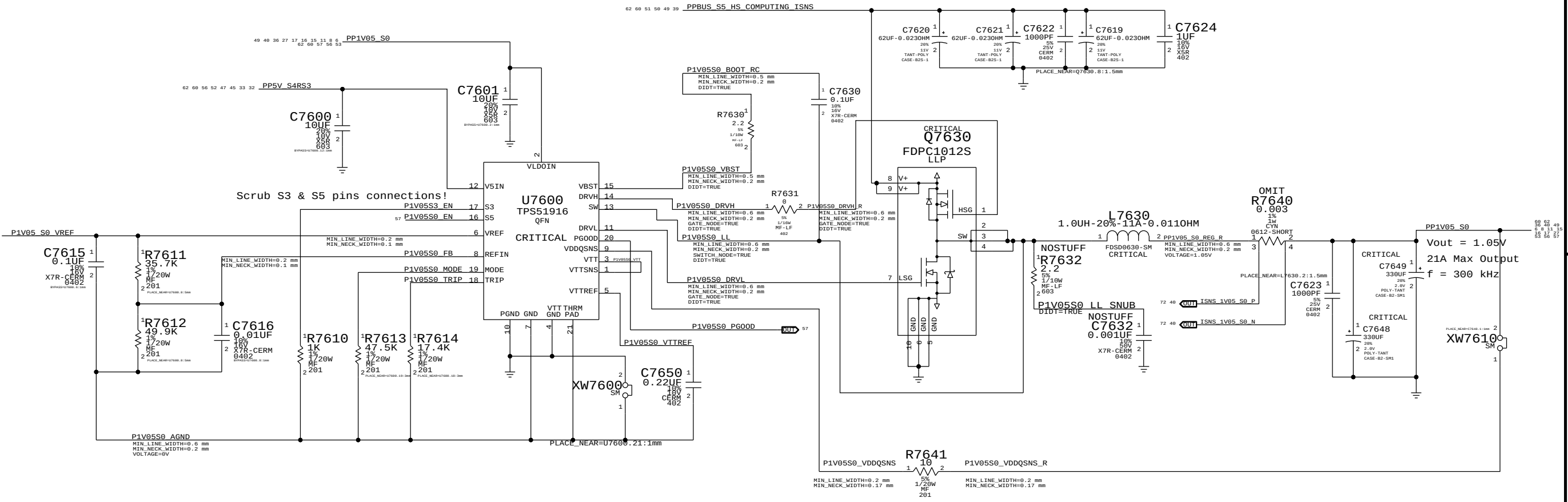


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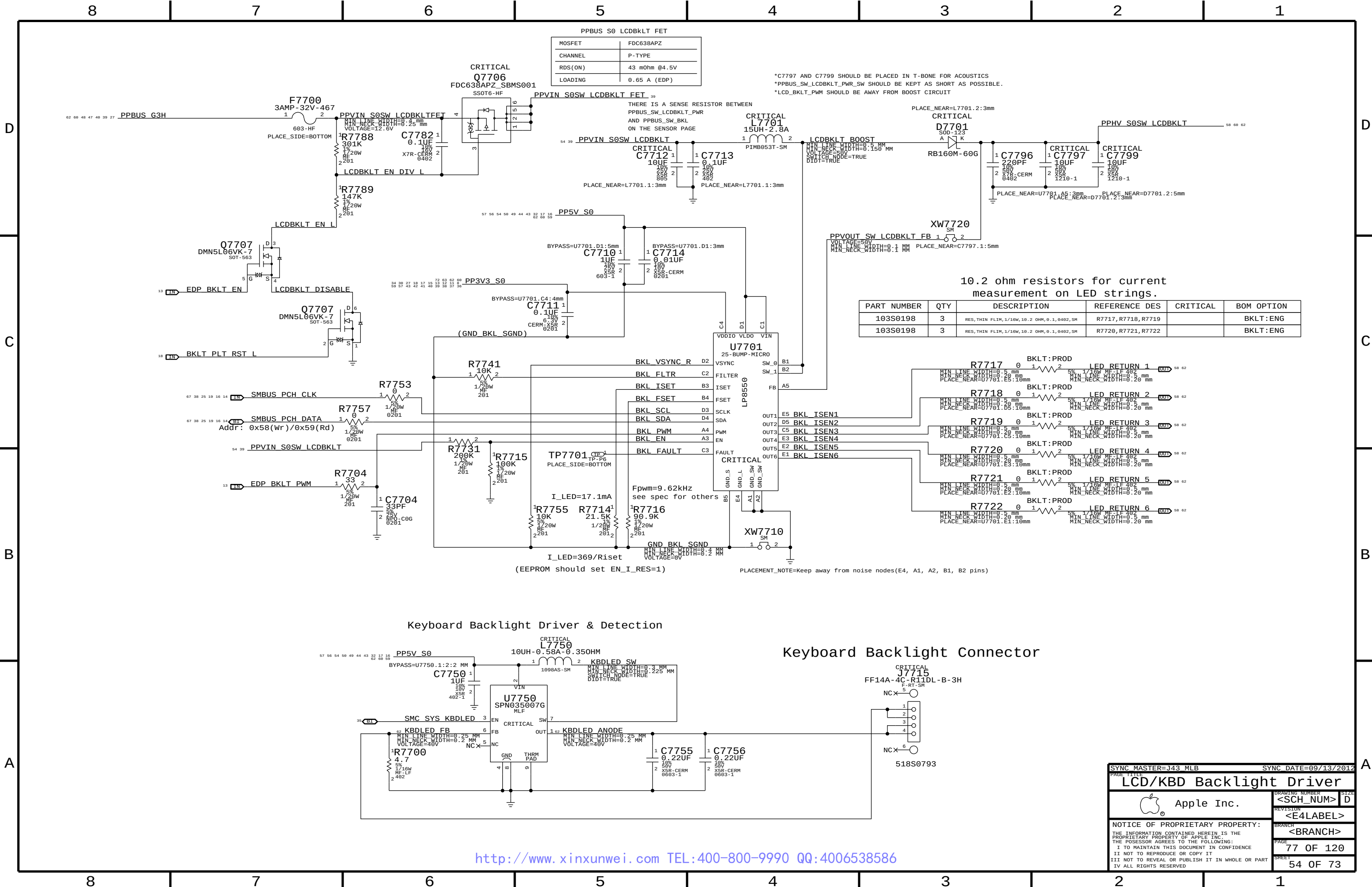
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CPU VR12.6 VCC Regulator IC			
 Apple Inc.		DRAWING NUMBER	SIZE
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		PAGE	72 OF 120
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1.05V S0 Regulator



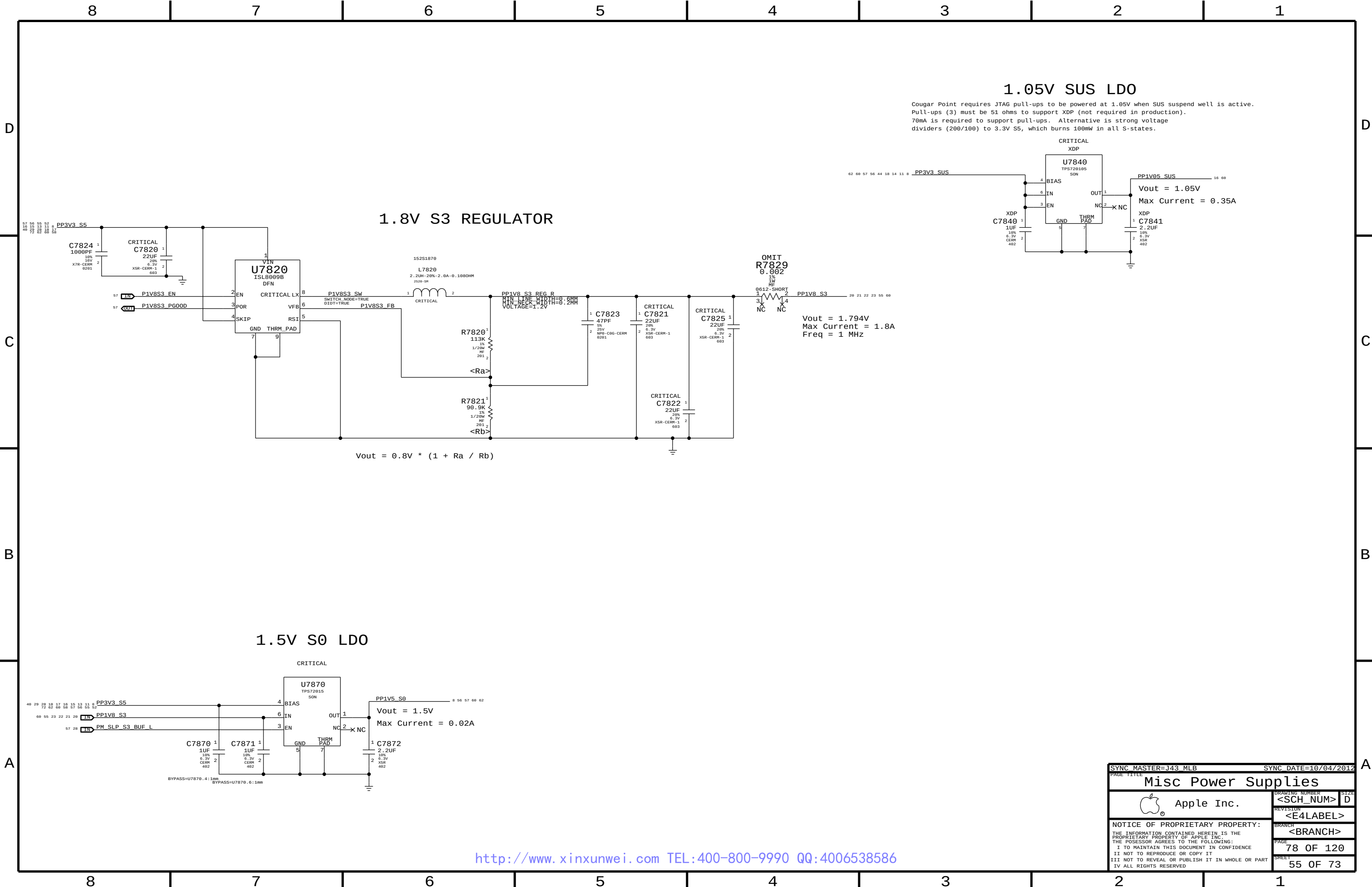
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PAGE TITLE			
1.05V S0 Power Supply			
 Apple Inc.	DRAWING NUMBER		SIZE
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Keyboard Backlight Driver & Detection

Keyboard Backlight Connector

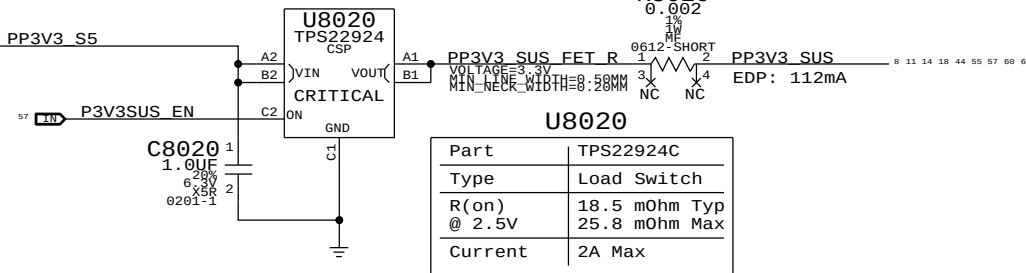
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PAGE TITLE		LCD/KBD Backlight Driver	
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SYNC MASTER=J43 MLB		SYNC DATE=10/04/2012	
PAGE TITLE			
Misc Power Supplies			
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Loading specs per J41/43_PowerBudget_Riviera_rev0.99e

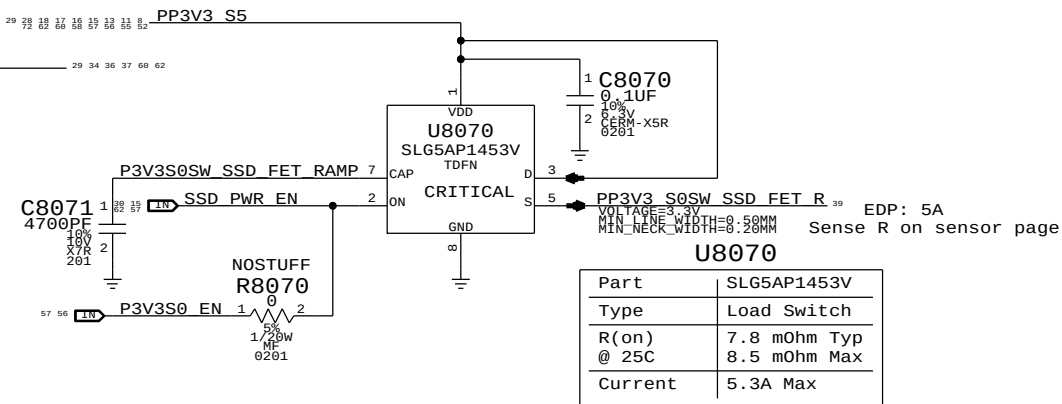
1.05V PCH HSI0 Switch



Part	TPS22924C
Type	Load Switch
R(on) @ 1.8V	19.6 mOhm Typ 21.8 mOhm Max
Current	2A Max

Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

3.3V SSD Switch



Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

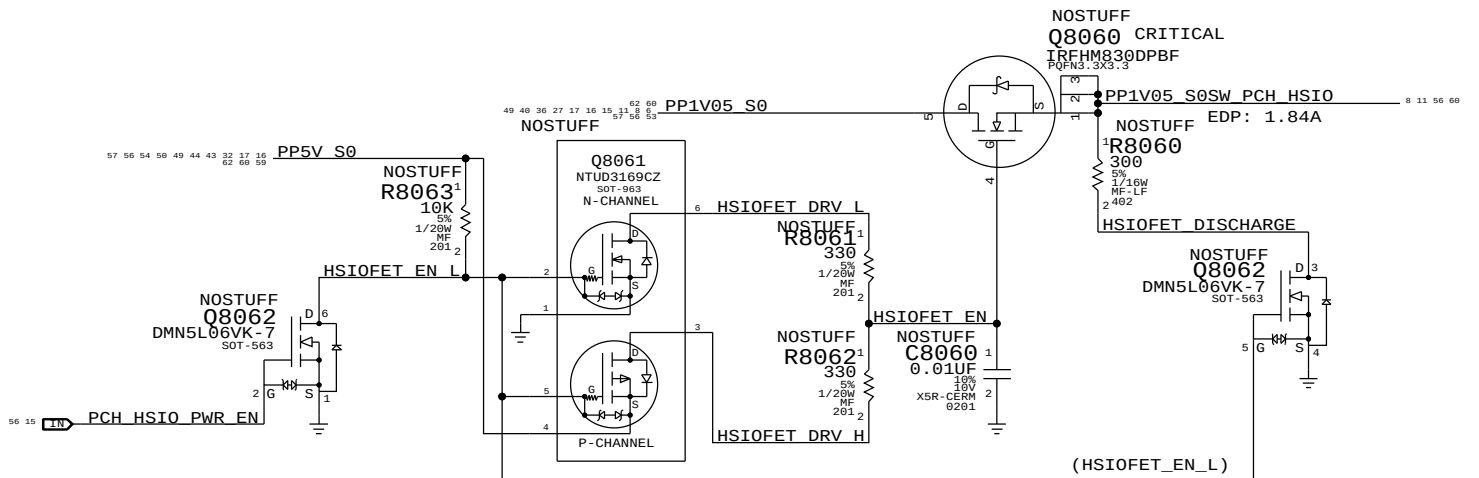
Part	SLG5AP1453V
Type	Load Switch
R(on) @ 25C	7.8 mOhm Typ 8.5 mOhm Max
Current	5.3A Max

page

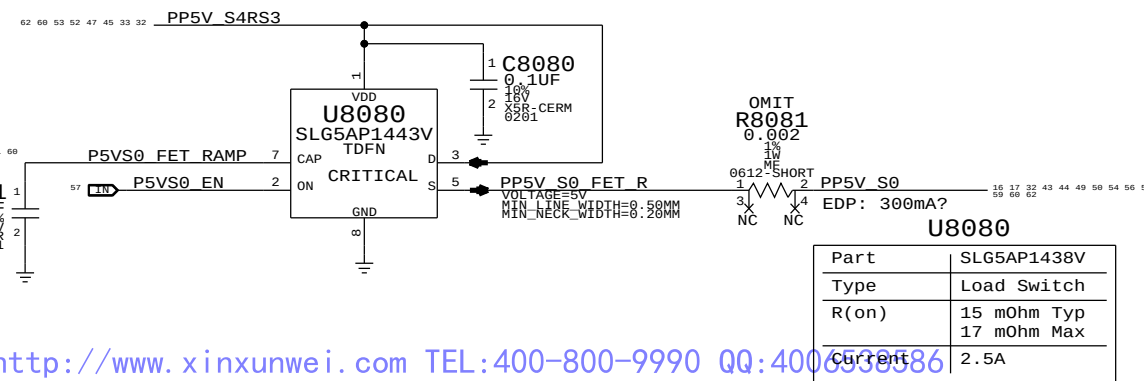
HSIO has turn-on requirement of
 <0.1V/ μ S ramp rate and
 <65 μ S from EN to 95% (1.05V)

Part	SLG5AP1471V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

Part	SLG5AP1417V
Type	Load Switch
R(on) @ 4V Vgs	9.8 mOhm Typ TBD mOhm Max
Current	6A Max

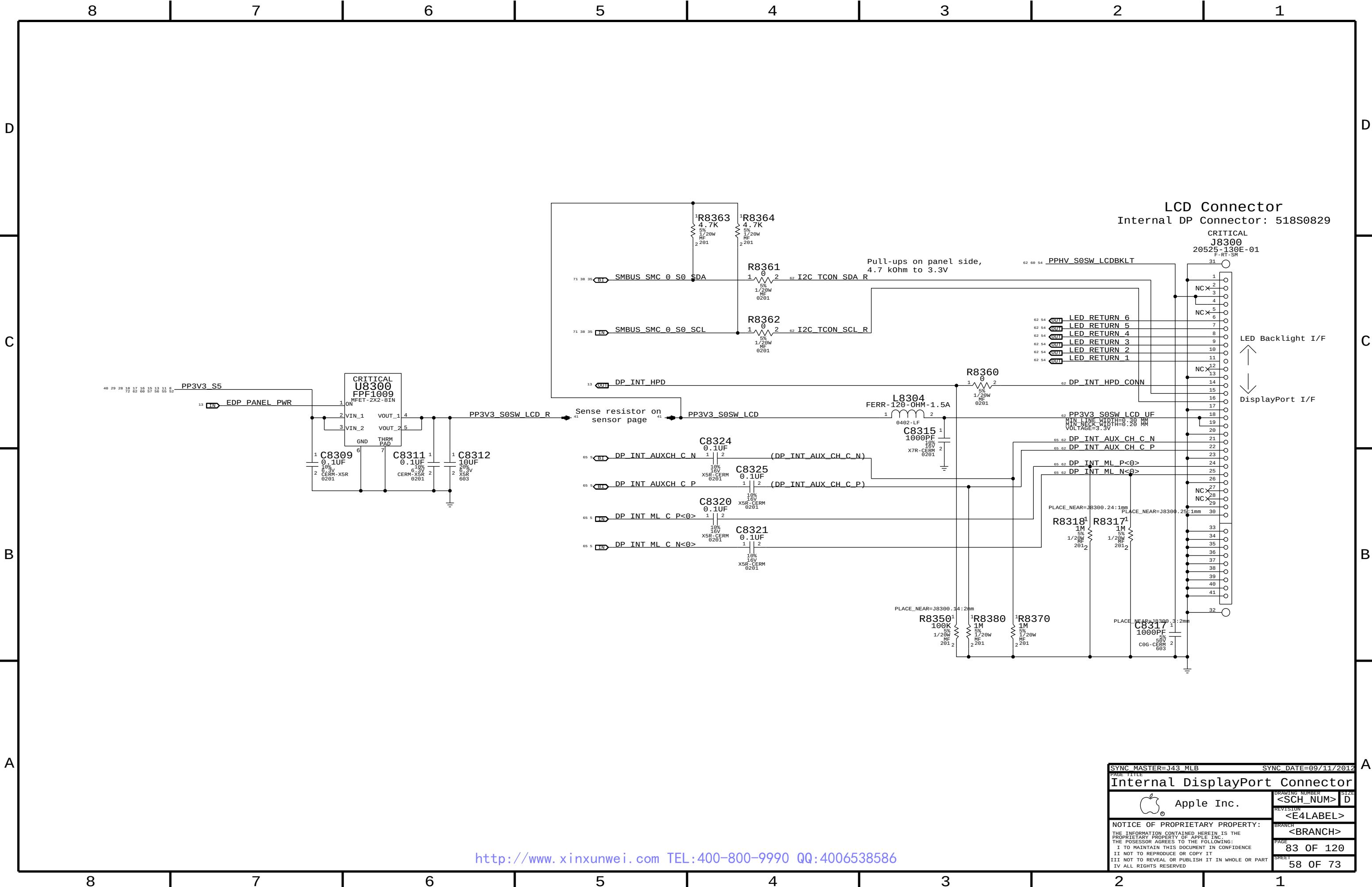


3.3V Sensor Switch

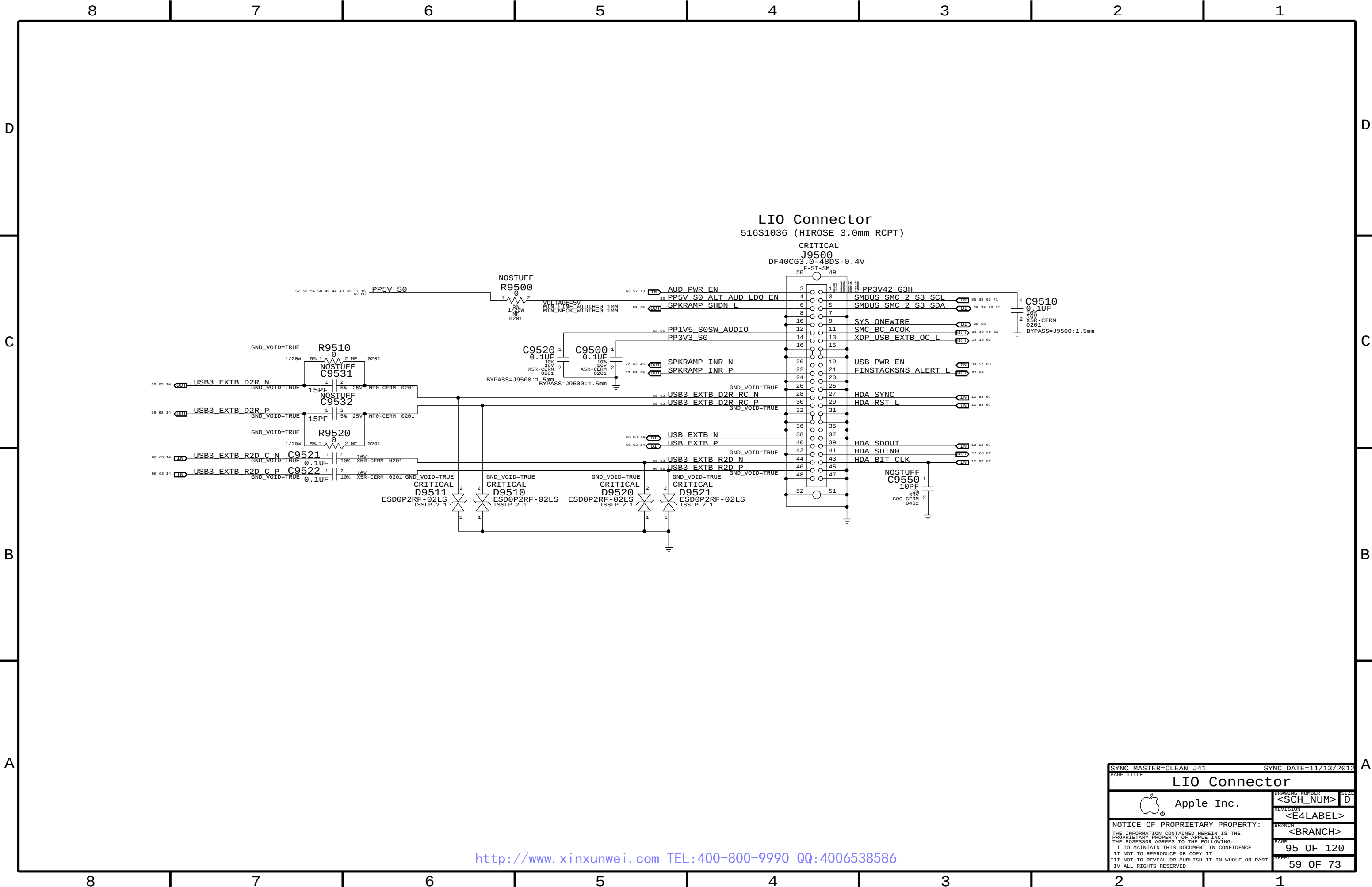


Part	TPS22924C
Type	Load Switch
R(on) @ 2.5V	18.5 mOhm Typ 25.8 mOhm Max
Current	2A Max

Part	SLG5AP1438V
Type	Load Switch
R(on)	15 mOhm Typ 17 mOhm Max
Current	2.5A



SYNC_MASTER=J43_MLB		SYNC_DATE=09/11/2012	
PAGE TITLE			
Internal DisplayPort Connector			
		DRAWING NUMBER	SIZE
Apple Inc.		<SCH_NUM>	D
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7

6

5

4

3

2

1

LPDDR3 Command/Address

Memory Bit/Byte Swizzle

D

C

B

A

MAKE_BASE			
7	=MEM A A<5>	TRUE	MEM A CAA<0>
7	=MEM A A<9>	TRUE	MEM A CAA<1>
7	=MEM A A<6>	TRUE	MEM A CAA<2>
7	=MEM A A<8>	TRUE	MEM A CAA<3>
7	=MEM A A<7>	TRUE	MEM A CAA<4>
7	=MEM A BA<2>	TRUE	MEM A CAA<5>
68 61 24 20 7	MEM A CAA<6>	TRUE	MEM A CAA<6>
7	=MEM A A<11>	TRUE	MEM A CAA<7>
7	=MEM A A<15>	TRUE	MEM A CAA<8>
7	=MEM A A<14>	TRUE	MEM A CAA<9>
7	=MEM A A<13>	TRUE	MEM A CAB<0>
7	=MEM A CAS L	TRUE	MEM A CAB<1>
7	=MEM A WE L	TRUE	MEM A CAB<2>
7	=MEM A RAS L	TRUE	MEM A CAB<3>
7	=MEM A BA<0>	TRUE	MEM A CAB<4>
7	=MEM A A<2>	TRUE	MEM A CAB<5>
68 61 24 21 7	MEM A CAB<6>	TRUE	MEM A CAB<6>
7	=MEM A A<10>	TRUE	MEM A CAB<7>
7	=MEM A A<1>	TRUE	MEM A CAB<8>
7	=MEM A A<0>	TRUE	MEM A CAB<9>
68 61 24 21 20 7	MEM A ODT<0>	TRUE	MEM A ODT<0>
61 7	TP LPDDR3 RSVD1	TRUE	TP LPDDR3 RSVD1
61 7	TP LPDDR3 RSVD2	TRUE	TP LPDDR3 RSVD2
7	=MEM B A<5>	TRUE	MEM B CAA<0>
7	=MEM B A<9>	TRUE	MEM B CAA<1>
7	=MEM B A<6>	TRUE	MEM B CAA<2>
7	=MEM B A<8>	TRUE	MEM B CAA<3>
7	=MEM B A<7>	TRUE	MEM B CAA<4>
7	=MEM B BA<2>	TRUE	MEM B CAA<5>
68 61 24 22 7	MEM B CAA<6>	TRUE	MEM B CAA<6>
7	=MEM B A<11>	TRUE	MEM B CAA<7>
7	=MEM B A<15>	TRUE	MEM B CAA<8>
7	=MEM B A<14>	TRUE	MEM B CAA<9>
7	=MEM B A<13>	TRUE	MEM B CAB<0>
7	=MEM B CAS L	TRUE	MEM B CAB<1>
7	=MEM B WE L	TRUE	MEM B CAB<2>
7	=MEM B RAS L	TRUE	MEM B CAB<3>
7	=MEM B BA<0>	TRUE	MEM B CAB<4>
7	=MEM B A<2>	TRUE	MEM B CAB<5>
68 61 24 23 7	MEM B CAB<6>	TRUE	MEM B CAB<6>
7	=MEM B A<10>	TRUE	MEM B CAB<7>
7	=MEM B A<1>	TRUE	MEM B CAB<8>
7	=MEM B A<0>	TRUE	MEM B CAB<9>
68 61 24 23 22 7	MEM B ODT<0>	TRUE	MEM B ODT<0>
61 7	TP LPDDR3 RSVD3	TRUE	TP LPDDR3 RSVD3
61 7	TP LPDDR3 RSVD4	TRUE	TP LPDDR3 RSVD4

MAKE_BASE			
20	=MEM A DQ<0>	TRUE	MEM A DQ<9>
20	=MEM A DQ<1>	TRUE	MEM A DQ<12>
20	=MEM A DQ<2>	TRUE	MEM A DQ<10>
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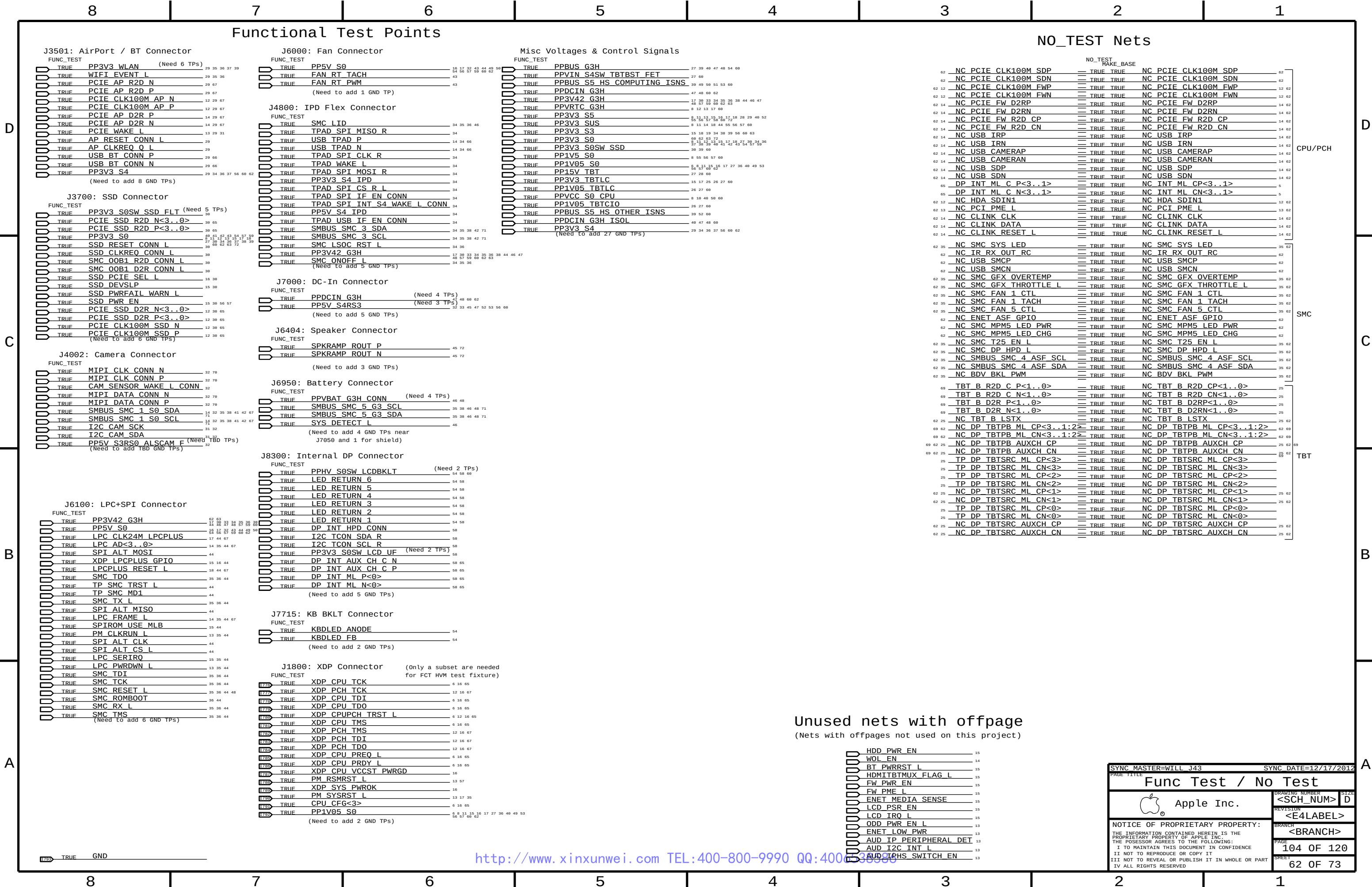
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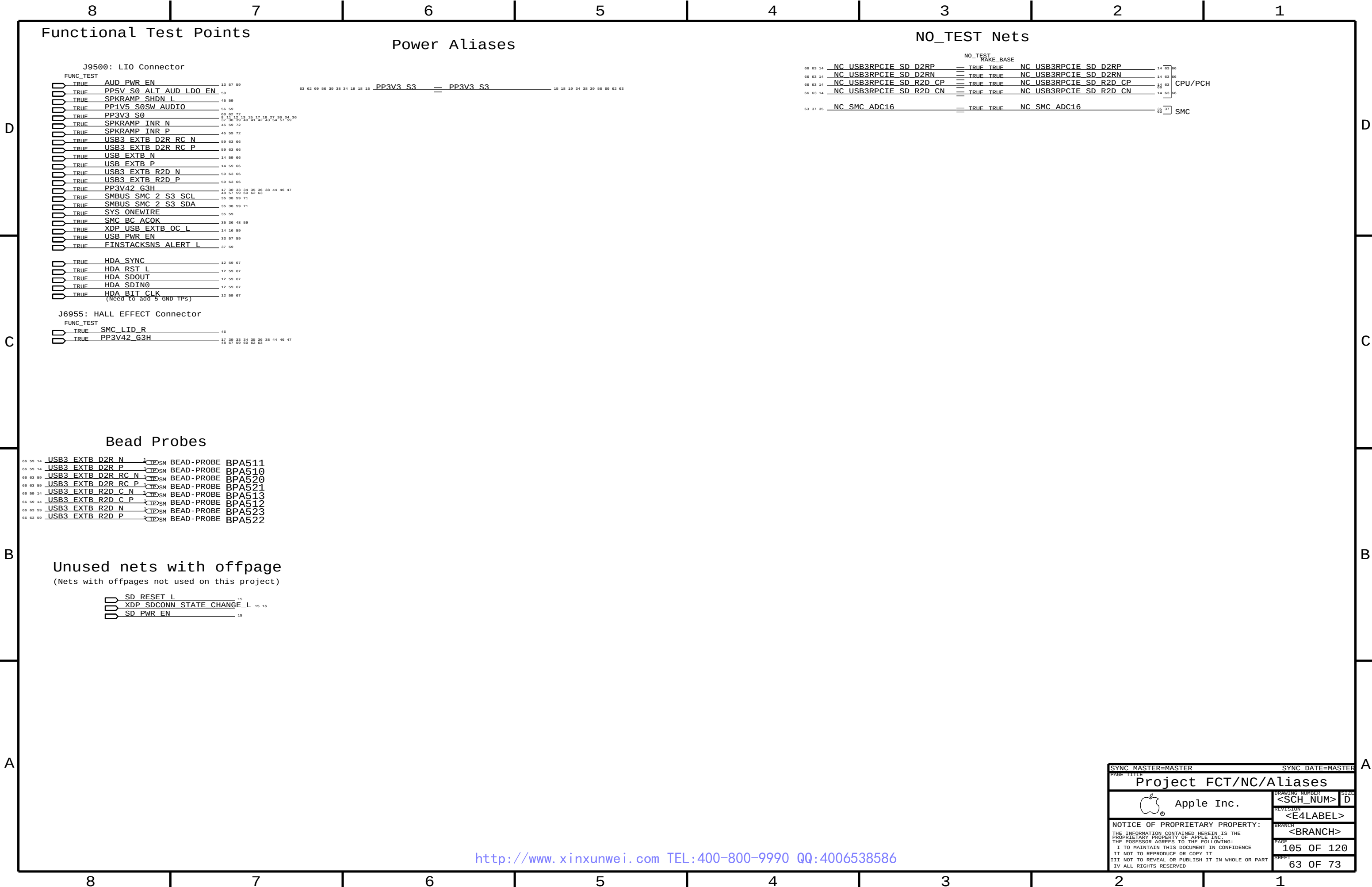
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IV ALL RIGHTS RESERVED		104 OF 120	
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J41/J43 Board-Specific Spacing & Physical Constraints

BOARD LAYERS	BOARD AREAS	BOARD UNITS (MIL or MM)	ALLEGRO VERSION
TOP, ISL2, ISL3, ISL4, ISL5, ISL6, ISL7, ISL8, ISL9, ISL10, ISL11, BOTTOM	NO_TYPE, BGA, MEM_TERM	MM	16.2

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DEFAULT	TOP,BOTTOM	Y	=50_OHM_SE	=50_OHM_SE			
DEFAULT	ISL2, ISL11	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	ISL3, ISL10	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	ISL4, ISL9	Y	=45_OHM_SE	=45_OHM_SE			
DEFAULT	*	N	100 MM	100 MM	10 MM	0 MM	0 MM
STANDARD	*	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT	=DEFAULT

Single-ended Physical Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
27P4_OHM_SE	TOP,BOTTOM	Y	0.310 MM	0.310 MM			
27P4_OHM_SE	ISL2,ISL11	Y	0.182 MM	0.182 MM			
27P4_OHM_SE	ISL3,ISL10	Y	0.182 MM	0.182 MM			
27P4_OHM_SE	ISL4,ISL9	Y	0.182 MM	0.182 MM			
27P4_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
35_OHM_SE	TOP,BOTTOM	Y	0.195 MM	0.195 MM			
35_OHM_SE	ISL2,ISL11	Y	0.125 MM	0.125 MM			
35_OHM_SE	ISL3,ISL10	Y	0.125 MM	0.125 MM			
35_OHM_SE	ISL4,ISL9	Y	0.125 MM	0.125 MM			
35_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
40_OHM_SE	TOP,BOTTOM	Y	0.170 MM	0.170 MM			
40_OHM_SE	ISL2,ISL11	Y	0.096 MM	0.096 MM			
40_OHM_SE	ISL3,ISL10	Y	0.096 MM	0.096 MM			
40_OHM_SE	ISL4,ISL9	Y	0.099 MM	0.099 MM			
40_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
45_OHM_SE	TOP,BOTTOM	Y	0.135 MM	0.135 MM			
45_OHM_SE	ISL2,ISL11	Y	0.075 MM	0.075 MM			
45_OHM_SE	ISL3,ISL10	Y	0.075 MM	0.075 MM			
45_OHM_SE	ISL4,ISL9	Y	0.080 MM	0.080 MM			
45_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
50_OHM_SE	TOP,BOTTOM	Y	0.110 MM	0.110 MM			
50_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
55_OHM_SE	TOP, BOTTOM	Y	0.090 MM	0.090 MM			
55_OHM_SE	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

Differential Pair Physical Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
70_OHM_DIFF	TOP,BOTTOM	Y	0.165 MM	0.165 MM		0.110 MM	0.110 MM
70_OHM_DIFF	ISL2,ISL11	Y	0.105 MM	0.105 MM		0.100 MM	0.100 MM
70_OHM_DIFF	ISL3,ISL10	Y	0.105 MM	0.105 MM		0.100 MM	0.100 MM
70_OHM_DIFF	ISL4,ISL9	Y	0.110 MM	0.110MM		0.095 MM	0.095 MM
70_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
80_OHM_DIFF	TOP,BOTTOM	Y	0.132 MM	0.132 MM		0.130 MM	0.130 MM
80_OHM_DIFF	ISL2,ISL11	Y	0.081 MM	0.081 MM		0.115 MM	0.115 MM
80_OHM_DIFF	ISL3,ISL10	Y	0.081 MM	0.081 MM		0.115 MM	0.115 MM
80_OHM_DIFF	ISL4,ISL9	Y	0.088 MM	0.088 MM		0.110 MM	0.110 MM
80_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
90_OHM_DIFF	TOP,BOTTOM	Y	0.115 MM	0.115 MM		0.200 MM	0.200 MM
90_OHM_DIFF	ISL2,ISL11	Y	0.070 MM	0.070 MM		0.180 MM	0.180 MM
90_OHM_DIFF	ISL3,ISL10	Y	0.070 MM	0.070 MM		0.180 MM	0.180 MM
90_OHM_DIFF	ISL4,ISL9	Y	0.076 MM	0.076 MM		0.180 MM	0.180 MM
90_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

Spacing Constraints

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1:1_SPACING	*	0.100 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
1X_DIELECTRIC	TOP, BOTTOM	0.071 MM	?
1X_DIELECTRIC	ISL3, ISL10	0.053 MM	?
1X_DIELECTRIC	ISL4, ISL9	0.050 MM	?
1X_DIELECTRIC	*	0.090 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DEFAULT	*	0.1 MM	?
STANDARD	*	=DEFAULT	?
BGA_P075MM	*	0.075 MM	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
*	*	BGA	BGA_P075MM

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
*	BGA	P070MM_BGA

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
P070MM_BGA	*			0.070 MM	5 MM		0.075 MM

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
73_OHM_DIFF	TOP, BOTTOM	Y	0.165 MM	0.165 MM		0.150 MM	0.150 MM
73_OHM_DIFF	ISL2, ISL11	Y	0.106 MM	0.106 MM		0.150 MM	0.150 MM
73_OHM_DIFF	ISL3, ISL10	Y	0.106 MM	0.106 MM		0.150 MM	0.150 MM
73_OHM_DIFF	ISL4, ISL9	Y	0.110 MM	0.110 MM		0.150 MM	0.150 MM
73_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
85_OHM_DIFF	TOP,BOTTOM	Y	0.120 MM	0.120 MM		0.150 MM	0.150 MM
85_OHM_DIFF	ISL2,ISL11	Y	0.078 MM	0.078 MM		0.160 MM	0.160 MM
85_OHM_DIFF	ISL3,ISL10	Y	0.078 MM	0.078 MM		0.160 MM	0.160 MM
85_OHM_DIFF	ISL4,ISL9	Y	0.082 MM	0.082 MM		0.140 MM	0.140 MM
85_OHM_DIFF	*	N	100 MM	100 MM	=STANDARD	=STANDARD	=STANDARD

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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	0.100 MM	0.100 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP,BOTTOM	=2x_DIELECTRIC	?
CPU_AGTL	*	=STANDARD	?

Note: CPU_8MIL and CPU_ITP can be converted back to TABLE_SPACING_RULE once rdar://10308147 is resolved

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_8MIL	*	*	CPU_8MIL_2ANY

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_8MIL_2ANY	*	8 MIL	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_ITP	*	*	CPU_ITP_2ANY

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_ITP_2ANY	*	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	CPU_COMP	*	CPU_COMP_2SELF
CPU_COMP	*	*	CPU_COMP_20THER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_COMP_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CPU_COMP_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_COMP_2SELF	*	=4x_DIELECTRIC	?
CPU_COMP_20THER	*	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	CPU_VCCSENSE	*	CPU_VCCSENSE_2SELF
CPU_VCCSENSE	*	*	CPU_VCCSENSE_20THER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CPU_VCCSENSE_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE_2SELF	*	=4x_DIELECTRIC	?
CPU_VCCSENSE_20THER	*	=6x_DIELECTRIC	?

PCI-Express Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
CLK_PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

PCIe Clock Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	CLK_PCIE	*	CLK_PCIE_2SELF
CLK_PCIE	*	*	CLK_PCIE_20THER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_2SELF	TOP,BOTTOM	=6x_DIELECTRIC	?
CLK_PCIE_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_2SELF	*	=4x_DIELECTRIC	?
CLK_PCIE_20THER	*	=6x_DIELECTRIC	?

CPU PCIe Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_CPU_TX	PCIE_CPU_TX	*	PCIE_TX2TX
PCIE_CPU_RX	PCIE_CPU_RX	*	PCIE_RX2RX
PCIE_CPU_TX	*_CPU_TX	*	PCIE_TX20THERTX
PCIE_CPU_RX	*_CPU_RX	*	PCIE_RX20THERRX
PCIE_CPU_TX	*_CPU_RX	*	PCIE_TX2RX
PCIE_CPU_RX	*_CPU_TX	*	PCIE_RX2TX
PCIE_CPU_TX	*_TX	*	PCIE_20THERHS
PCIE_CPU_RX	*_TX	*	PCIE_20THERHS
PCIE_CPU_TX	*_RX	*	PCIE_20THERHS
PCIE_CPU_RX	*_RX	*	PCIE_20THERHS
PCIE_CPU_TX	*	*	PCIE_20THER
PCIE_CPU_RX	*	*	PCIE_20THER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_TX2TX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_RX2RX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_TX20THERTX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_RX20THERRX	TOP,BOTTOM	=5x_DIELECTRIC	?
PCIE_TX2RX	TOP,BOTTOM	=7x_DIELECTRIC	?
PCIE_RX2TX	TOP,BOTTOM	=7x_DIELECTRIC	?
PCIE_20THERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
PCIE_20THER	TOP,BOTTOM	=5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCIE_TX2TX	*	=2.5x_DIELECTRIC	?
PCIE_RX2RX	*	=2.5x_DIELECTRIC	?
PCIE_TX20THERTX	*	=4x_DIELECTRIC	?
PCIE_RX20THERRX	*	=4x_DIELECTRIC	?
PCIE_TX2RX	*	=6x_DIELECTRIC	?
PCIE_RX2TX	*	=6x_DIELECTRIC	?
PCIE_20THERHS	*	=4x_DIELECTRIC	?
PCIE_20THER	*	=3x_DIELECTRIC	?

PCH PCIe Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_PCH_TX	PCIE_PCH_TX	*	PCIE_TX2TX
PCIE_PCH_RX	PCIE_PCH_RX	*	PCIE_RX2RX
PCIE_PCH_TX	*_PCH_TX	*	PCIE_TX20THERTX
PCIE_PCH_RX	*_PCH_RX	*	PCIE_RX20THERRX
PCIE_PCH_TX	*_PCH_RX	*	PCIE_TX2RX
PCIE_PCH_RX	*_PCH_TX	*	PCIE_RX2TX
PCIE_PCH_TX	*_TX	*	PCIE_20THERHS
PCIE_PCH_RX	*_TX	*	PCIE_20THERHS
PCIE_PCH_TX	*_RX	*	PCIE_20THERHS
PCIE_PCH_RX	*_RX	*	PCIE_20THERHS
PCIE_PCH_TX	*	*	PCIE_20THER
PCIE_PCH_RX	*	*	PCIE_20THER

Note: DisplayPort tables are on Page 113

<http://www.xinxunwei.com> TEL:400-800-9990 QQ:4006538586

CPU Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE	
		PHYSICAL	SPACING
	CPU_PECT	CPU_45S	CPU_COMP
	PM_SYNC	CPU_45S	CPU_AGTL
	PM_MEM_PWRGD	CPU_45S	CPU_AGTL
		CPU_45S	CPU_ITP
		CPU_45S	CPU_ITP
		CPU_45S	CPU_ITP
		CPU_27P4S	CPU_COMP
		CPU_27P4S	CPU_COMP
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP
	CPU_SM_RCOMP	CPU_27P4S	CPU_COMP
		CPU_45S	CPU_ITP
	CPU_CATERR_I	CPU_45S	CPU_AGTL
		CPU_45S	CPU_AGTL
	CPU_PROCHOT_I	CPU_45S	CPU_AGTL
	CPU_PWRGD	CPU_45S	CPU_AGTL
	PM_THRMTRIP_I	CPU_45S	CPU_RMTI
	DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE
	DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE
	DP_L1_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE
	DP_L1_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE
	ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE
	XDP_TDI	CPU_45S	CPU_ITP
	XDP_TDO	CPU_45S	CPU_ITP
	XDP_TMS	CPU_45S	CPU_ITP
	XDP_TCK	CPU_45S	CPU_ITP
	XDP_TRST_I	CPU_45S	CPU_ITP
	XDP_BPM_I	CPU_45S	CPU_ITP
		CPU_45S	CPU_ITP
		CPU_45S	CPU_ITP
	(FSB_CPURST_I)	CPU_45S	CPU_ITP
	CPU_VCCSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE
	CPU_VCCSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE
	CPU_VCCIOSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE
	CPU_VCCIOSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE
	CPU_AXG_SENSE	SENSE_1T01_P2MM	CPU_VCCSENSE
	CPU_AXG_SENSE	SENSE_1T01_P2MM	CPU_VCCSENSE
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE
	CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE
	CPU_VIDALERT_I	CPU_45S	CPU_COMP
	CPU_SVIDSCLK	CPU_45S	CPU_COMP
	CPU_SVIDSOUT	CPU_45S	CPU_COMP
	PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX
	PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX
		PCIE_80D	PCIE_CPU_TX
		PCIE_80D	PCIE_CPU_TX
		PCIE_80D	PCIE_CPU_RX
		PCIE_80D	PCIE_CPU_RX
	PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX
	PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX
	PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE
	PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE
	DP_TBT_ML	DP_80D	DP_TX
	DP_TBT_ML	DP_80D	DP_TX
		DP_80D	DP_TX
		DP_80D	DP_TX
	DP_TBT_AUXCH	DP_80D	DP_AUX
	DP_TBT_AUXCH	DP_80D	DP_AUX
		DP_80D	DP_AUX
		DP_80D	DP_AUX
	DP_TBT_ML	DP_80D	DP_TX
	DP_TBT_ML	DP_80D	DP_TX
		DP_80D	DP_TX
		DP_80D	DP_TX
	DP_TBT_AUXCH	DP_80D	DP_AUX
	DP_TBT_AUXCH	DP_80D	DP_AUX
		DP_80D	DP_AUX
		DP_80D	DP_AUX
	DP_INT_ML	DP_80D	DP_TX
	DP_INT_ML	DP_80D	DP_TX
		DP_80D	DP_TX
		DP_80D	DP_TX
	DP_INT_AUXCH	DP_80D	DP_AUX
	DP_INT_AUXCH	DP_80D	DP_AUX
	DP_INT_AUXCH	DP_80D	DP_AUX
	DP_INT_AUXCH	DP_80D	DP_AUX
		DP_80D	DP_AUX
		DP_80D	DP_AUX

PCie SSD

DP

SYNC MASTER=J43 MLB		SYNC DATE=09/21/2012	
PAGE TITLE		CPU Constraints	
 Apple Inc.		DRAWING NUMBER	SIZE
		<SCH_NUM>	D
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SATA Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_ICOMP	*	=4x_DIELECTRIC	?

PCH Net Properties

[illegible]

USB Hucopyb nets

TP SPI nets

USB EXTA nets (Right USB port)

USB EXTB nets (Left USB port)

UART Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
UART_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
UART	*	=2x_DIELECTRIC	?

USB 2.0 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIAS	*	=STANDARD	8 MIL	8 MIL	=STANDARD	=STANDARD	=STANDARD
USB_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=2x_DIELECTRIC	?	USB	TOP, BOTTOM	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.8

USB 3.0 Interface Constraints

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_PCH_TX	USB3_PCH_TX	*	USB3_TX2TX	USB3_TX2TX	TOP, BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_RX	USB3_PCH_RX	*	USB3_RX2RX	USB3_RX2RX	TOP, BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_TX	*_PCH_TX	*	USB3_TX20THERTX	USB3_TX20THERTX	TOP, BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_RX	*_PCH_RX	*	USB3_RX20THERRX	USB3_RX20THERRX	TOP, BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_TX	*_PCH_RX	*	USB3_TX2RX	USB3_TX2RX	TOP, BOTTOM	=7x_DIELECTRIC	?
USB3_PCH_RX	*_PCH_TX	*	USB3_RX2TX	USB3_RX2TX	TOP, BOTTOM	=7x_DIELECTRIC	?
USB3_PCH_TX	*_TX	*	USB3_20THERHS	USB3_20THERHS	TOP, BOTTOM	=6x_DIELECTRIC	?
USB3_PCH_RX	*_TX	*	USB3_20THERHS	USB3_20THER	TOP, BOTTOM	=5x_DIELECTRIC	?
USB3_PCH_TX	*_RX	*	USB3_20THERHS				
USB3_PCH_RX	*_RX	*	USB3_20THERHS				
USB3_PCH_TX	*	*	USB3_20THER				
USB3_PCH_RX	*	*	USB3_20THER				

SOURCE: 471984_Cheif_River_MS_PDG_1.0 and the spacing rule is adjusted per SI team feedback.

8			7			6					
LPC Bus Constraints											
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP				
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD				
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD				

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=3x_DIELECTRIC	?
CLK_LPC	*	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibox Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S_R_50S	TOP,BOTTOM	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE		
SMB_45S_R_50S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HDAudio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibox Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=4x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=4x_DIELECTRIC	?

XDP Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_ITP	*	=2:1_SPACING	?

DisplayPort

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2DP	*	=3x_DIELECTRIC	?
DP_20THERHS	*	=4x_DIELECTRIC	?
DP_20THER	*	=3x_DIELECTRIC	?
DP_AUX	*	=3x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DP_TX	DP_TX	*	DP_2DP
DP_TX	*_TX	*	DP_20THERHS
DP_TX	*_RX	*	DP_20THERHS
DP_TX	*	*	DP_20THER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		
		PHYSICAL	SPACING	
IPC_AD	IPC_45S	IPC	LPC_AD<3..0>	14 35 44 62
IPC_FRAME_L	IPC_45S	IPC	LPC_FRAME_L	14 35 44 62
IPC_45S	IPC_45S	IPC	LPCPLUS_RESET_L	18 44 62
IPC_CLK33M	CLK_IPC_45S	CLK_IPC	LPC_CLK24M_SMC	17 35
IPC_CLK33M	CLK_IPC_45S	CLK_IPC	LPC_CLK24M_SMC_R	12 17
IPC_CLK33M	CLK_IPC_45S	CLK_IPC	LPC_CLK24M_LPCPLUS	17 44 62
IPC_CLK33M	CLK_IPC_45S	CLK_IPC	LPC_CLK24M_LPCPLUS_R	12 17
SMBUS_PCH_CLK	SMB_45S_R_50S	SMB	SMBUS_PCH_CLK	14 16 19 25 38 54
SMBUS_PCH_DATA	SMB_45S_R_50S	SMB	SMBUS_PCH_DATA	14 16 19 25 38 54
SMBUS_PCH_0_CLK	SMB_45S_R_50S	SMB	SML_PCH_0_CLK	14 38
SMBUS_PCH_0_DATA	SMB_45S_R_50S	SMB	SML_PCH_0_DATA	14 38
SMBUS_SMC_1_S0_SCL	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SCL	11 32 35 38 41 42 62
SMBUS_SMC_1_S0_SDA	SMB_45S_R_50S	SMB	SMBUS_SMC_1_S0_SDA	11 32 35 38 41 42 62
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	12 59 63
HDA_BIT_CLK_R	HDA_45S	HDA	HDA_BIT_CLK_R	12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12 59 63
HDA_SYNC_R	HDA_45S	HDA	HDA_SYNC_R	12
HDA_RST_L	HDA_45S	HDA	HDA_RST_R_L	12
HDA_RST_L	HDA_45S	HDA	HDA_RST_L	12 59 63
HDA_SDIN0	HDA_45S	HDA	HDA_SDIN0	12 59 63
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	12 59 63
HDA_SDOUT_R	HDA_45S	HDA	HDA_SDOUT_R	12 17
PM_SUS_CLK	CLK_SLOW_45S	CLK_SLOW	PM_CLK32K_SUSCLK_R	13 36
CLK_SLOW_45S	CLK_SLOW_45S	CLK_SLOW	SMC_CLK32K	35 36
SPT_CLK	SPT_45S	SPT	SPI_CLK_R	14 44
SPT_45S	SPT_45S	SPT	SPI_CLK	44
SPT_MOSI	SPT_45S	SPT	SPI_MOSI_R	14 44
SPT_MOSI	SPT_45S	SPT	SPI_MOSI	44
SPT_MISO	SPT_45S	SPT	SPI_MISO	14 44
SPT_MISO	SPT_45S	SPT	SPI_MISO_R	44
SPT_CS0	SPT_45S	SPT	SPT_CS0_R_L	14 44
SPT_CS0	SPT_45S	SPT	SPT_CS0_L	44
SPT_45S	SPT_45S	SPT	SPI_SMC_CLK	35 44
SPT_45S	SPT_45S	SPT	SPI_SMC_MOSI	35 44
SPT_45S	SPT_45S	SPT	SPI_SMC_MISO	35 44
SPT_45S	SPT_45S	SPT	SPI_SMC_CS_L	35 44
SPT_45S	SPT_45S	SPT	SPI_MLB_CLK	44
SPT_45S	SPT_45S	SPT	SPI_MLB_MOSI	44
SPT_45S	SPT_45S	SPT	SPI_MLB_MISO	44
SPT_45S	SPT_45S	SPT	SPI_MLB_CS_L	44
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_P	29 62
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_N	29 62
PCIE_80D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_C_P	14 29
PCIE_80D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_C_N	14 29
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_P	14 29 62
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_N	14 29 62
PCIE_CLK100M_AP	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_P	12 29 62
PCIE_CLK100M_AP	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_AP_N	12 29 62
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_P<3..0>	25
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_N<3..0>	25
PCIE_80D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_P<3..0>	14 25
PCIE_80D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_N<3..0>	14 25
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_P<3..0>	14 25
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_N<3..0>	14 25
PCIE_80D	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_P<3..0>	25
PCIE_80D	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_N<3..0>	25
PCIE_CLK100M_TBT	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_TBT_P	12 25
PCIE_CLK100M_TBT	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_TBT_N	12 25
CLK_PCIE_80D	CLK_PCIE_80D	CLK_PCIE	PEG_CLK100M_P	
CLK_PCIE_80D	CLK_PCIE_80D	CLK_PCIE	PEG_CLK100M_N	
XDP_TDI	PCH_45S	PCH_ITP	XDP_PCH_TDI	12 16 62
XDP_TDO	PCH_45S	PCH_ITP	XDP_PCH_TDO	12 16 62
XDP_TMS	PCH_45S	PCH_ITP	XDP_PCH_TMS	12 16 62
XDP_TCK	PCH_45S	PCH_ITP	XDP_PCH_TCK	12 16 62
PCIE_CAM	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_P	31 32
PCIE_CAM	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_N	31 32
PCIE_80D	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_C_P	14 32
PCIE_80D	PCIE_80D	PCIE_PCH_TX	PCIE_CAMERA_R2D_C_N	14 32
PCIE_CAM	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_P	14 32
PCIE_CAM	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_N	14 32
PCIE_80D	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_C_P	31 32
PCIE_80D	PCIE_80D	PCIE_PCH_RX	PCIE_CAMERA_D2R_C_N	31 32
PCIE_CLK100M_CAM	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_P	12 32
PCIE_CLK100M_CAM	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_N	12 32
CLK_PCIE_80D	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_P	31 32
CLK_PCIE_80D	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_CAMERA_C_N	31 32

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTCX1	
	SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA	17 32
		CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP	31 32
		CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R	32
		CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP	32
		CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN	32
		CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN	31 32
	SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	17 25
		CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	25
	SYSCLK_CLK25M_XTAL	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1	17
		CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2	17
		CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R	17
		CLK_25M_45S	CLK_25M	SDCLK_CLK25M_X2	
		CLK_25M_45S	CLK_25M	SDCLK_CLK25M_X2_R	
		CLK_25M_45S	CLK_25M	SDSCLK_CLK25M_X1	

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8	7	6	5	4	3	2	1
Memory Bus Constraints							
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_73D	*	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DATA2OTHERMEM	*	=8x_DIELECTRIC	?
MEM_QS2OWNDATA	*	=3x_DIELECTRIC	?
MEM_CMD2CMD	*	=3x_DIELECTRIC	?
MEM_CMD2CTRL	*	=3x_DIELECTRIC	?
MEM_CTRL2CTRL	*	=3x_DIELECTRIC	?
MEM_CLK2CLK	*	=6x_DIELECTRIC	?
MEM_20THERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	10000
MEM_2GND	*	=2x_DIELECTRIC	10000
MEM_20THER	*	=6x_DIELECTRIC	?

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_70D	MEM_TERM	MEM_73D
MEM_40S	MEM_TERM	MEM_50S

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DATA_0	*	MEM_QS2OWNDATA
MEM_A_DQS_1	MEM_A_DATA_1	*	MEM_QS2OWNDATA
MEM_A_DQS_2	MEM_A_DATA_2	*	MEM_QS2OWNDATA
MEM_A_DQS_3	MEM_A_DATA_3	*	MEM_QS2OWNDATA
MEM_A_DQS_4	MEM_A_DATA_4	*	MEM_QS2OWNDATA
MEM_A_DQS_5	MEM_A_DATA_5	*	MEM_QS2OWNDATA
MEM_A_DQS_6	MEM_A_DATA_6	*	MEM_QS2OWNDATA
MEM_A_DQS_7	MEM_A_DATA_7	*	MEM_QS2OWNDATA
MEM_B_DQS_0	MEM_B_DATA_0	*	MEM_QS2OWNDATA
MEM_B_DQS_1	MEM_B_DATA_1	*	MEM_QS2OWNDATA
MEM_B_DQS_2	MEM_B_DATA_2	*	MEM_QS2OWNDATA
MEM_B_DQS_3	MEM_B_DATA_3	*	MEM_QS2OWNDATA
MEM_B_DQS_4	MEM_B_DATA_4	*	MEM_QS2OWNDATA
MEM_B_DQS_5	MEM_B_DATA_5	*	MEM_QS2OWNDATA
MEM_B_DQS_6	MEM_B_DATA_6	*	MEM_QS2OWNDATA
MEM_B_DQS_7	MEM_B_DATA_7	*	MEM_QS2OWNDATA

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DATA_*	MEM_*	*	MEM_DATA2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTRL	*	MEM_CMD2CTRL
MEM_CTRL	MEM_CTRL	*	MEM_CTRL2CTRL

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*	MEM_*	*	MEM_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	*	*	MEM_20THER
MEM_A_DQS_1	*	*	MEM_20THER
MEM_A_DQS_2	*	*	MEM_20THER
MEM_A_DQS_3	*	*	MEM_20THER
MEM_A_DQS_4	*	*	MEM_20THER
MEM_A_DQS_5	*	*	MEM_20THER
MEM_A_DQS_6	*	*	MEM_20THER
MEM_A_DQS_7	*	*	MEM_20THER
MEM_B_DQS_0	*	*	MEM_20THER
MEM_B_DQS_1	*	*	MEM_20THER
MEM_B_DQS_2	*	*	MEM_20THER
MEM_B_DQS_3	*	*	MEM_20THER
MEM_B_DQS_4	*	*	MEM_20THER
MEM_B_DQS_5	*	*	MEM_20THER
MEM_B_DQS_6	*	*	MEM_20THER
MEM_B_DQS_7	*	*	MEM_20THER

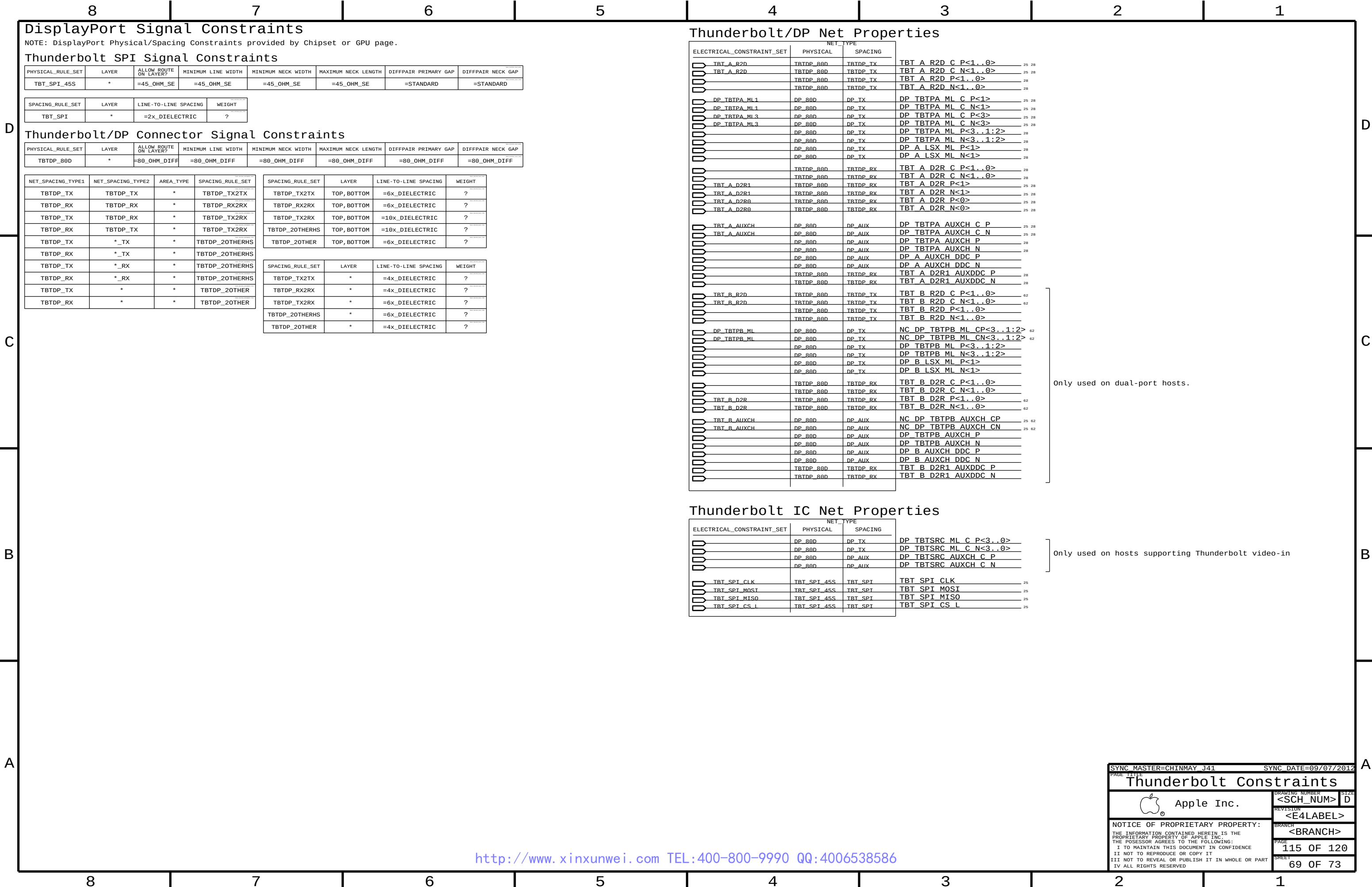
MEM_A_DATA_0	*	*	MEM_20THER
MEM_A_DATA_1	*	*	MEM_20THER
MEM_A_DATA_2	*	*	MEM_20THER
MEM_A_DATA_3	*	*	MEM_20THER
MEM_A_DATA_4	*	*	MEM_20THER
MEM_A_DATA_5	*	*	MEM_20THER
MEM_A_DATA_6	*	*	MEM_20THER
MEM_A_DATA_7	*	*	MEM_20THER
MEM_B_DATA_0	*	*	MEM_20THER
MEM_B_DATA_1	*	*	MEM_20THER
MEM_B_DATA_2	*	*	MEM_20THER
MEM_B_DATA_3	*	*	MEM_20THER
MEM_B_DATA_4	*	*	MEM_20THER
MEM_B_DATA_5	*	*	MEM_20THER
MEM_B_DATA_6	*	*	MEM_20THER
MEM_B_DATA_7	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_CTRL	*	*	MEM_20THER
MEM_CLK	*	*	MEM_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DATA_0	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_1	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_2	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_3	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_4	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_5	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_6	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_A_DATA_7	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_0	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_1	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_2	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_3	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_4	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_5	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_6	MEM_*_DATA_*	*	MEM_20THERMEM
MEM_B_DATA_7	MEM_*_DATA_*	*	MEM_20THERMEM

Memory Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM A CLK P<0>
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM A CLK N<0>
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM A CLK P<1>
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM A CLK N<1>
MEM_A_CTRL	MEM_40S	MEM_CTRL	MEM A CS L<1..0>
MEM_A_CTRL	MEM_40S	MEM_CTRL	MEM A ODT<0>
MEM_A_CKE0	MEM_40S	MEM_CMD	MEM A CKE<1..0>
MEM_A_CKE1	MEM_40S	MEM_CMD	MEM A CKE<3..2>
MEM_A_CMD0	MEM_40S	MEM_CMD	MEM A CAA<9..0>
MEM_A_CMD1	MEM_40S	MEM_CMD	MEM A CAB<9..0>
MEM_A_DQ_BYTE0	MEM_40S	MEM_A_DATA_0	MEM A DQ<7..0>
MEM_A_DQ_BYTE1	MEM_40S	MEM_A_DATA_1	MEM A DQ<15..8>
MEM_A_DQ_BYTE2	MEM_40S	MEM_A_DATA_2	MEM A DQ<23..16>
MEM_A_DQ_BYTE3	MEM_40S	MEM_A_DATA_3	MEM A DQ<31..24>
MEM_A_DQ_BYTE4	MEM_40S	MEM_A_DATA_4	MEM A DQ<39..32>
MEM_A_DQ_BYTE5	MEM_40S	MEM_A_DATA_5	MEM A DQ<47..40>
MEM_A_DQ_BYTE6	MEM_40S	MEM_A_DATA_6	MEM A DQ<55..48>
MEM_A_DQ_BYTE7	MEM_40S	MEM_A_DATA_7	MEM A DQ<63..56>
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM A DQS P<0>
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM A DQS N<0>
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM A DQS P<1>
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM A DQS N<1>
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM A DQS P<2>
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM A DQS N<2>
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM A DQS P<3>
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM A DQS N<3>
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM A DQS P<4>
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM A DQS N<4>
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM A DQS P<5>
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM A DQS N<5>
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM A DQS P<6>
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM A DQS N<6>
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM A DQS P<7>
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM A DQS N<7>
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM B CLK P<0>
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM B CLK N<0>
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM B CLK P<1>
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM B CLK N<1>
MEM_B_CTRL	MEM_40S	MEM_CTRL	MEM B CS L<1..0>
MEM_B_CTRL	MEM_40S	MEM_CTRL	MEM B ODT<0>
MEM_B_CKE0	MEM_40S	MEM_CMD	MEM B CKE<1..0>
MEM_B_CKE1	MEM_40S	MEM_CMD	MEM B CKE<3..2>
MEM_B_CMD0	MEM_40S	MEM_CMD	MEM B CAA<9..0>
MEM_B_CMD1	MEM_40S	MEM_CMD	MEM B CAB<9..0>
MEM_B_DQ_BYTE0	MEM_40S	MEM_B_DATA_0	MEM B DQ<7..0>
MEM_B_DQ_BYTE1	MEM_40S	MEM_B_DATA_1	MEM B DQ<15..8>
MEM_B_DQ_BYTE2	MEM_40S	MEM_B_DATA_2	MEM B DQ<23..16>
MEM_B_DQ_BYTE3	MEM_40S	MEM_B_DATA_3	MEM B DQ<31..24>
MEM_B_DQ_BYTE4	MEM_40S	MEM_B_DATA_4	MEM B DQ<39..32>
MEM_B_DQ_BYTE5	MEM_40S	MEM_B_DATA_5	MEM B DQ<47..40>
MEM_B_DQ_BYTE6	MEM_40S	MEM_B_DATA_6	MEM B DQ<55..48>
MEM_B_DQ_BYTE7	MEM_40S	MEM_B_DATA_7	MEM B DQ<63..56>
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM B DQS P<0>
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM B DQS N<0>
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM B DQS P<1>
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM B DQS N<1>
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM B DQS P<2>
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM B DQS N<2>
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM B DQS P<3>
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM B DQS N<3>
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM B DQS P<4>
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM B DQS N<4>
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM B DQS P<5>
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM B DQS N<5>
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM B DQS P<6>
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM B DQS N<6>
MEM_B_DQS7	MEM_70D	MEM_B_DQS_7	MEM B DQS P<7>
MEM_B_DQS7	MEM_70D	MEM_B_DQS_7	MEM B DQS N<7>
		MEM_PWR	PP1V2 S3
		MEM_PWR	PP0V6 S3 MEM VREFCA A
		MEM_PWR	PP0V6 S3 MEM VREFDQ A
		MEM_PWR	PP0V6 S3 MEM VREFCA B
		MEM_PWR	PP0V6 S3 MEM VREFDQ B

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MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MIPI_850	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MIPI_20THER	*	=4X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?
MIPICKL_20THER	*	=70X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICKL_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?
S2_DQS20WNDATA	*	=2X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?
S2MEM_2PWR	*	=2X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_DQS20WNDATA	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CMD	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CTRL2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_20THERMEM	TOP,BOTTOM	=6X_DIELECTRIC	?
S2MEM_2PWR	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_2GND	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER
S2_MEM_DQS*	*	*	S2MEM_20THER
S2_MEM_CMD	*	*	S2MEM_20THER
S2_MEM_CTRL	*	*	S2MEM_20THER
S2_MEM_CLK	*	*	S2MEM_20THER
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WNDATA
S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WNDATA

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE			
	PHYSICAL	SPACING		
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_P	31 32
S2_MEM_CLK	S2_MEM_85D	S2_MEM_CLK	MEM_CAM_CLK_N	31 32
S2_MEM_CTRL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CKE	31 32
S2_MEM_CTRL	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_ODT	32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_CAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CTRL	MEM_CAM_RAS_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_WE_L	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<0>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<1>	31 32
S2_MEM_CMD	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_BA<2>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_P<0>	31 32
S2_MEM_DQS0	S2_MEM_85D	S2_MEM_DQS0	MEM_CAM_DQS_N<0>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_P<1>	31 32
S2_MEM_DQS1	S2_MEM_85D	S2_MEM_DQS1	MEM_CAM_DQS_N<1>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DM<0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DM<1>	31 32
S2_MEM_A	S2_MEM_45S	S2_MEM_CMD	MEM_CAM_A<14..0>	31 32
S2_MEM_DATA_0	S2_MEM_45S	S2_MEM_DATA0	MEM_CAM_DQ<7..0>	31 32
S2_MEM_DATA_1	S2_MEM_45S	S2_MEM_DATA1	MEM_CAM_DQ<15..8>	31 32
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_P	31 32
MIPI_DATA_S2	MIPI_85D	MIPI_DATA	MIPI_DATA_N	31 32
	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_P	32 62
	MIPI_85D	MIPI_DATA	MIPI_DATA_CONN_N	32 62
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_P	31 32
MIPI_CLK_S2	MIPI_85D	CLK_MIPI	MIPI_CLK_N	31 32
	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_P	32 62
	MIPI_85D	CLK_MIPI	MIPI_CLK_CONN_N	32 62
		S2_MEM_PWR	PP1V35_CAM	31 32
		S2_MEM_PWR	PP0V675_CAM_VREF	31 32
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFCA	32
		S2_MEM_PWR	PP0V675_MEM_CAM_VREFDQ	32

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	Kismet: afp:///kismet.apple.com/Kismet-Projects/J41-J43						
	Useful Wiki Links: Schematic Conventions - https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design						
C	MobileMac HW Radar: <rdar:///component/497591> MobileMac HW Task <rdar:///component/497587> MobileMac HW Schematic <rdar:///component/497585> MobileMac HW New Bugs <rdar:///component/497588> MobileMac HW Layout <rdar:///component/497590> MobileMac HW Investigation <rdar:///component/497589> MobileMac HW Architecture						C
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